

APPROVED as to form only:
By Stefanie Boster at 4:48 pm, Feb 01 2024

Date Prepared: January 29, 2024

Consultant/Contractor: _____ Date: _____

City of Cheyenne

DATE: _____

Mayor Patrick Collins

/_____
/ATTEST: City Clerk Kristina F. Jones



January 19, 2024

Mr. Samuele Quattrini
Landfill Manager
City of Cheyenne
1461 Happy Jack Rd.
Cheyenne, WY 82009

RE: 2024 Proposal for Conducting Routine Environmental Monitoring Activities for the City of Cheyenne Sanitary Landfill (Happy Jack Landfill), SHWD File #10.080, Cheyenne, Wyoming

Dear Mr. Quattrini:

Trihydro Corporation appreciates the opportunity to provide this cost proposal to the City of Cheyenne (City) for the 2024 environmental monitoring at the Cheyenne Sanitary Landfill (Landfill). This proposal is for a one-year extension of the existing Happy Jack Landfill Environmental Monitoring Contract, #6798, between the City of Cheyenne and Trihydro Corporation, set to expire on March 1, 2024. Work will fulfill requirements permitted under the Sanitary Landfill Environmental Monitoring Plan (SLEMP) originally dated April 12, 2017, and revised March 14, 2019.

SCOPE OF WORK

Our project understanding and the scope of work for each proposed task is outlined below for your review and consideration. The scope of work was developed based on our experience conducting the environmental monitoring at the Landfill since 2002 and from our experience working with the Wyoming Department of Environmental Quality, Solid & Hazardous Waste Division (WDEQ/SHWD) on dozens of sites across Wyoming. A detailed cost estimate for each task and a description of related activities and assumptions is provided as Attachment A.

Task 1A and 2A – April and August Environmental Monitoring

Semiannual groundwater monitoring will be conducted in April and August whereby static water levels will be gauged in up to 40 wells and 2 leachate sumps, and groundwater samples collected from up to 36 wells and 1 leachate sump. Samples will be submitted for laboratory analysis of the Wyoming Solid Waste Rules and Regulations (WSWRR), Chapter 2, Appendix A, B, and/or C constituents, as designated by the SLEMP. Methane monitoring of structures and perimeter wells will also be conducted during these semiannual events.

Our 2024 cost estimate for the April and August semiannual monitoring events is \$59,283, which includes an estimated \$28,875 for laboratory services. Despite our aggressive and beneficial negotiations



Mr. Sam Quattrini
January 19, 2024
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with three competitive laboratories over the last year, analytical costs are almost double what they were six years ago; costs ballooned in the wake of COVID during a period of large laboratory mergers, supply issues, inflation, and courier costs. Based on our most recent experience and negotiations, Trihydro believes it most advantageous to remain with Eurofins Denver (formerly TestAmerica Denver) as they are currently providing reliable and economical service. A copy of the related quote and annual extension is provided as Attachment B. The cost at similarly capable laboratories would be, at a minimum, 20% higher, which is equivalent to \$6,000 or more.

Based on our relationship with Eurofins paired with thoughtful and effective negotiations, Eurofins has agreed to honor last year's quote through 2024, including a flat rate for standard overnight shipping at \$60/cooler. Many laboratories are no longer providing pre-paid shipping due to the erratic market. If Eurofins had not extended their shipping agreement, shipping costs would have increased by at least \$190/cooler, or \$4,750 per year (assuming 25 coolers). We are pleased that our negotiations are resulting in a substantial cost savings to the City.

Task 1B and 2B – April and August Environmental Reporting

Semiannual reports will be prepared for the April and August environmental monitoring events and include data management, data validation, and report preparation. Reports will provide a narrative of field, laboratory, validation activities, and results as well as observations, conclusions, and recommendations. The report will also provide relevant tables, figures, and appendices, as outlined in Attachment A. Our 2024 cost estimate for the April and August semiannual reporting is \$14,320.

Task 3 – Two Quarterly Methane Monitoring Events

Under Task 3, two quarterly methane monitoring events will be conducted in February and December. During each event, 14 perimeter methane probes, five passive gas vents, and four structures (scale house, office, shop area, and equipment building) will be visited to measure and verify that methane concentrations are within permissible limits. (The other two quarterly events will be covered under Tasks 1A and 1B, in April and August.) Our 2024 cost estimate for this task is \$4,006.

Task 4 – Annual Environmental Reporting

Following collection of all data required for monitoring during 2024, an Annual Report will be prepared to summarize findings over the course of the year. Statistical analysis of groundwater in each of the three water-bearing zones will also be conducted under this task. Our proposed budget for this task is \$11,633, including approximately \$4,800 for the statistical analysis of the three water-bearing zones.



Mr. Sam Quattrini
January 19, 2024
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Task 5 – Optional Technical Assistance

For the coming year, we have included a task dedicated to optional technical assistance. For many of our Wyoming landfill projects, Trihydro includes a budget for technical and regulatory assistance for unanticipated tasks that may arise during the year. If agreeable, Trihydro proposes a \$5,000 technical and regulatory assistance budget in the event work is requested to address issues that are outside of existing work orders (e.g., responses to new regulatory issues, additional site activities, well repairs, etc.). No work would be completed under this task without written approval from the City. For any activities exceeding the \$5,000 allocated for this task, Trihydro will prepare a separate cost proposal.

CLOSING

Our overall cost estimate for the 2024 Environmental Monitoring and Reporting project is \$94,242 and our Standard Schedule of Charges is included as Attachment C. As always, Trihydro will strive to complete the requested activities as efficiently and cost-effectively as possible, and we are available to start the work upon receipt of a purchase order. If you have any questions or require additional information regarding this proposal and/or contracts, please do not hesitate to contact us at (307) 745-7474.

Sincerely,
Trihydro Corporation

Caroline Brewer, P.G.
Project Manager

Marjorie Bedessem, Ph.D, P.E.
Project Director

P9999-024-0095

Attachments

ATTACHMENT A

**ACTIVITIES, ASSUMPTIONS, AND COSTS
2024 ENVIRONMENTAL MONITORING AND REPORTING**

**ATTACHMENT A. ACTIVITIES, ASSUMPTIONS, AND COSTS FOR
2024 ENVIRONMENTAL MONITORING AND REPORTING
CHEYENNE SANITARY LANDFILL, LARAMIE COUNTY, WYOMING**



Activity	Schedule of Charges	Hourly Rate or Charge	Hours or Unit Estimate	Subtotal
Task 1A April Environmental Monitoring	Professional Level 9	\$178	2	\$356
	Professional Level 8	\$164	4	\$656
	Professional Level 5	\$125	4	\$500
	Professional Level 4	\$113	4	\$452
	Professional Level 3	\$102	100	\$10,200
	Technical Level 2	\$76	1	\$76
<u>Activities</u>				
- Project coordination with City of Cheyenne (City), Laboratory, and Wyoming Department of Environmental Quality (WDEQ).				
- Update and implement site-specific health & safety activities.				
- Perform semiannual monitoring activities in accordance with the Sanitary Landfill Environmental Monitoring Plan (SLEMP).				
- Gauge static water levels as indicated in up to 40 wells and 2 leachate sumps.				
- Provide laboratory analysis for Wyoming Solid Waste Rules & Regulations (WSWRR), Chapter 2, Appendix A & C parameters for up to 28 wells and 1 leachate sump.				
- Provide laboratory analysis for WSWRR, Chapter 2, Appendix B (VOCs and inorganics) and C for up to 8 wells.				
- Collect and provide analysis for 1 equipment blank, 2 blind duplicates, and up to 8 trip blanks (for VOCs).				
- Monitor 14 perimeter methane probes and landfill structures, including the office, the shop area, and the equipment building.				
	Labor			\$12,240
	Equipment	\$2,010	1	\$2,010
	Travel	0.670	1000	\$670
	Shipping	\$300	1	\$300
	Supplies	\$200	1	\$200
<u>Assumptions</u>				
- City will provide access to groundwater monitoring well locations.				
- Purge water may be discharged on-site.				
- Laboratory costs will be billed through Trihydro.				
- In the event that commercial carriers are unable to deliver samples within required holding times, or laboratory receives damaged sample containers, the City will be notified, but wells will not be resampled unless the City approves a change order for additional sampling costs.				
- Sampling activities will be completed in 10 days.				
- Sampling staff will mobilize from Laramie or Cheyenne, WY.				
- Equipment costs include, pump controller, water quality meter, electronic down-rigger, and bailers.				
- Supply costs include cord, ice, paper towels, etc.				
- The City will provide the air compressor for sampling.				
- Trihydro will coordinate equipment, bottle, and sample shipping through Eurofins Denver.				
	Expenses			\$3,180
	Eurofins Denver	\$17,500	1.1	\$19,250
	Subcontractors			\$19,250
Activity Task 1A Subtotal				\$34,670

**ATTACHMENT A. ACTIVITIES, ASSUMPTIONS, AND COSTS FOR
2024 ENVIRONMENTAL MONITORING AND REPORTING
CHEYENNE SANITARY LANDFILL, LARAMIE COUNTY, WYOMING**



Activity	Schedule of Charges	Hourly Rate or Charge	Hours or Unit Estimate	Subtotal
Task 1B April Environmental Reporting	Technical Specialist 1	\$225	2	\$450
	Professional Level 10	\$192	0	\$0
	Professional Level 8	\$164	10	\$1,640
	Professional Level 7	\$149	12	\$1,788
	Professional Level 5	\$125	20	\$2,500
	Technical Level 5	\$105	6	\$630
	Administrative 3	\$76	2	\$152
	Labor			\$7,160
	Equipment			\$0
	Travel			\$0
	Shipping			\$0
	Supplies			\$0
	Expenses			\$0
	Laboratory			\$0
	Subcontractors			\$0
	Activity Task 1B Subtotal			\$7,160

**ATTACHMENT A. ACTIVITIES, ASSUMPTIONS, AND COSTS FOR
2024 ENVIRONMENTAL MONITORING AND REPORTING
CHEYENNE SANITARY LANDFILL, LARAMIE COUNTY, WYOMING**



Activity	Schedule of Charges	Hourly Rate or Charge	Hours or Unit Estimate	Subtotal
Task 2A August Environmental Monitoring	Professional Level 9	\$178	0	\$0
	Professional Level 8	\$164	4	\$656
	Professional Level 5	\$125	4	\$500
	Professional Level 4	\$113	4	\$452
	Professional Level 3	\$102	100	\$10,200
	Technical Level 2	\$76	0	\$0
<u>Activities</u> - Task activities are the same as those listed in Task 1A with the following exceptions. - Provide laboratory analysis for WSWRR, Chapter 2, Appendix C parameters for up to 36 wells.				
<u>Assumptions</u> - Assumptions for this task are the same as those listed in Task 1A. - Costs assume all required Appendix A & B samples will be collected in April. If they are not and there is sufficient water in August, samples will be collected in August. - Trip blanks will only be collected if an Appendix A or B sample is collected during August that could not be collected during April.				
	Labor			\$11,808
	Equipment	\$2,010	1	\$2,010
	Travel	\$0.670	1000	\$670
	Shipping	\$300	1	\$300
	Supplies	\$200	1	\$200
	Expenses			\$3,180
	Eurofins Denver	\$8,750	1.1	\$9,625
	Subcontractors			\$9,625
	Task 2A Subtotal			\$24,613

ATTACHMENT A. ACTIVITIES, ASSUMPTIONS, AND COSTS FOR
2024 ENVIRONMENTAL MONITORING AND REPORTING
CHEYENNE SANITARY LANDFILL, LARAMIE COUNTY, WYOMING



Activity	Schedule of Charges	Hourly Rate or Charge	Hours or Unit Estimate	Subtotal
Task 2B August Environmental Reporting	Technical Specialist 1	\$225	2	\$450
	Professional Level 10	\$192	0	\$0
	Professional Level 8	\$164	10	\$1,640
	Professional Level 7	\$149	12	\$1,788
	Professional Level 5	\$125	20	\$2,500
	Technical Level 5	\$105	6	\$630
	Administrative 3	\$76	2	\$152
	Labor			\$7,160
	Equipment			\$0
	Travel			\$0
	Per Diem			\$0
	Miscellaneous			\$0
	Expenses			\$0
				\$0
	Subcontractors			\$0
	Task 2B Subtotal			\$7,160

**ATTACHMENT A. ACTIVITIES, ASSUMPTIONS, AND COSTS FOR
2024 ENVIRONMENTAL MONITORING AND REPORTING
CHEYENNE SANITARY LANDFILL, LARAMIE COUNTY, WYOMING**



Activity	Schedule of Charges	Hourly Rate or Charge	Hours or Unit Estimate	Subtotal
Task 3 Two Quarterly Methane Monitoring Events				
<u>Activities</u>				
- Project management and coordination with the City and WDEQ.	Professional Level 9	\$178	0	\$0
- Update and implement site-specific health & safety activities.	Professional Level 8	\$164	4	\$656
- Conduct monitoring in February and December.	Professional Level 5	\$125	4	\$500
- Monitor 14 perimeter methane probes and landfill structures, including the office, the shop area, and the equipment building.	Professional Level 4	\$113	4	\$452
	Professional Level 3	\$102	12	\$1,224
	Technical Level 2	\$76	0	\$0
<u>Assumptions</u>				
- District will provide access to structures and methane monitoring probes.				
- April and August monitoring events will be conducted during routine groundwater monitoring events.				
- Vendor will pre-calibrate methane monitor in accordance with manufacturer's instructions.				
	Labor			\$2,832
	Equipment	\$420	2	\$840
	Travel	0.670	200	\$134
	Shipping	100	2	\$200
	Supplies	\$0	2	\$0
	Expenses			\$1,174
	Laboratory			\$0
	Subcontractors			\$0
	Task 3 Subtotal			\$4,006

**ATTACHMENT A. ACTIVITIES, ASSUMPTIONS, AND COSTS FOR
2024 ENVIRONMENTAL MONITORING AND REPORTING
CHEYENNE SANITARY LANDFILL, LARAMIE COUNTY, WYOMING**



Activity		Schedule of Charges	Hourly Rate or Charge	Hours or Unit Estimate	Subtotal
Task 4 Annual Environmental Reporting		Technical Specialist 1	\$225	4	\$900
		Professional Level 10	\$192	25	\$4,800
		Professional Level 8	\$164	10	\$1,640
		Professional Level 5	\$125	30	\$3,750
		Technical Level 5	\$105	3	\$315
		Administrative 3	\$76	3	\$228
		Labor			\$11,633
		Equipment			\$0
		Travel			\$0
		Per Diem			\$0
		Miscellaneous			\$0
		Expenses			\$0
					\$0
		Subcontractors			\$0
					\$0
		Task 4 Subtotal			\$11,633

Activities

- Prepare report and conduct statistical analysis in accordance with SLEMP.
- The report will contain:
 - Narrative of site background, synopsis of hydrogeologic properties, environmental procedures, and a summary of findings.
 - Narrative of December methane monitoring results.
 - Narrative of statistical analysis.
 - Narrative of observations, conclusions, recommendations, and future schedule.
 - Tables for sample collection summary; fluid levels; annual leachate volumes; groundwater, methane, and leachate results; and a summary of the statistical evaluation.
 - Figures of site location, site layout, historical water level data, and 3 potentiometric surface maps for each water-bearing zone.
 - Attachments of any additional field forms for year (e.g., December methane), statistical outputs, and electronic data used for statistical analysis.

Assumptions

- No revisions will be required by WDEQ/SHWD.

**ATTACHMENT A. ACTIVITIES, ASSUMPTIONS, AND COSTS FOR
2024 ENVIRONMENTAL MONITORING AND REPORTING
CHEYENNE SANITARY LANDFILL, LARAMIE COUNTY, WYOMING**



Activity	Schedule of Charges	Hourly Rate or Charge	Hours or Unit Estimate	Subtotal
Task 5 Optional Technical Assistance	Professional Level 8	\$164	30.49	\$5,000
<u>Activities</u> - Provide technical & regulatory assistance on an as-needed basis. - Possible activities may include: - Assistance repairing pumps, completions, or casings at groundwater monitoring wells. - Non-routine correspondence with WDEQ.				
<u>Notes</u> - Variable billing rates may be applied to this task, although not shown here. - Work under this task would only be conducted with client written authorization.				
Labor				\$5,000
Expenses				\$0
Subcontractors				\$0
Task 5 Subtotal				\$5,000

TOTAL
Task 1A April Environmental Monitoring
Task 1B April Environmental Reporting
Task 2A August Environmental Monitoring
Task 2B August Environmental Reporting
Task 3 Two Quarterly Methane Monitoring Events
Task 4 Annual Environmental Reporting
Task 5 Optional Technical Assistance
\$94,242

Abbreviations:
 EDD = Electronic data deliverable
 SLEMP = Sanitary Landfill Environmental Monitoring Plan
 WDEQ = Wyoming Department of Environmental Quality
 WSWRR = Wyoming Solid Waste Rules and Regulations
 VOC = Volatile Organic Compound

ATTACHMENT B

LABORATORY QUOTE AND EXTENSION FROM EUROFINS DENVER

December 16, 2022

Caroline Brewer
Trihydro Corporation
388 Main, Suite C
Lander, WY 82520
cbrewer@trihydro.com
Tel: (307) 745-7474

Subject: Analytical Services Proposal - Wyoming Landfills 2023 Multi-Sites
Eurofins Environment Testing America Quotation Number 28024157

Dear Caroline Brewer:

We appreciate the opportunity to provide your company with a quotation for your Wyoming Landfills 2023 Multi-Sites project. Eurofins Environment Testing America has a unique combination of full service capabilities, technical expertise, local service options, and online resources necessary to ensure successful project outcomes.

At Eurofins Environment Testing America, quality is the hallmark of our business. To ensure your project's data quality objectives are met, we offer experienced personnel who are trained and committed to completing your analytical project on time, a fully documented QA/QC program, and state-of-the-art laboratory equipment and facilities. In addition to being a full service laboratory, we are part of the nation's largest environmental laboratory network. This provides access to an unparalleled spectrum of capabilities and turnaround time options, all through a single point of contact. Janice Winn-Shilling has been assigned as your Project Manager for this work and can be reached by phone at 303-736-0119 or via email at Janice.Winn-Shilling@ET.eurofinsUS.com.

- **MyEOL®**, a web portal offering you customizable, real time access to data. With 24 hour access you can perform data trending, compare data to industry or project limits, track CoCs, invoices, reports and much more.
- **Level IV Deliverables/Customizable EDDs**: high resolution, text searchable reports, available in virtually any format.
- **Extensive Experience**: Project Managers with in-depth knowledge of regulatory protocols and procedures.
- **Nationwide Logistical Support**: bringing you an extensive courier network, service centers and shipping options throughout the U.S. and abroad.
- **PFAS, Dioxins/Furans, Air, Radiochemistry, IH** and other specialty analyses are offered alongside routine soil and water methods with seamless reports and consolidated EDDs.

The following quotation includes a detailed price breakdown, as well as any notes and clarifications pertaining to your project, and is subject to Eurofins Environment Testing America's Standard Terms and Conditions, unless otherwise agreed upon in writing.

We thank you for choosing Eurofins Environment Testing America, and we look forward to working with you on this project.

Sincerely,

Joy Chang
Client Relations Manager
Joy.Chang@et.eurofinsus.com

cc: Kim Johnson; Janice Winn-Shilling

Eurofins Denver
4955 Yarrow Street
Arvada, CO 80002

Prepared by Chang, Joy B
Date 12/16/2022
Expiration Date 12/31/2023
Est. Start Date

Prepared for:
Caroline Brewer
Trihydro Corporation
388 Main, Suite C
Lander, WY 82520
cbrewer@trihydro.com
Tel: (307) 745-7474

Project: Wyoming Landfills 2023 Multi-Sites

Quote Number: 28024157 - 0

Appendix A

TAT: 10_Days (Business Days)

Matrix	*	Method	Test Description	Quantity	Unit Price	Extended Price
Water		6020B	Total Metals -- 15 specific analyte	1	\$ 58.00	\$ 58.00
Water		8260D	Volatile Organic Compounds, 47 specific analyte	1	\$ 68.00	\$ 68.00
Water		8011	EDB, DBCP and 1,2,3-TCP -- Low Level	1	\$ 41.00	\$ 41.00
Total Appendix A						\$ 167.00

Appendix B

TAT: 10_Days (Business Days)

Matrix	*	Method	Test Description	Quantity	Unit Price	Extended Price
Water		6020B	Total Metals -- 16 specific analyte	1	\$ 58.00	\$ 58.00
Water		7470A	Total Mercury	1	\$ 17.00	\$ 17.00
Water		9012B	Cyanide, Total	1	\$ 35.00	\$ 35.00
Water		9034	Sulfide, Total	1	\$ 35.00	\$ 35.00
Water		8260D	Volatile Organic Compounds, 64 specific analyte	1	\$ 68.00	\$ 68.00
Water		8011	EDB, DBCP and 1,2,3-TCP -- Low Level	1	\$ 41.00	\$ 41.00
Water	1	8270E	SVOCs, specific analyte with p-Phenylene Diamine as TIC	1	\$ 160.00	\$ 160.00
Water		8081B	Organochlorine Pesticides, specific analyte list	1	\$ 90.00	\$ 90.00
Water	1	8321A	Herbicides, specific analyte list	1	\$ 155.00	\$ 155.00
Water		8082A	PCBs, specific analyte list	1	\$ 55.00	\$ 55.00
Total Appendix B						\$ 714.00

Appendix C

TAT: 10_Days (Business Days)

Matrix	*	Method	Test Description	Quantity	Unit Price	Extended Price
Water		9040C	pH	1	\$ 10.00	\$ 10.00
Water		350.1	Nitrogen, Ammonia	1	\$ 23.00	\$ 23.00
Water		300.0A	Chloride, Fluoride and Sulfate	1	\$ 27.00	\$ 27.00
Water		6010D	6 Total Metals -- Ca, Fe, Mn, Mg, K, Na	1	\$ 36.00	\$ 36.00
Water		SM 2540C	Solids, Total Dissolved (TDS)	1	\$ 10.00	\$ 10.00
Water		SM 2320B	Alkalinity - Bicarbonate & Carbonate	1	\$ 15.00	\$ 15.00
Water		353.2	Nitrogen, Nitrate-Nitrite	1	\$ 21.00	\$ 21.00
Water		410.4	COD	1	\$ 12.00	\$ 12.00
Water		SM 5310B	TOC	1	\$ 31.00	\$ 31.00
Total Appendix C						\$ 185.00



Environment Testing

Eurofins Denver
4955 Yarrow Street
Arvada, CO 80002

Prepared by Chang, Joy B
Date 12/16/2022
Expiration Date 12/31/2023
Est. Start Date

Prepared for:
Caroline Brewer
Trihydro Corporation
388 Main, Suite C
Lander, WY 82520
cbrewer@trihydro.com
Tel: (307) 745-7474

Project: Wyoming Landfills 2023 Multi-Sites

Quote Number: 28024157 - 0

Additional Tests

TAT: 10_Days (Business Days)

Matrix	*	Method	Test Description	Quantity	Unit Price	Extended Price
Water		SM 2540D	Solids, Total Suspended (TSS)	1	\$ 10.00	\$ 10.00
Water		SM5210B	BOD	1	\$ 40.00	\$ 40.00
Water		8015D	TPH-GRO (C6-C10)	1	\$ 40.00	\$ 40.00
Water		8015D	WY TPH-DRO (C10-C32)	1	\$ 55.00	\$ 55.00
Total Additional Tests						\$ 145.00

* See Footnotes in Detail Section

Quote Other Charges

Description	Quantity	Unit Price	Extended Price
Safe and Environmentally Responsible Waste Management (per sample)	1	\$ 2.50	\$ 2.50
Minimum Total Invoice per analytical receipt (for details see T&Cs)	0	\$ 150.00	\$ 0.00
WYDEQ SHWD EDD, email/Total Access	1	\$ 0.00	\$ 0.00
Trihydro EDD, email/Total Access	0.0% of Total	\$ 0.00	\$ 0.00
Level II Report PDF, email/Total Access	0.0% of Total	\$ 0.00	\$ 0.00
Sample Return Shipping via Fed Ex Standard Overnight	1	\$ 60.00	\$ 60.00
Total Other Charge			\$62.50

Total Other Charges \$ 62.50

Total Analysis Charges \$ 1,211.00

Grand Total for Quote 28024157 \$ 1,273.50

****Quoted charges do not include sales tax. Applicable sales tax will be added to invoices where required by law.**

Eurofins Denver
4955 Yarrow Street
Arvada, CO 80002

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Date 12/16/2022
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PROJECT DETAILS

Clarifications and Exceptions

Footnote 1:

Eurofins Denver laboratory does not report p-Phenylene diamine by 8270E as target analyte. This compound will be reported as Tentatively Identified Compound (TIC).

Sample Receiving Hours/Hold Time/TAT

Sample Receiving Hours

8 am to 5 pm Monday - Friday

Samples received at the laboratory after 11 AM are considered received on the following business day (Monday-Friday, no holidays). We will do our best to meet rush received after 3 pm, but to better ensure success this should be arranged with your PM prior to sample submittal. Eurofins Environment Testing America Denver accepts Saturday deliveries via Fed Ex and client drop off every Saturday from 9-12. You will need to request Saturday delivery when you ship or Fed Ex will hold the samples until Monday. **Please note that samples received on Saturday will not be processed until the next business day.**

Hold Time

Eurofins Environment Testing America, Inc. requests that samples be hand delivered the same day of collection or shipped to the laboratory via overnight priority service on the day of collection. For samples received with less than one half the holding time remaining, Eurofins Environment Testing America will make our best effort to extract and analyze these samples within the holding time; however, **Eurofins Environment Testing America will not be held responsible for meeting holding times if samples are received with less than one half the holding time remaining.** For methods that require a short holding time (less than 7 calendar days from collection), Eurofins TestAmerica requests prior notification before samples are shipped to the laboratory. **Samples that have to be analyzed and/or extracted on expedited turnaround time to meet hold time will incur a surcharge rate.**

Turnaround Time

- Quoted Turnaround Time - Data will be delivered at the proposed turnaround time in Business Days from Sample Receipt unless otherwise agreed upon. TAT begins the day the laboratory performing analyses receives the samples (day of lab receipt = day zero).

Samples received after 11 AM will be considered received the next business day.

- Expedited Turnaround Time: Expedited turnaround times may be available and must be pre-approved by the laboratory. Expedited turnaround delivery is contingent upon meeting the agreed upon delivery date/time and number of samples. Samples received after 11 AM will be considered received the next business day. Results will be provided via e-mail or TotalAccess by close of business in the lab's time zone unless another time has been agreed to in advance.

Expedited turnaround time surcharges for standard analyses are:

- 5 Business Days TAT = 30%
- 4 Business Days TAT = 45%
- 3 Business Days TAT = 60%
- 2 Business Days TAT = 75%
- 1 Business Day TAT = 100%

Eurofins Denver
4955 Yarrow Street
Arvada, CO 80002

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Date 12/16/2022
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Tel: (307) 745-7474

Project: Wyoming Landfills 2023 Multi-Sites

Quote Number: 28024157 - 0

Different surcharges may apply for specialty analyses. These will be provided in your quotation. Weekend TAT can be arranged on a project-specific basis at an additional cost. Please contact your PM to inquire about availability and cost.

Eurofins TestAmerica Business T&Cs

Confidentiality -

This quote has been prepared by TestAmerica Laboratories, Inc. dba Eurofins TestAmerica (Eurofins TestAmerica), solely for the use of the customer to whom it is addressed in evaluating Eurofins TestAmerica's qualifications and capabilities in connection with a particular project. The user of this document agrees by its acceptance to return it to Eurofins TestAmerica upon request and not to reproduce, copy, lend, or otherwise disclose its contents, directly or indirectly, and not to use it for any purpose other than that for which it was specifically provided. The user also agrees that where consultants or other outside parties are involved in the evaluation process, access to this document shall not be given to said parties unless those parties also specifically agree to these conditions. In the absence of signed acceptance, submittal of samples will indicate acceptance of this quotation.

Terms and Conditions -

This quotation is based solely upon Eurofins TestAmerica's standard product (routine QA/QC, detection limits, deliverables and standard turnaround times) and noted exceptions herein. The discounts incorporated into the pricing are based upon the sample quantity, test method, and schedule quoted. Any deviations may impact pricing and/or the acceptance of work. Final acceptance of this work is contingent upon a mutually agreed Sample Delivery Schedule. All sales are subject to Eurofins TestAmerica's Terms and Conditions **unless alternative terms have been agreed to in writing**. Submittal of samples will indicate acceptance of this quotation.

Quote Expiration -

Pricing listed in the proposal will expire **90 days** from the quote date unless the project is awarded/confirmed within that time period. Unless otherwise set forth in this quotation, Eurofins TestAmerica reserves the right to re-evaluate pricing for extended length projects on an annual basis.

SERWM -

A fee, notated as Safe and Environmentally Responsible Waste Management (SERWM), will be applied to all invoices for each sample processed by the laboratory.

PROJECT SETUP

Field Sampling Products -

Eurofins TestAmerica can provide field sampling products upon request and these products are billable to the client at the time of shipment.

Field sampling products may include, but are not limited to:

- VOA sampling devices, preservation kits, ISM supplies, tubing and filters
- Industrial Hygiene media, surface sampling wipes, source sampling traps

The cost of the soil sampling devices is not included in the quoted rates for volatile analysis. Core Samplers

- 5 gram Core N' One and EnCore samplers are available at \$10 each (three required per sample).
- Disposable Core N' One T-Handles are available at \$15 each.
- Eurofins TestAmerica does not supply EnCore T-Handles.

Field Preservation

Eurofins Denver
4955 Yarrow Street
Arvada, CO 80002

Prepared by Chang, Joy B
Date 12/16/2022
Expiration Date 12/31/2023
Est. Start Date

Prepared for:
Caroline Brewer
Trihydro Corporation
388 Main, Suite C
Lander, WY 82520
cbrewer@trihydro.com
Tel: (307) 745-7474

Project: Wyoming Landfills 2023 Multi-Sites

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- TerraCore Kits (varied) - quoted based on client required configurations
- Disposable Lock N' Load Handles are available at \$12 each.

If any sampling supplies for soil VOCs are requested, they will be included in the quote under Other Charges.

Coolers and Sampling Supplies -

- Sampling Supplies: Eurofins TestAmerica will provide sample containers and coolers to support the sampling of water and soil samples. Extra bottles may be provided (up to 10%) in case of breakage. Eurofins TestAmerica expects that samples and supplies will be returned to the lab, including empty coolers and a reasonable percentage of the projected sample load - 70% or higher of the expected/quoted sample number. Coolers not received back by the projected deadline or as agreed with the PM may be charged at \$30 per cooler. Similarly, if the sample containers received as samples are less than 90% of the containers provided, the sample containers not received as samples will be charged at a flat rate of \$1 per container.
- Sample Container Shipping: The containers and preservatives required by the project shall be delivered via ground transportation. A minimum of **5 business days** advance notice is required in order to achieve shipment by ground transportation. Supply shipments requiring priority delivery due to insufficient lead time for ground transportation shall be charged to the client at Eurofins TestAmerica's cost. Alternatively, Eurofins TestAmerica can ship the supplies via carrier of choice by the client using the client's shipping account.

Eurofins TestAmerica does not supply wet ice or blue ice for shipments. If shipping containers are not returned within 60 days, a charge for the containers will be billed at the current market rate. Please contact your PM for the current charges.

Courier Services and Sample Pick-Up -

Courier Services are offered by some Eurofins TestAmerica facilities. Where offered, the cost of the services will vary based on the distance traveled, the scope of the project being supported, and whether sufficient notice (typically 48 hours) is provided to facilitate efficient scheduling. If no details are described in this quotation and you are interested in learning more about courier options, please contact your Client Relations Manager or Project Manager to inquire about availability and cost.

Minimum Log-In Charges -

Eurofins TestAmerica's minimum charge is **\$150** for a group of samples received together for analysis. Our minimum invoice value is \$150.

QC Limit Disclaimer -

The laboratory's reporting limits, detection limits, and control limits are subject to change as these values are updated periodically to reflect analytical sensitivity and capability.

Turnaround Time -

- Quoted Turnaround Time - Data will be delivered at the proposed turnaround time in Business Days from Sample Receipt unless otherwise agreed upon. TAT begins the day the laboratory performing analyses receives the samples (day of lab receipt = day zero).

Samples received after 11 AM will be considered received the next business day.

- Expedited Turnaround Time: Expedited turnaround times may be available and must be

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pre-approved by the laboratory. Expedited turnaround delivery is contingent upon meeting the agreed upon delivery date/time and number of samples. Samples received after 11 AM will be considered received the next business day. Results will be provided via e-mail or TotalAccess by close of business in the lab's time zone unless another time has been agreed to in advance.

Expedited turnaround time surcharges for standard analyses are:

- 5 Business Days TAT = 30%
- 4 Business Days TAT = 45%
- 3 Business Days TAT = 60%
- 2 Business Days TAT = 75%
- 1 Business Day TAT = 100%

Different surcharges may apply for specialty analyses. These will be provided in your quotation. Weekend TAT can be arranged on a project-specific basis at an additional cost. Please contact your PM to inquire about availability and cost.

PROJECT DELIVERABLES

Eurofins TestAmerica will provide two analytical report formats, a final report in PDF format and a standard Eurofins TestAmerica EDD. Both electronic report formats will be delivered via email or web portal. If additional formats or retroactive deliverables are requested, costs of report generation will be billable. Charges will be based on labor and materials cost of report generation and data retrieval. Please contact your PM to inquire about availability and the price of additional deliverables.

- Report Format: Unless a level III or IV deliverable is specifically listed on the pricing page, this quotation includes delivery of a Level I or II report. Level III or IV reports are available at an additional charge.
- Electronic Data Deliverable Format: Eurofins TestAmerica has many EDD formats available to our clients including the most widely used commercial formats. Eurofins TestAmerica offers data using a standard EDD. Other EDD formats are available for a minimal cost of \$25 per format (if not included as part of the report options listed in the quotation). The development of EDD formats that are not already available, including modification to existing formats to fit client specific needs, can be provided for a fee starting at \$100. Additional fees will be estimated by the lab and approved by the client. Additional programming fees are billed at \$50/hour.

PROJECT SPECIFICATIONS

Cancellation Fee -

A fee will be charged for cancellation of samples/analyses after a project is received in the laboratory. The fee will be based on the status of analysis at the time of cancellation in accordance with the following categories:

- Received - 35%
- Prepped - 50%
- Analyzed - 95%

Changes in Scope and Work Revisions -

Project requirements must be agreed upon prior to sample receipt. Samples will be logged according to the chain of custody received with the samples. Changes after initiation of the project will be subject to additional charges, including labor time required to reset project, communicate changes to laboratory staff, and rework data. Turnaround time will be reset or

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rush surcharges will be assessed where applicable. Analyses added with less than 1/2 of the analytical hold time remaining will incur rush turnaround charges. Your project manager will evaluate project specific charges at the time a change order is received.

Held Samples -

- Held samples not analyzed: Samples submitted on hold will be billed at 35% of the analysis fee (minimum \$10/sample). If samples are later analyzed, the handling fee will be waived and only the analysis price will be charged. Samples taken off hold with less than 1/2 of the analytical hold time remaining may incur rush turnaround charges. Samples will be disposed of 30 days after the report for analyzed samples in the same job is issued, unless alternate archival arrangements are made in advance.
- Extracted/Prepped and Held samples: Samples submitted for prep and hold will be billed at 60% of the analysis fee for each prepped sample (minimum \$30/sample). Samples taken off hold with less than 1/2 of the analytical hold time remaining may incur rush turnaround charges. Samples will be disposed of 30 days after the report for analyzed samples in the same job is issued, unless alternate archival arrangements are made in advance.
- Extended archival of samples: Extended archival of samples (including held samples) may be available for a fee starting at \$2 per container per month (minimum \$10/sample). This fee will be billed in advance on a quarterly basis for every quarter after the standard sample retention time of 30 days after the report is issued. Fees for larger volumes, non-standard matrices or cold storage will be negotiated on a case-by-case basis. Please contact your PM to inquire about availability and pricing for samples that are sent to the lab and archived.

Matrix Spike/Spike Duplication (MS/MSD) Samples -

When MS/MSDs are not specifically requested, Eurofins TestAmerica will strive to perform the required QC using whatever sample is available but will not report the QC results unless the client requests it. The reporting of client requested MS/MSD results will be charged at applicable unit rates. If MS/MSDs are specifically required or requested, the client must provide additional sample volume.

Multiple Dilutions Analyzed -

Eurofins TestAmerica strives to analyze samples without dilution or with the minimum dilution required. Samples are diluted to bring the primary analyte within the calibration range of the instrument, to compensate for matrix co-extractives, or to prevent instrument contamination. Eurofins TestAmerica will report the analytical run containing the highest concentration component/analyte in the sample within the calibrated (quantifiable) range of the method. Analytical screening runs are not reported. If project specific data quality objectives require additional runs, analyses will be billable unless otherwise noted in this quote. Please contact your PM to inquire about the availability of this service for your project.

Sample Disposal -

Eurofins TestAmerica will dispose of non-hazardous samples, sample extracts and digestates 30 days after the final report is issued. Charges for disposal of non-routine or uniquely hazardous samples will be billed to the client. Alternatively, samples can be returned to the client for disposal. Cost of return shipping will be billable to the client.

Special Sample Handling Fees -

Unit prices assume that samples are a single-phase matrix and that analyses can be performed

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in accordance with the laboratory's standard analytical procedures. If additional handling is required, additional fees may apply. Examples of special handling include (but are not limited to):

- Matrices requiring additional dilutions or special clean up steps
- Multiphasic samples requiring separate preparations and/or analyses
- Particle size reduction or special sub-sampling procedures
- Extra disposal costs for unique waste streams

Trip Blanks -

Eurofins TestAmerica typically provides trip blanks with our sample kits containing volatile analysis. When samples are received at the laboratory with trip blanks, the lab will analyze, report and charge the unit rate for the analysis. Please add this sample to your chain of custody. If you do not want the trip blank analyzed, please note this on the COC.

ADDITIONAL BUSINESS TERMS AND CONDITIONS

Environmental Management Fee

If this fee is included in the Other Charges section of the pricing page, a fee equal to the listed percentage of the total invoice amount will be applied to all work done under this quotation. The Fee will appear as a separate line item on each invoice. In the absence of any other firm pricing agreements, your sending work to us under this quotation will signify your acceptance of responsibility for payment of the Fee.

Field Parameters -

pH, Temperature, and Dissolved Oxygen analyses, along with Residual Chlorine screening, are properly performed and treated in the field at the time of sample collection. Laboratory analysis may result in a holding time exceedance qualifier.

Network or Subcontract Labs -

- Networking: Eurofins TestAmerica reserves the right to perform the services at any laboratory in the Eurofins TestAmerica network, unless the Client has required a particular location for the work.
- Subcontracting: Eurofins TestAmerica reserves the right to subcontract services ordered by the Client to another laboratory or laboratories, if, in Eurofins TestAmerica's sole judgment, it is reasonably necessary, appropriate or advisable to do so. Eurofins TestAmerica will make every effort to notify the client prior to delivering samples to an out-of-network laboratory. Eurofins TestAmerica will in no way be liable for any subcontracted services (outside the Eurofins TestAmerica network) except for work performed at laboratories which have been audited and approved by Eurofins TestAmerica.

Price Surcharge Due To Sample Volume -

Unless dictated by contract, this quotation is based on the scope of work defined in the quote request. If the volume of samples submitted is less than 70% of the projected volume, a surcharge of 10% of the total project cost may be assessed.

Professional and Administrative Services -

A variety of professional and administrative services are available. Prices for services not specifically detailed in this quotation will be billed in accordance with Eurofins TestAmerica's Professional Rate Schedule.

Taxes -

Where reports are issued in or delivered to a state which assesses sales tax on Eurofins

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TestAmerica's services, applicable sales taxes will be added to the invoice as required by law, unless an appropriate sales tax exemption form is on file with Eurofins TestAmerica.

Rev: 04/04/2019

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Appendix A

Matrix	Method	Test Description	Analyte	RL	MDL	Units
Water	6020B	Total Metals -- 15 specific analyte	Antimony	1.00	0.400	ug/L
			Arsenic	1.00	0.330	ug/L
			Barium	10.0	0.290	ug/L
			Beryllium	1.00	0.0800	ug/L
			Cadmium	1.00	0.265	ug/L
			Chromium	1.00	0.500	ug/L
			Cobalt	1.00	0.0923	ug/L
			Copper	1.00	0.560	ug/L
			Lead	1.00	0.180	ug/L
			Nickel	1.00	0.300	ug/L
			Selenium	1.00	0.373	ug/L
			Silver	1.00	0.0330	ug/L
			Thallium	1.00	0.0890	ug/L
			Vanadium	10.0	1.24	ug/L
			Zinc	10.0	2.00	ug/L
Water	8260D	Volatile Organic Compounds, 47 specific analyte	Acetone	20	6.60	ug/L
			Acrylonitrile	20.0	4.47	ug/L
			Benzene	1.00	0.308	ug/L
			Bromochloromethane	1.00	0.403	ug/L
			Bromodichloromethane	1.00	0.386	ug/L
			Bromoform	2.00	1.21	ug/L
			Methyl bromide	1	2.36	ug/L
			Methyl ethyl ketone (MEK)	20	5.95	ug/L
			Carbon disulfide	5	0.631	ug/L
			Carbon tetrachloride	1.00	0.566	ug/L
			Chlorobenzene	1.00	0.422	ug/L
			Dibromochloromethane	2.00	0.618	ug/L
			Chloroethane	1	1.37	ug/L
			Chloroform	1	0.358	ug/L
			Methyl chloride	1	0.753	ug/L
			1,2-Dibromo-3-Chloropropane	5.00	1.76	ug/L
			Methylene bromide	1.00	0.343	ug/L
			1,2-Dichlorobenzene	1.00	0.372	ug/L
			1,4-Dichlorobenzene	1.0	0.389	ug/L
			trans-1,4-Dichloro-2-butene	1	1.38	ug/L
			1,1-Dichloroethane	1.00	0.220	ug/L
			1,2-Dichloroethane	1.00	0.541	ug/L
			cis-1,2-Dichloroethene	1.00	0.321	ug/L
			trans-1,2-Dichloroethene	1.00	0.368	ug/L
			1,1-Dichloroethene	1.00	0.230	ug/L
			1,2-Dichloropropane	1.00	0.515	ug/L

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Trihydro Corporation
388 Main, Suite C
Lander, WY 82520
cbrewer@trihydro.com
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Appendix A

Matrix	Method	Test Description	Analyte	RL	MDL	Units
Continued						
			cis-1,3-Dichloropropene	2.00	0.626	ug/L
			trans-1,3-Dichloropropene	1	0.646	ug/L
			Ethylbenzene	1.00	0.303	ug/L
			2-Hexanone	20	1.70	ug/L
			Methyl iodide	5.00	2.58	ug/L
			Methylene Chloride	1	0.938	ug/L
			4-Methyl-2-pentanone (MIBK)	20	0.980	ug/L
			Styrene	1.00	0.356	ug/L
			1,1,1,2-Tetrachloroethane	1.00	0.577	ug/L
			1,1,2,2-Tetrachloroethane	1.00	0.210	ug/L
			Tetrachloroethene	1.00	0.403	ug/L
			Toluene	1.00	0.322	ug/L
			1,1,1-Trichloroethane	1.00	0.390	ug/L
			1,1,2-Trichloroethane	1.00	0.270	ug/L
			Trichloroethene	1.00	0.300	ug/L
			Trichlorofluoromethane	1	0.566	ug/L
			1,2,3-Trichloropropane	1	0.858	ug/L
			Vinyl acetate	1	0.940	ug/L
			Vinyl chloride	2.00	0.505	ug/L
			Xylenes, Total	1	0.331	ug/L
			1,2-Dibromoethane	1.00	0.404	ug/L
		Surrogate Cpnd				
			1,2-Dichloroethane-d4 (Surr)	2.00	0.200	
			Toluene-d8 (Surr)	2.00	0.500	
			4-Bromofluorobenzene (Surr)	2.00	0.500	
			Dibromofluoromethane (Surr)	2.00	0.500	
				RL	MDL	Units
Water	8011	EDB, DBCP and 1,2,3-TCP -- Low Level	1,2-Dibromo-3-Chloropropane	0.0200	0.00680	ug/L
			1,2-Dibromoethane	0.0200	0.00370	ug/L
			1,2,3-Trichloropropane	0.0200	0.00500	ug/L
		Surrogate Cpnd				
			1,2-Dibromopropane	0.0200	0.00370	

Appendix B

Matrix	Method	Test Description	Analyte	RL	MDL	Units
Water	6020B	Total Metals -- 16 specific analyte	Antimony	1.00	0.400	ug/L
			Arsenic	1.00	0.330	ug/L
			Barium	10.0	0.290	ug/L
			Beryllium	1.00	0.0800	ug/L

Eurofins Denver
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Lander, WY 82520
cbrewer@trihydro.com
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Appendix B

Matrix	Method	Test Description	Analyte	RL	MDL	Units
Continued						
			Cadmium	1.00	0.265	ug/L
			Chromium	1.00	0.500	ug/L
			Cobalt	1.00	0.0923	ug/L
			Copper	1.00	0.560	ug/L
			Lead	1.00	0.180	ug/L
			Nickel	1.00	0.300	ug/L
			Selenium	1.00	0.373	ug/L
			Silver	1.00	0.0330	ug/L
			Thallium	1.00	0.0890	ug/L
			Tin	1.00	0.770	ug/L
			Vanadium	10.0	1.24	ug/L
			Zinc	10.0	2.00	ug/L
				RL	MDL	Units
Water	7470A	Total Mercury	Mercury	1.00	0.0610	ug/L
				RL	MDL	Units
Water	9012B	Cyanide, Total	Cyanide, Total	10.0	5.00	ug/L
				RL	MDL	Units
Water	9034	Sulfide, Total	Sulfide	4000	1600	ug/L
				RL	MDL	Units
Water	8260D	Volatile Organic Compounds, 64 specific analyte	Acetone	15.0	6.60	ug/L
			Acetonitrile	30.0	9.56	ug/L
			Acrolein	20.0	4.86	ug/L
			Acrylonitrile	20.0	4.47	ug/L
			Benzene	1.00	0.308	ug/L
			Bromochloromethane	1.00	0.403	ug/L
			Bromodichloromethane	1.00	0.386	ug/L
			Bromoform	2.00	1.21	ug/L
			Methyl bromide	5.00	2.36	ug/L
			Methyl ethyl ketone (MEK)	15.0	5.95	ug/L
			Carbon disulfide	2.00	0.631	ug/L
			Carbon tetrachloride	1.00	0.566	ug/L
			Chlorobenzene	1.00	0.422	ug/L
			Chloroprene	3.00	1.17	ug/L
			Dibromochloromethane	2.00	0.618	ug/L
			Chloroethane	4.00	1.37	ug/L
			Chloroform	1.00	0.358	ug/L
			Methyl chloride	2.00	0.753	ug/L
			Allyl chloride	2.00	0.170	ug/L

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Matrix	Method	Test Description	Analyte	RL	MDL	Units
Continued						
			1,2-Dibromo-3-Chloropropane	5.00	1.76	ug/L
			Methylene bromide	1.00	0.343	ug/L
			1,2-Dichlorobenzene	1.00	0.372	ug/L
			1,3-Dichlorobenzene	1.00	0.334	ug/L
			1,4-Dichlorobenzene	1.00	0.389	ug/L
			trans-1,4-Dichloro-2-butene	3.00	1.38	ug/L
			Dichlorodifluoromethane	3.00	0.962	ug/L
			1,1-Dichloroethane	1.00	0.220	ug/L
			1,2-Dichloroethane	1.00	0.541	ug/L
			cis-1,2-Dichloroethene	1.00	0.321	ug/L
			trans-1,2-Dichloroethene	1.00	0.368	ug/L
			1,1-Dichloroethene	1.00	0.230	ug/L
			1,2-Dichloropropane	1.00	0.515	ug/L
			1,3-Dichloropropane	1.00	0.379	ug/L
			2,2-Dichloropropane	1.00	0.380	ug/L
			cis-1,3-Dichloropropene	2.00	0.626	ug/L
			trans-1,3-Dichloropropene	2.00	0.646	ug/L
			1,1-Dichloropropene	1.00	0.416	ug/L
			Ethylbenzene	1.00	0.303	ug/L
			Ethyl methacrylate	3.00	0.860	ug/L
			2-Hexanone	5.00	1.70	ug/L
			Methyl iodide	5.00	2.58	ug/L
			Isobutyl alcohol	110	36.5	ug/L
			Methacrylonitrile	15.0	5.31	ug/L
			Methylene Chloride	2.00	0.938	ug/L
			Methyl methacrylate	4.00	1.11	ug/L
			4-Methyl-2-pentanone (MIBK)	5.00	0.980	ug/L
			Naphthalene	2.00	0.634	ug/L
			Propionitrile	20.0	3.70	ug/L
			Styrene	1.00	0.356	ug/L
			1,1,1,2-Tetrachloroethane	1.00	0.577	ug/L
			1,1,2,2-Tetrachloroethane	1.00	0.210	ug/L
			Tetrachloroethene	1.00	0.403	ug/L
			Toluene	1.00	0.322	ug/L
			1,2,4-Trichlorobenzene	1.00	0.584	ug/L
			1,1,1-Trichloroethane	1.00	0.390	ug/L
			1,1,2-Trichloroethane	1.00	0.270	ug/L
			Trichloroethene	1.00	0.300	ug/L
			Trichlorofluoromethane	2.00	0.566	ug/L
			1,2,3-Trichloropropane	2.50	0.858	ug/L
			Vinyl acetate	3.00	0.940	ug/L
			Vinyl chloride	2.00	0.505	ug/L
			Xylenes, Total	1.00	0.331	ug/L
			1,2-Dibromoethane	1.00	0.404	ug/L

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Appendix B

Matrix	Method	Test Description	Analyte				
Surrogate Cpnd							
				1,2-Dichloroethane-d4 (Surr)	2.00	0.200	
				Toluene-d8 (Surr)	2.00	0.500	
				4-Bromofluorobenzene (Surr)	2.00	0.500	
				Dibromofluoromethane (Surr)	2.00	0.500	
					RL	MDL	Units
Water	8011	EDB, DBCP and 1,2,3-TCP -- Low Level	1,2-Dibromo-3-Chloropropane	0.0200	0.00680	ug/L	
			1,2-Dibromoethane	0.0200	0.00370	ug/L	
			1,2,3-Trichloropropane	0.0200	0.00500	ug/L	
Surrogate Cpnd							
				1,2-Dibromopropane	0.0200	0.00370	
					RL	MDL	Units
Water	8270E	SVOCs, specific analyte with p-Phenylene Diamine as TIC	p-Dimethylamino azobenzene	20.0	0.879	ug/L	
			5-Nitro-o-toluidine	20.0	0.634	ug/L	
			N-Nitrosomethylethylamine	10.0	1.76	ug/L	
			N-Nitrosodimethylamine	10.0	0.290	ug/L	
			Methyl methanesulfonate	10.0	1.00	ug/L	
			N-Nitrosodiethylamine	10.0	0.823	ug/L	
			Ethyl methanesulfonate	10.0	0.943	ug/L	
			Phenol	10.0	2.00	ug/L	
			2-Chlorophenol	10.0	0.971	ug/L	
			Benzyl alcohol	10.0	0.230	ug/L	
			o-Cresol	10.0	0.980	ug/L	
			p-Cresol	10.0	0.379	ug/L	
			m-Cresol	10.0	0.400	ug/L	
			3 & 4 Methylphenol	10.0	0.250	ug/L	
			Acetophenone	10.0	0.426	ug/L	
			N-Nitrosodi-n-propylamine	10.0	0.350	ug/L	
			N-Nitrosopyrrolidine	10.0	0.804	ug/L	
			N-Nitrosomorpholine	10.0	2.00	ug/L	
			Hexachloroethane	10.0	0.983	ug/L	
			Nitrobenzene	10.0	0.810	ug/L	
			N-Nitrosopiperidine	10.0	2.00	ug/L	
			Isophorone	10.0	0.210	ug/L	
			2,4-Dimethylphenol	10.0	0.580	ug/L	
			2-Nitrophenol	10.0	1.39	ug/L	
			o,o',o"-Triethylphosphorothioate	50.0	2.00	ug/L	
			Bis(2-chloroethoxy)methane	10.0	0.970	ug/L	
			2,4-Dichlorophenol	10.0	0.640	ug/L	
			1,2,4-Trichlorobenzene	4.00	1.26	ug/L	
			4-Chloroaniline	10.0	2.14	ug/L	

Eurofins Denver
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Arvada, CO 80002

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Prepared for:
Caroline Brewer
Trihydro Corporation
388 Main, Suite C
Lander, WY 82520
cbrewer@trihydro.com
Tel: (307) 745-7474

Project: Wyoming Landfills 2023 Multi-Sites

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Appendix B

Matrix	Method	Test Description	Analyte	RL	MDL	Units
Continued						
			2,6-Dichlorophenol	10.0	1.35	ug/L
			Hexachloropropene	100	0.663	ug/L
			Hexachlorobutadiene	10.0	3.30	ug/L
			N-Nitrosodi-n-butylamine	10.0	1.22	ug/L
			p-Phenylene diamine	100	5.00	ug/L
			4-Chloro-3-methylphenol	10.0	2.41	ug/L
			Safrole	20.0	2.34	ug/L
			2-Methylnaphthalene	4.00	1.51	ug/L
			Isosafrole	20.0	1.00	ug/L
			1,2,4,5-Tetrachlorobenzene	10.0	1.73	ug/L
			Hexachlorocyclopentadiene	50.0	3.10	ug/L
			2,4,6-Trichlorophenol	10.0	0.562	ug/L
			2,4,5-Trichlorophenol	10.0	1.19	ug/L
			2-Chloronaphthalene	4.00	0.531	ug/L
			2-Nitroaniline	10.0	1.73	ug/L
			1,4-Naphthoquinone	50.0	13.8	ug/L
			Dimethyl phthalate	4.00	0.210	ug/L
			m-Dinitrobenzene	10.0	2.00	ug/L
			2,6-Dinitrotoluene	10.0	1.89	ug/L
			Acenaphthylene	4.00	0.490	ug/L
			3-Nitroaniline	10.0	2.00	ug/L
			Acenaphthene	4.00	0.627	ug/L
			2,4-Dinitrophenol	30.0	10.0	ug/L
			4-Nitrophenol	10.0	3.17	ug/L
			Dibenzofuran	4.00	0.650	ug/L
			2,4-Dinitrotoluene	10.0	1.66	ug/L
			Pentachlorobenzene	10.0	2.00	ug/L
			1-Naphthylamine	10.0	3.10	ug/L
			2,3,4,6-Tetrachlorophenol	50.0	2.00	ug/L
			2-Naphthylamine	10.0	0.527	ug/L
			Diethyl phthalate	4.00	0.380	ug/L
			Thionazin	50.0	0.864	ug/L
			4-Chlorophenyl phenyl ether	10.0	1.66	ug/L
			Fluorene	4.00	0.310	ug/L
			4-Nitroaniline	10.0	2.00	ug/L
			4,6-Dinitro-2-methylphenol	50.0	9.10	ug/L
			N-Nitrosodiphenylamine	10.0	0.440	ug/L
			Diphenylamine	10.0	1.06	ug/L
			1,3,5-Trinitrobenzene	50.0	19.2	ug/L
			Diallate	20.0	0.560	ug/L
			Phorate	50.0	2.00	ug/L
			Phenacetin	20.0	1.08	ug/L
			4-Bromophenyl phenyl ether	10.0	0.430	ug/L
			Dimethoate	20.0	2.64	ug/L
			Hexachlorobenzene	10.0	0.660	ug/L

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Continued						
			4-Aminobiphenyl	50.0	4.50	ug/L
			Pronamide	20.0	2.00	ug/L
			Pentachlorophenol	50.0	20.0	ug/L
			Pentachloronitrobenzene	50.0	2.00	ug/L
			Disulfoton	50.0	2.76	ug/L
			Phenanthrene	4.00	0.686	ug/L
			Anthracene	4.00	0.420	ug/L
			Methyl parathion	50.0	9.70	ug/L
			Di-n-butyl phthalate	4.00	1.16	ug/L
			Methapyrilene	50.0	20.0	ug/L
			Fluoranthene	4.00	0.900	ug/L
			Pyrene	10.0	0.370	ug/L
			Butyl benzyl phthalate	4.00	1.00	ug/L
			Famphur	100	1.54	ug/L
			3,3'-Dimethylbenzidine	20.0	4.00	ug/L
			2-Acetylaminofluorene	100	6.99	ug/L
			Bis(2-ethylhexyl) phthalate	10.0	2.45	ug/L
			3,3'-Dichlorobenzidine	50.0	3.12	ug/L
			Benzo[a]anthracene	4.00	0.904	ug/L
			Chrysene	4.00	0.540	ug/L
			Di-n-octyl phthalate	10.0	4.00	ug/L
			Benzo[b]fluoranthene	4.00	1.30	ug/L
			Benzo[k]fluoranthene	4.00	0.460	ug/L
			7,12-Dimethylbenz(a)anthracene	20.0	1.56	ug/L
			Benzo[a]pyrene	4.00	0.730	ug/L
			3-Methylcholanthrene	20.0	5.11	ug/L
			Dibenz(a,h)anthracene	10.0	2.18	ug/L
			Indeno[1,2,3-cd]pyrene	10.0	3.08	ug/L
			Benzo[g,h,i]perylene	4.00	0.500	ug/L
			Bis(2-chloroethyl)ether	10.0	0.831	ug/L
			o-Toluidine	10.0	1.40	ug/L
			Chlorobenzilate	10.0	0.657	ug/L
			Ethyl Parathion	50.0	5.07	ug/L
			bis (2-chloroisopropyl) ether	10.0	0.280	ug/L
Surrogate Cpnd						
			2-Fluorophenol	5.00	1.00	
			Phenol-d5	5.00	3.00	
			Nitrobenzene-d5	5.00	1.00	
			2-Fluorobiphenyl	5.00	1.00	
			2,4,6-Tribromophenol	5.00	1.00	
			Terphenyl-d14	5.00	1.00	

RL MDL Units

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Appendix B

Matrix	Method	Test Description	Analyte					
				RL	MDL	Units		
Continued Water	8081B	Organochlorine Pesticides, specific analyte list	4,4'-DDD	0.0500	0.00420	ug/L		
			4,4'-DDE	0.0500	0.00420	ug/L		
			4,4'-DDT	0.0500	0.0240	ug/L		
			Aldrin	0.0500	0.00620	ug/L		
			alpha-BHC	0.0500	0.00965	ug/L		
			beta-BHC	0.0500	0.00910	ug/L		
			DNU Chlordane (n.o.s.)	0.500	0.120	ug/L		
			delta-BHC	0.0500	0.00780	ug/L		
			Dieldrin	0.0500	0.00460	ug/L		
			Endosulfan I	0.0500	0.00584	ug/L		
			Endosulfan II	0.0500	0.00660	ug/L		
			Endosulfan sulfate	0.0500	0.00490	ug/L		
			Endrin	0.0500	0.00860	ug/L		
			Endrin aldehyde	0.0500	0.00865	ug/L		
			gamma-BHC (Lindane)	0.0500	0.0103	ug/L		
			Heptachlor	0.0500	0.0100	ug/L		
			Heptachlor epoxide	0.0500	0.00320	ug/L		
			Kepone	0.348	0.891	ug/L		
			Methoxychlor	0.100	0.0140	ug/L		
			Toxaphene	5	1.50	ug/L		
			Isodrin	0.05	0.0124	ug/L		
			Surrogate Cpnd					
					DCB Decachlorobiphenyl	0.0500	0.0150	
					Tetrachloro-m-xylene	0.0500	0.0120	
							RL	MDL
Water	8321A	Herbicides, specific analyte list	2,4,5-T	5.00	1.45	ug/L		
			2,4-D	5.00	1.58	ug/L		
			Silvex (2,4,5-TP)	5.00	0.970	ug/L		
			Dinoseb	5.00	0.225	ug/L		
			Surrogate Cpnd					
		DCAA	5.00	1.00				
				RL	MDL	Units		
Water	8082A	PCBs, specific analyte list	PCB-1016	1.00	0.170	ug/L		
			PCB-1221	1.00	0.180	ug/L		
			PCB-1232	1.00	0.130	ug/L		
			PCB-1242	1.00	0.104	ug/L		
			PCB-1248	1.00	0.170	ug/L		
			PCB-1254	1.00	0.140	ug/L		
			PCB-1260	1.00	0.0890	ug/L		

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Caroline Brewer
Trihydro Corporation
388 Main, Suite C
Lander, WY 82520
cbrewer@trihydro.com
Tel: (307) 745-7474

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Appendix B

Matrix	Method	Test Description	Analyte	RL	MDL	Units
Continued						
			Aroclors (Total)	1.00	0.0730	ug/L
		Surrogate Cpnd				
			DCB Decachlorobiphenyl	0.100	0.0500	
			Tetrachloro-m-xylene	0.100	0.0500	

Appendix C

Matrix	Method	Test Description	Analyte	RL	RL	Units
Water	9040C	pH	pH adj. to 25 deg C Temperature	0.100 1.00	0.100 1.00	SU Degrees
Water	350.1	Nitrogen, Ammonia	Ammonia	0.05	0.0290	mg/L
Water	300.0A	Chloride, Fluoride and Sulfate	Chloride Fluoride Sulfate	3.00 0.1 2.5	1.02 0.165 1.03	mg/L mg/L mg/L
Water	6010D	6 Total Metals -- Ca, Fe, Mn, Mg, K, Na	Calcium Iron Magnesium Manganese Potassium Sodium	1.0 0.3 1.0 10.0 1.0 1.00	0.0241 0.00910 0.00420 0.450 0.237 0.0972	mg/L mg/L mg/L ug/L mg/L mg/L
Water	SM 2540C	Solids, Total Dissolved (TDS)	Total Dissolved Solids	10.0	4.70	mg/L
Water	SM 2320B	Alkalinity - Bicarbonate & Carbonate	Bicarbonate Alkalinity as CaCO3 Carbonate Alkalinity as CaCO3	1.07 10.0	3.08 3.08	mg/L mg/L
Water	353.2	Nitrogen, Nitrate-Nitrite	Nitrate Nitrite as N	0.100	0.0440	mg/L

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Trihydro Corporation
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cbrewer@trihydro.com
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Appendix C

Matrix	Method	Test Description	Analyte	RL	MDL	Units
Water	410.4	COD	Chemical Oxygen Demand	4.06	8.68	mg/L
Water	SM 5310B	TOC	Total Organic Carbon - Quad	1.00	0.345	mg/L

Additional Tests

Matrix	Method	Test Description	Analyte	RL	MDL	Units
Water	SM 2540D	Solids, Total Suspended (TSS)	Total Suspended Solids	4.00	1.10	mg/L
Water	SM5210B	BOD	Biochemical Oxygen Demand	2.00	0.236	mg/L
Water	8015D	TPH-GRO (C6-C10)	GRO (C6-C10)	25.0	10.0	ug/L
		Surrogate Cpnd	a,a,a-Trifluorotoluene	1.00	0.500	
Water	8015D	WY TPH-DRO (C10-C32)	DRO (C10-C32)	0.250	0.0326	mg/L
		Surrogate Cpnd	o-Terphenyl	0.00200	0.000823	
			n-Octacosane	0.00200	0.00151	

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Analytical Sample Information

Analysis Method	Matrix	Preservative	Client Sub List Desc Container	Volume Required	Holding Time
Alkalinity 2320B	Water	None	Alkalinity - Bicarbonate & Carbonate Plastic 250ml - unpreserved	50 mL	14 Days
Anions, Ion Chromatography 300.0_28D	Water	None	Chloride, Fluoride and Sulfate Plastic 250ml - unpreserved	100 mL	28 Days
BOD, 5 Day SM5210B_BODCalc	Water	None	BOD Plastic 1 liter - unpreserved	1000 mL	48 Hours
COD 410.4	Water	Sulfuric Acid	COD Amber Glass 500mL - Sulfuric Acid	500 mL	28 Days
Diesel Range Organics (DRO) (GC) 8015D_DRO	Water	None	WY TPH-DRO (C10-C32) Amber Glass 1 liter - unpreserved	2000 mL	7 Days
EDB, DBCP, and 1,2,3-TCP (GC) 8011	Water	Sodium Thiosulfate	EDB, DBCP and 1,2,3-TCP -- Low Level Voa Vial 40ml - with Sodium Thiosulfate	120 mL	14 Days
Gasoline Range Organics (GRO) (GC) 8015D_GRO	Water	Hydrochloric Acid	TPH-GRO (C6-C10) Voa Vial 40ml - Hydrochloric Acid	120 mL	14 Days
Herbicides (LC/MS) 8321A_Herb	Water	None	Herbicides, specific analyte list Voa Vial 40ml - unpreserved	80 mL	7 Days
Mercury (CVAA) 7470A	Water	Nitric Acid	Total Mercury Plastic 500ml - with Nitric Acid	150 mL	28 Days
Metals (ICP) 6010D	Water	Nitric Acid	6 Total Metals -- Ca, Fe, Mn, Mg, K, Na Plastic 500ml - with Nitric Acid	150 mL	180 Days
Metals (ICP/MS) 6020B	Water	Nitric Acid	Total Metals -- 15 specific analyte Plastic 500ml - with Nitric Acid	150 mL	180 Days
Metals (ICP/MS) 6020B	Water	Nitric Acid	Total Metals -- 16 specific analyte Plastic 500ml - with Nitric Acid	150 mL	180 Days
Nitrogen, Ammonia 350.1	Water	Sulfuric Acid	Nitrogen, Ammonia Amber Glass 250ml - Sulfuric Acid	50 mL	28 Days
Nitrogen, Nitrate-Nitrite 353.2_Pres	Water	Sulfuric Acid	Nitrogen, Nitrate-Nitrite Amber Glass 500mL - Sulfuric Acid	500 mL	28 Days
Organic Carbon, Total (TOC) SM5310B	Water	Sulfuric Acid	TOC Amber Glass 250ml - Sulfuric Acid	100 mL	28 Days
Organochlorine Pesticides (GC)			Organochlorine Pesticides, specific analyte list		

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8081B	Water	None	Amber Glass 250ml - unpreserved	500 mL	7 Days
pH			pH		
9040C	Water	None	Plastic 250ml - unpreserved	50 mL	IMMEDIATELY
Polychlorinated Biphenyls (PCBs) by Gas Chromatography			PCBs, specific analyte list		
8082A	Water	None	Amber Glass 250ml - unpreserved	500 mL	1 Year
Semivolatile Organic Compounds (GC/MS)			SVOCs, specific analyte with p-Phenylene Diamine as TIC		
8270E	Water	None	Amber Glass 1 liter - unpreserved	2000 mL	7 Days
Solids, Total Dissolved (TDS)			Solids, Total Dissolved (TDS)		
2540C_Calcd	Water	None	Plastic 1 liter - unpreserved	200 mL	7 Days
Solids, Total Suspended (TSS)			Solids, Total Suspended (TSS)		
2540D	Water	None	Plastic 1 liter - unpreserved	500 mL	7 Days
Sulfide, Acid Soluble and Insoluble (Titrimetric)			Sulfide, Total		
9034	Water	Zinc Acetate and Sodium Hydroxide	Plastic 250ml - with Zinc Acetate & NaOH	250 mL	7 Days
Volatile Organic Compounds by GC/MS			Volatile Organic Compounds, 47 specific analyte		
8260D	Water	Hydrochloric Acid	Voa Vial 40ml - Hydrochloric Acid	120 mL	14 Days
Volatile Organic Compounds by GC/MS			Volatile Organic Compounds, 64 specific analyte		
8260D	Water	Hydrochloric Acid	Voa Vial 40ml - Hydrochloric Acid	120 mL	14 Days

Hold Times listed above represent the minimum allotted time between sampling and lab extraction, prep or analysis.

Multiple analyses may be consolidated into fewer containers. Please contact your Project Manager for clarification when requesting sample containers.

Except for some special tests, all samples should be kept cold at 6 degrees C.

EUROFINS TESTAMERICA TERMS AND CONDITIONS OF SALE (Short Form)

When a purchaser (Client) places an order for laboratory, consulting or sampling services from TestAmerica Laboratories, Inc., d/b/a Eurofins TestAmerica a Delaware corporation, Eurofins TestAmerica shall provide the ordered services pursuant to these Terms and Conditions and the related Quotation or Price Schedule, or as agreed in a negotiated contract. In the absence of a written agreement to the contrary, a client order constitutes an acceptance by the Client of Eurofins TestAmerica's offer to do business under these Terms and Conditions, and an agreement to be bound by these Terms and Conditions. Receipt of a Client's samples at a Eurofins TestAmerica laboratory constitutes acceptance of these Terms and Conditions (in the absence of any other negotiated contract). No contrary or additional terms and conditions expressed in a Client's document shall be deemed to become a part of the contract created upon acceptance of these Terms and Conditions, unless accepted by Eurofins TestAmerica in writing.

1. ORDERS AND RECEIPT OF SAMPLES

1.1 A Client may place an order (i.e., specify a Scope of Work) either by submitting a purchase order to Eurofins TestAmerica in writing or by telephone subsequently confirmed in writing, or by negotiated contract. Whichever option the Client selects for placing an order, the order shall not be valid unless it contains sufficient specification to enable Eurofins TestAmerica to carry out the Client's requirements. In particular, samples must be accompanied by: a) adequate instruction on type of analysis requested, and b) complete written disclosure of the known or suspected presence of any hazardous substances, as defined by applicable federal or state law. If a Client fails to provide these required disclosures accompanying the submission of samples, and such failure results in an interruption in the lab's ability to process work due to contamination of instruments or work areas, the Client will be responsible for the costs of clean-up and recovery.

1.2 The Client shall provide one week's advance notice of the sample delivery schedule, or any changes to the schedule, whenever possible. Upon timely delivery of samples, Eurofins TestAmerica will use its best efforts to meet mutually agreed turnaround times. All turnaround times will be calculated from the point in time when Eurofins TestAmerica has determined that it can proceed with defined work following receipt, inspection of samples, and resolution of any discrepancies in Chain-of-Custody forms and project guidance regarding work to be done (Sample Delivery Acceptance). Rush turnaround times not requested in advance of the delivery of samples and specifically agreed to by the lab are not guaranteed. If the Client changes the sample delivery schedule prior to Sample Delivery Acceptance, Eurofins TestAmerica reserves its rights to modify its turnaround time commitment, change the date upon which Eurofins TestAmerica will accept samples, or refuse Sample Delivery Acceptance for the affected samples.

1.3 Eurofins TestAmerica reserves the right, exercisable at any time, to refuse or revoke Sample Delivery Acceptance for any sample which in the sole judgment of Eurofins TestAmerica: a) is of unsuitable volume; b) may pose a risk or become unsuitable for handling, transport, or processing for any health, safety, environmental or other reason, whether or not due to the presence of any hazardous substance in the sample and whether or not such presence has been disclosed to Eurofins TestAmerica by the Client; or c) holding times cannot be met, due to passage of more than 48 hours from the time of sampling or 1/2 the holding time for the requested test, whichever is less.

1.4 Prior to Sample Delivery Acceptance, the entire risk of loss or damage to samples remains with the Client, except where Eurofins TestAmerica provides courier services. In no event will Eurofins TestAmerica have any responsibility or liability for the action or inaction of any carrier shipping or delivering any sample to or from Eurofins TestAmerica's premises. Client is responsible for assuring that any sample that contains or may contain any hazardous substance to be delivered to Eurofins TestAmerica's premises is properly packaged, labeled, transported and delivered, all in accordance with applicable laws.

1.5 Eurofins TestAmerica reserves the right to begin processing samples upon receipt, unless the Client specifically notifies Eurofins TestAmerica in writing prior to sample receipt that the samples are to be held without preparation or other processing or pending receipt of a purchase order. Eurofins TestAmerica shall under no circumstances be responsible for missed holding times or turnaround times or for re-sampling costs if samples are released from hold with less than 48 hours or 1/2 the holding time for the requested test remaining, whichever is less.

2. PAYMENT TERMS

2.1 Services performed by Eurofins TestAmerica will be in accordance with prices quoted and later confirmed in writing or as stated in the Price Schedule. Quoted prices do not include sales tax. Applicable sales tax will be added to invoices where required by law.

2.2 Invoices may be submitted to Client upon completion of any sample delivery group. Billing corrections must be requested within 30 days of invoice date. Payment in advance is required for all clients except those whose credit has been established with Eurofins TestAmerica. For clients with approved credit, payment terms are net 30 days from the date of invoice by Eurofins TestAmerica, unless alternative terms have been agreed in a separate written agreement. Payment shall be made without retainage, and shall not be contingent upon the receipt of funds from third parties. All overdue payments are subject to an additional interest and service charge of one and one half percent (1.5%) (or the maximum rate permissible by law, whichever is less) per month or portion thereof from the due date until the date of payment. All fees are charged or billed directly to the Client. The billing of a third party will not be accepted without a statement, signed by the third party, acknowledging and accepting payment responsibility in accordance with these payment terms.

2.3 If Client fails to make timely payment of its invoices, Eurofins TestAmerica reserves the right to pursue all appropriate remedies, including withdrawing certifications, suspending work and withholding delivery of data under this order without recourse. Client shall be responsible for all reasonable fees, expenses, and costs of collection including but not limited to arbitrator's and attorney's fees. Eurofins TestAmerica reserves the right to refuse to proceed with work at any time based upon an unfavorable Client credit report.

3. CHANGE ORDERS, TERMINATION

3.1 Changes to the Scope of Work, price, or result delivery date may be initiated by Eurofins TestAmerica after Sample Delivery Acceptance due to any condition which conflicts with analytical, QA or other protocols warranted in these Terms and Conditions. Eurofins TestAmerica will not proceed with such changes until an agreement with the Client is reached on the amount of any cost, schedule change or technical change to the Scope of Work, and such agreement is documented in writing.

3.2 Changes to the Scope of Work, including but not limited to increasing or decreasing the work, changing test and analysis specification, or acceleration in the performance of the work may be initiated by the Client after Sample Delivery Acceptance. Such change must be documented in writing and may result in a change in cost and turnaround time commitment. Eurofins TestAmerica's acceptance of such changes is contingent upon technical feasibility and operational capacity.

3.3 Suspension or termination of all or any part of the work may be initiated by the Client. Eurofins TestAmerica will be compensated consistent with Section 2 of these Terms and Conditions. Eurofins TestAmerica will complete all work in progress and be paid in full for all work completed.

4. WARRANTIES AND LIABILITY

4.1 Where applicable, Eurofins TestAmerica will use appropriate and approved analytical test methods. Eurofins TestAmerica has referenced these methods in its Laboratory Quality Manuals and has documented them in Standard Operating Procedures. Eurofins TestAmerica reserves the right based on its reasonable judgment to deviate from these methodologies as necessary or appropriate to the extent required by the nature or composition of the sample, which deviations, if any, will be made on a basis consistent with recognized standards of the industry and/or Eurofins TestAmerica's Laboratory Quality Manuals. Client may request that Eurofins TestAmerica perform according to a mutually agreed Quality Assurance Project Plan (QAPP). If samples arrive prior to agreement on a QAPP, Eurofins TestAmerica will proceed with analyses under its standard Quality Manuals then in effect. Eurofins TestAmerica will not be responsible for any resampling or other charges if work must be repeated to comply with a subsequently finalized QAPP.

4.2 Eurofins TestAmerica shall start preparation and/or analysis within holding times provided that Sample Delivery Acceptance occurs within 48 hours of sampling or 1/2 of the holding time for the test, whichever is less,

unless the Client has specifically requested that Eurofins TestAmerica hold the samples without preparation or other processing or pending receipt of a purchase order. Where resolution of inconsistencies leading to Sample Delivery Acceptance does not occur within this period, Eurofins TestAmerica will use its best efforts to meet holding times and will proceed with the work provided that, in Eurofins TestAmerica's judgment, the chain-of-custody or definition of the Scope of Work provide sufficient guidance. Reanalysis of samples to comply with Eurofins TestAmerica's Quality Manuals will be deemed to have met holding times provided the initial analysis was performed within the applicable holding time. Where reanalysis demonstrates that sample matrix interference is the cause of failure to meet any Quality Manual requirements, the warranty will be deemed to have been met.

4.3 Eurofins TestAmerica warrants that it possesses and maintains all licenses and certifications that are required to perform services under these Terms and Conditions provided that such requirements are specified in writing to Eurofins TestAmerica prior to Sample Delivery Acceptance. Eurofins TestAmerica will notify the Client in writing of any decertification or revocation of any license, or notice of either, that affects work in progress.

4.4 The warranty obligations set forth in Sections 4.1, 4.2 and 4.3 are the sole and exclusive warranties given by Eurofins TestAmerica in connection with any services performed by Eurofins TestAmerica or any results generated from such services, and Eurofins TestAmerica gives and makes **NO OTHER REPRESENTATION OR WARRANTY OF ANY KIND, EXPRESS OR IMPLIED**. No representative of Eurofins TestAmerica is authorized to give or make any other representation or warranty or modify this warranty in any way.

4.5 Client's sole and exclusive remedy for breach of warranty in connection with any services performed by Eurofins TestAmerica will be limited to repeating any services performed, contingent on the Client's providing, at the request of Eurofins TestAmerica and at the Client's expense, additional sample(s) if necessary. Any reanalysis requested by the Client generating results consistent with the original results will be at the Client's expense. If resampling is necessary, Eurofins TestAmerica's liability for resampling costs will be limited to actual cost or one hundred and fifty dollars (\$150) per sample, whichever is less.

4.6 Eurofins TestAmerica's liability for any and all causes of action arising hereunder, whether based in contract, tort, warranty, negligence or otherwise, shall be limited to the lesser amount of compensation for the services performed or \$100,000. All claims, including those for negligence, shall be deemed waived unless suit thereon is filed within one year after Eurofins TestAmerica's completion of the services. Under no circumstances, whether arising in contract, tort (including negligence), or otherwise, shall Eurofins TestAmerica be responsible for loss of use, loss of profits, or for any special, indirect, incidental or consequential damages occasioned by the services performed or by application or use of the reports prepared.

4.7 In no event shall Eurofins TestAmerica have any responsibility or liability to the Client for any failure or delay in performance by Eurofins TestAmerica that results, directly or indirectly, in whole or in part, from any cause or circumstance beyond the reasonable control of Eurofins TestAmerica. Such causes and circumstances include, but are not limited to, acts of God, acts of Client, acts or orders of any governmental authority, strikes or other labor disputes, natural disasters, accidents, wars, civil disturbances, equipment breakdown, matrix interference or unknown highly contaminated samples that impact instrument operation, unavailability of supplies from usual suppliers, difficulties or delays in transportation, mail or delivery services, or any other cause beyond Eurofins TestAmerica's reasonable control.

5. RESULTS, WORK PRODUCT

5.1 Data or information provided to Eurofins TestAmerica or generated by services performed under this agreement shall only become the property of the Client upon receipt in full by Eurofins TestAmerica of payment for the entire order. Ownership of any analytical method, QA/QC protocols, software programs or equipment developed by Eurofins TestAmerica for performance of work will be retained by Eurofins TestAmerica. Client shall not disclose such information to any third party without Eurofins TestAmerica's express prior consent.

5.2 Data and sample materials provided by Client or at Client's request, and the result obtained by Eurofins TestAmerica shall be held in confidence

(unless such information is generally available to the public or is in the public domain or Client has failed to pay Eurofins TestAmerica for all services rendered or is otherwise in breach of these Terms and Conditions), subject to any disclosure required by law or legal process.

5.3 Should the results delivered by Eurofins TestAmerica be used by the Client or Client's client, even though subsequently determined not to meet the warranties described in these Terms and Conditions, then the compensation will be adjusted based upon mutual agreement. In no case shall the Client unreasonably withhold Eurofins TestAmerica's right to independently defend its data.

5.4 Eurofins TestAmerica reserves the right to perform the services at any laboratory in the Eurofins TestAmerica network. If a Client has requested a particular location for the work, Eurofins TestAmerica will inform the Client when operational constraints require the work to be performed at another Eurofins TestAmerica location. In addition, Eurofins TestAmerica reserves the right to subcontract services ordered by the Client to another laboratory or laboratories, if, in Eurofins TestAmerica's sole judgment, it is reasonably necessary, appropriate or advisable to do so. Eurofins TestAmerica will in no way be liable for any subcontracted services (outside the Eurofins TestAmerica network) except for work performed at laboratories which have been audited and approved by Eurofins TestAmerica.

5.5 Eurofins TestAmerica will dispose of non-hazardous samples, sample extracts and digestates 30 days after the final analytical report is issued, unless instructed to store them for an alternate period of time or to return such samples to the Client, in a manner consistent with U.S. Environmental Protection Agency regulations or other applicable federal, state or local requirements. Charges for disposal will be billed to the client. Alternatively, samples can be returned to the client for disposal. Cost of return shipping will be billable to the client. Air samples in Summa canisters and tedlar bags are used and the containers cleaned immediately after testing, such that those samples are not retained. Longer storage periods may be requested and may be accommodated if space allows, and for an additional charge. Any samples for projects that are canceled or not accepted, or for which return was requested, will be returned to the Client at its own expense. Eurofins TestAmerica reserves the right to return to the Client any sample or unused portion of a sample that is not within Eurofins TestAmerica's permitted capability or the capabilities of Eurofins TestAmerica's designated waste disposal vendor(s). **ALL DIOXIN, MIXED WASTE, AND RADIOACTIVE SAMPLES WILL BE RETURNED TO THE CLIENT**, unless prior arrangements for disposal are made.

5.6 Unless a different time period is agreed to in an order under these Terms and Conditions, Eurofins TestAmerica agrees to retain all records for five (5) years.

5.7 If Eurofins TestAmerica is required to respond to legal process related to services for Client, Client agrees to reimburse Eurofins TestAmerica for hourly charges for personnel involved in the response and attorney's fees reasonably incurred in obtaining advice concerning the response, preparation to testify, and appearances related to the legal process, travel and all reasonable expenses associated with the litigation. Additional consulting beyond that normally associated with lab reports will be billed at Eurofins TestAmerica's current published rates.

6. INSURANCE

6.1 During the performance of services under these Terms and Conditions, Eurofins TestAmerica shall maintain in force Workers' Compensation and Employer's Liability Insurance in accordance with the laws of the states having jurisdiction over Eurofins TestAmerica's employees who are engaged in the performance of the work. Eurofins TestAmerica shall also maintain during such period Comprehensive General and Contractual Liability (limit of \$1,000,000 per occurrence/ \$2,000,000 aggregate), Comprehensive Automobile Liability, owned and hired, (\$1,000,000 combined single limit), Professional Liability Insurance (limit of \$5,000,000 per claim/ aggregate and Pollution Liability insurance (limit of \$1,000,000 per claim/aggregate).

7. MISCELLANEOUS PROVISIONS

7.1 These Terms and Conditions, together with any additions or revisions which may be agreed to in writing by Eurofins TestAmerica, embody the whole agreement of the parties and provide the only remedies available. There are no promises, terms, conditions, understandings, obligations or

agreements other than those contained herein, and these Terms and Conditions shall supersede all previous communications, representations, or agreements, either verbal or written, between the Client and Eurofins TestAmerica. These Terms and Conditions, and any transactions or agreements to which they apply, shall be governed both as to interpretation and performance by the laws of the state where Eurofins TestAmerica's services are performed.

7.2 The invalidity or unenforceability, in whole or in part, of any provision, term or condition hereof shall not affect in any way the validity or enforceability of the remainder of these Terms and Conditions, the intent of the parties being that the provisions be severable. The section headings of these Terms and Conditions are intended solely for convenient reference and shall not define, limit or affect in any way these Terms and Conditions or their interpretations. No waiver by either party of any provision, term or condition hereof or of any obligation of the other party hereunder shall constitute a waiver of any subsequent breach or other obligation.

7.3 The obligations, liabilities, and remedies of the parties, as provided herein, are exclusive and in lieu of any others available at law or in equity. Indemnifications, releases from liability and limitations of liability shall apply, notwithstanding the fault, negligence or strict liability of the party to be indemnified, released, or whose liability is limited, except to the extent of sole negligence or willful misconduct.

Caroline Brewer

To: Caroline Brewer
Subject: RE: Wyoming Landfills: 2024 Quote Review/Request

From: Joy Dela rosa-chang <Joy.Chang@et.eurofinsus.com>
Sent: Tuesday, January 9, 2024 5:00 PM
To: Caroline Brewer <cbrewer@trihydro.com>
Cc: Nella Dagnillo <ndagnillo@trihydro.com>; Maura Hanning <mhanning@trihydro.com>; Anthony Grimaldi <Anthony.Grimaldi@et.eurofinsus.com>; Patrick McEntee <Patrick.McEntee@et.eurofinsus.com>
Subject: RE: Wyoming Landfills: 2024 Quote Review/Request

Caution: This email is from an external sender. Please report suspicious emails using the **Report Message** button in Outlook.

Carolline,
Eurofins Denver laboratory will extend pricing based on our Quotation Number 2802415 thru December 31, 2024. We will provide a new quote/pricing in 2025. Kindly let us know if you have any questions.

Joy Chang
Client Relations Manager
Phone: 303-467-7396
Joy.Chang@ET.EurofinsUS.com
www.EurofinsUS.com/Env

ATTACHMENT C

2024 STANDARD SCHEDULE OF CHARGES

TRIHYDRO STANDARD SCHEDULE OF CHARGES

JANUARY 1, 2024 - DECEMBER 31, 2024 ^{2, 3, 4}

<u>PERSONNEL</u>	<u>UNIT RATE</u> ^{1, 7}
Senior Principal-----	260.00/hour
Principal-----	235.00/hour
Project Principal-----	220.00/hour
Technical Specialist 4-----	270.00/hour
Technical Specialist 3-----	255.00/hour
Technical Specialist 2-----	240.00/hour
Technical Specialist 1-----	225.00/hour
Professional Level 12-----	220.00/hour
Professional Level 11-----	206.00/hour
Professional Level 10-----	192.00/hour
Professional Level 9-----	178.00/hour
Professional Level 8-----	164.00/hour
Professional Level 7-----	149.00/hour
Professional Level 6-----	135.00/hour
Professional Level 5-----	125.00/hour
Professional Level 4-----	113.00/hour
Professional Level 3-----	102.00/hour
Professional Level 2-----	93.00/hour
Professional Level 1-----	83.00/hour
Technician Level 8-----	132.00/hour
Technician Level 7-----	123.00/hour
Technician Level 6-----	113.00/hour
Technician Level 5-----	105.00/hour
Technician Level 4-----	95.00/hour
Technician Level 3-----	86.00/hour
Technician Level 2-----	76.00/hour
Technician Level 1-----	67.00/hour
Administrative 4-----	87.00/hour
Administrative 3-----	76.00/hour
Administrative 2-----	68.00/hour
Administrative 1-----	57.00/hour
<u>EXPENSES</u>	
Subcontracts (Labor, Equipment and Services)-----	Cost + 10%
Shipping (i.e. Documents, Equipment, Supplies)-----	Cost
<u>TRAVEL EXPENSES</u>	
Meal Per Diem ⁶ -----	\$59/day/person
Airline Tickets-----	Cost
Hotel/Motel-----	Cost
Rental Vehicle-----	Cost
<u>FIELD EXPENSES AND EQUIPMENT</u>	
Consumable Field Supplies-----	Cost + 10%
Rental Equipment-----	Cost + 10%
Purchased Equipment-----	Cost + 10%
Company Field Instruments, Equipment, Vehicles, etc.	See Project-Specific Cost Estimate
Consumable Field Supplies and PPE-----	See Project-Specific Cost Estimate
Company Vehicles (daily) ⁵ -----	\$95/day min or 65.5 cents/mile
Company Vehicles (monthly)-----	Cost + fuel cost

1. The above charges include fringe benefits, overhead and profit. No multiplier is used for billing.
2. An annual escalation rate less than or equal to 5% will be applied to these rates for multi-year projects and contracts.
3. Payment of invoices shall be due within thirty days; delinquent amounts due shall accrue a late charge of 1 1/2% per month from date of invoice.
4. The rates in this Schedule of Charges are subject to change on December 31, 2024.
5. Minimum charge of \$95/day. Daily mileage exceeding 145 miles is charged at the current IRS rate per mile. Mileage rates are subject to change throughout the year.
6. Any international travel meal per diem will be at cost.
7. Expert testimony services, including but not limited to preparing for and time spent in depositions, arbitration or trial testimony, shall be charged at 3.0 times the individual's billing level. Other expert technical consulting services, including but not limited to research, review, evaluation, and preparation of expert technical opinions and deliverables, shall be charged at 2.0 times the individual's billing level.

John

Date: 3/1/18

City Contract # 6798

PROFESSIONAL SERVICES AGREEMENT
for

HAPPY JACK LANDFILL ENVIRONMENTAL MONITORING

THIS AGREEMENT is made and entered into this 12th day of March, 2018, by and between the City of Cheyenne, located at 2101 O'Neil Avenue, Cheyenne, Wyoming 82001 ("City"), and Trihydro Corporation, located at 1252 Commerce Drive, Laramie, Wyoming 82070 ("Consultant").

RECITALS

In consideration of the mutual covenants and obligations herein expressed, it is agreed by and between the parties as follows:

1. SCOPE OF WORK

The Consultant agrees to provide services as described for the above-mentioned project, in accordance with the Scope of Work attached hereto as Exhibits A, B, and the attached Sanitary Landfill Environmental Plan (SLEMP), incorporated by reference and expressly made a part hereof.

2. TIME OF PERFORMANCE

The work included in this Scope of Work will be completed when the final annual report of 2020 is turned into WDEQ by March 1, 2021, as stated in Exhibit A.

3. CONSULTANT RESPONSIBILITY

In providing services under this Agreement, the Consultant shall perform in a manner consistent with that degree of care and skill ordinarily exercised by members of the same profession currently practicing under similar circumstances at the same time and in the same or similar locality. Upon notice to the Consultant and by mutual agreement between the parties, the Consultant will, without additional compensation, correct those services not meeting such a standard.

4. COMPLIANCE WITH LAWS

The Parties agree that they will perform their obligations as provided in this Agreement in accordance with all applicable laws and ordinances.

5. DRUG-FREE WORKPLACE

In compliance with the Drug Free Work Place Act of November 1988, the City of Cheyenne has established an Alcohol and Controlled Substance Policy that pertains to alcohol and drug usage by City Employees. All architects, engineers and other consultants under contract with City of Cheyenne, and their employees and sub-consultants, are

required to comply with the provisions of the City's Alcohol and Controlled Substance Policy for drug and/or alcohol usage on City property or other sites occupied by the Consultant while performing the duties and responsibilities of the Agreement. It is the responsibility of the Consultant to familiarize themselves with the requirements of this policy and to inform all their employees and sub-consultants of the requirements and ensure their compliance. If the Consultant, their employees or sub-consultants are found in violation of this policy, the Agreement may be terminated.

6. INDEMNIFICATION/HOLD HARMLESS

The Consultant agrees to indemnify, hold harmless and defend the City from and against any and all liabilities, claims, penalties, forfeitures and suits, and the cost and expenses incident thereto, including reasonable attorney's fees, which may hereafter arise as a result of death or bodily injury to any person, destruction or damage to any property, contamination of or adverse effects on the environment, or any violation of governmental laws, regulations or orders to the extent caused by (1) the Consultant's breach of any term or provision of this Agreement; or (2) any negligent or wrongful act, error or omission by the Consultant, or its employees or sub-consultants in the performance of this Agreement. The Consultant acknowledges that it may incur a financial obligation to the City pursuant to the terms of this paragraph.

7. INSURANCE REQUIREMENTS

The Consultant must provide proof of the following insurance coverages:

Commercial General Liability Insurance

For claims arising out of bodily injury, illness or death, or from damage to or destruction of property of others, including loss or use thereof, with minimum limits of \$1,000,000 per occurrence, \$2,000,000 aggregate for the entire term of the Agreement.

Business Automobile Insurance

Including owned, non-owned and hired vehicles with minimum limits for bodily injury and property damage of \$1,000,000 per accident for the entire term of the Agreement.

Workers' Compensation

Workers' compensation coverage shall be in effect for the entire term of the Agreement, as required by Wyoming law, for all employees or agents providing services under this Agreement. Consultant shall provide the City with proof of workers' compensation or employer's liability insurance coverage.

Professional Liability Insurance

The Consultant shall provide proof of professional liability insurance or errors and omissions liability insurance in an amount not less than \$500,000 to protect the City from any and all claims arising from the Consultant's negligence in the performance of duties under this Agreement. The City prefers that this liability insurance coverage be provided pursuant to an "occurrence" policy.

If this coverage is provided pursuant to a "claims made" policy:

- a. Consultant shall, concurrently with the execution of this Agreement, provide the City with a certificate of insurance demonstrating that such coverage is or shall be in effect at the time the Consultant begins the provision of services under this Agreement; and
- b. In the event the Consultant's services extend into a future policy period, the Consultant shall, prior to the policy expiration date, provide the City with a new certificate of insurance demonstrating that such coverage is or shall be in effect during all periods of time that Consultant will provide services under this Agreement; and
- c. Consultant shall maintain said "claims made" coverage for a period of five (5) years following the last date that Consultant has provided services under this Agreement; and
- d. In the event the Consultant or the insurer terminates "claims made" coverage prior to the expiration of the periods provided in subparagraphs a., b., or c. of this paragraph, the Consultant shall provide to the City advance written notification of the termination of said coverage and shall provide the City with an endorsement for an extended reporting period ("tail coverage") which shall be in effect for a period of time not less than five (5) years following the last date that Consultant has provided services under this Agreement.

Additional Insurance Information

The Consultant shall name the City of Cheyenne as **an Additional Insured** by endorsement on its insurance policies, and shall provide the City with a copy of the endorsements. This requirement does not apply to workers' compensation and professional liability insurance policies.

Consultant shall provide the City with certificates of insurance acknowledging the above-stated coverages prior to beginning any work under this Agreement.

It is understood and agreed that these policies are primary and not contributory. All policies required under this Agreement shall be in effect for the duration of the Agreement. It shall be an affirmative obligation upon Consultant to immediately notify in writing the city risk manager, city clerk, and city attorney of any fact, circumstance, or occurrence that has resulted in or may result in the cancellation of, or substantive change to any insurance coverage required by this Agreement, and failure to do so shall be construed to be a breach of this Agreement.

If requested by the City, the Consultant shall provide the City with copies of insurance policies and/or policy endorsements listing the City of Cheyenne as an additional insured. The City's failure to request or review such policies, endorsements, or certificates shall not affect the City's rights or Consultant's obligation hereunder.

Any insurance company providing coverage under this Agreement shall have a minimum A. M. Best rating of A- (excellent).

8. MINORITY AND DISADVANTAGED BUSINESS ENTERPRISES 49 C.F.R. Part 26
All parties to this Agreement assure that no person will be excluded from participation in, denied the benefits of, or otherwise discriminated against, in connection with the award and performance of this Agreement on the grounds of age, race, color, disability, national origin or sex.

9. COMPENSATION

In consideration of the services to be performed pursuant to this Agreement, the Consultant will bill the City and the City agrees to pay Consultant a not to exceed amount of one hundred and seventy one thousand nine hundred and twelve Dollars (\$171,912.00) as stated in Exhibit B.

10. MONTHLY REPORT

With every monthly billing, Consultant shall provide the City Representative with a written statement of the status of the work with respect to the Scope of Work, time sheets, and work schedule. Failure to provide the required monthly report will delay processing of any payment request until the report is submitted.

11. INDEPENDENT CONSULTANT

The Consultant shall function as an independent consultant for the purposes of this Agreement. The Consultant shall assume sole responsibility for any debts or liabilities that may be incurred by the Consultant in fulfilling the terms of this Agreement. Nothing in this Agreement shall be interpreted as authorizing the Consultant or its agents or employees to act as an agent or representative of or on behalf of the City or to incur any obligation of any kind on the behalf of the City.

12. TAXES

The Consultant agrees to pay all valid taxes, excises, license fees, permit fees, bills, debts and obligations incurred by and in connection with its operations under this Agreement.

13. DEFAULT

Each and every term and condition in this Agreement shall be deemed to be a material element of the Agreement. In the event either party should fail or refuse to perform according to the terms of this Agreement, such party may be declared in default.

14. REMEDIES

In the event a party has been declared in default, such defaulting party shall be allowed a period of fifteen (15) days within which to cure the default. In the event the default remains uncorrected, the non-defaulting party declaring default may elect to:

- a. Terminate the Agreement and seek damages, which damages shall not exceed the contract amount; or
- b. Treat the Agreement as continuing and require specific performance.

15. TERM AND TERMINATION

This Agreement shall be for a period of three years. The parties may by mutual written agreement renew or extend this Agreement.

The City may, without cause, and upon thirty days' written notice to the Consultant, terminate this Agreement in whole or in part at any time, for the City's convenience. Upon receipt of such notice, the Consultant shall:

- a. Discontinue all services affected; and
- b. Deliver to the City within five (5) days all data, drawings, specifications, reports, estimates, summaries, and such other information and materials as may have been accumulated by the Consultant in performing this Agreement, whether completed or in process.
- c. In the event of termination for convenience, the City will pay the Consultant for accepted work done and expenses incurred to the date of termination. Such acceptance shall not be unreasonably withheld.

16. ADDITIONAL REMEDIES

In the event the Consultant fails to strictly perform in accordance with this Agreement, the City may elect to correct the deficiencies and charge the Consultant. In the event of default of any of the conditions by either party which shall require the party not in default to commence legal or equitable action against the defaulting party each party shall bear its own costs and expenses, including without limitation, attorney's fees.

17. GOVERNING LAW, JURISDICTION AND VENUE

The construction, interpretation and enforcement of this Agreement shall be governed by the laws of the State of Wyoming. The courts of the State of Wyoming shall have jurisdiction over any action arising out of this Agreement and over the parties, and the venue shall be the First Judicial District, Laramie County, Wyoming.

18. GOVERNMENTAL IMMUNITY

The City of Cheyenne, along with its officials and employees, does not waive its governmental immunity by entering into this Agreement, except to the extent necessary for the parties to pursue a contract action to clarify or enforce the written terms of the Agreement. Furthermore, the City specifically retain all immunities and defenses available to it as a sovereign or governmental entity pursuant to Wyo. Stat. § 1-39-101, *et seq.*, and all other relevant state and federal law. Designations of venue, choice of law, enforcement actions, and similar provisions should not be constructed as a waiver of governmental immunity. The parties agree that any ambiguity in this Agreement shall not be strictly construed, either against or for either party, except that any ambiguity as to governmental or sovereign immunity shall be construed in favor of governmental immunity.

19. MONITORING ACTIVITIES

The City shall have the right to monitor all activities related to this Agreement that are performed by the Consultant or its sub-consultants. This shall include, but not be limited to, the right to make site inspections at any time and with reasonable notice; to bring experts and consultants on site to examine or evaluate completed work or work in progress; to examine the books, ledgers, documents, papers, and records pertinent to this Agreement; and to observe personnel in every phase of performance of the related work.

20. NON-DISCRIMINATION

The Consultant shall comply with the Civil Rights Act of 1964, the Wyoming Fair Employment Practices Act (Wyo. Stat. § 27-9-105, *et seq.*), the Americans With Disabilities Act (ADA), as amended, 42 U.S.C. § 12101, *et seq.*, and the Age

Discrimination Act of 1975 and any properly promulgated rules and regulations thereto, and shall not discriminate against any individual on the grounds of age, sex, creed, color, race, religion, national origin, ancestry, pregnancy or qualifying disability in connection with the performance under this Agreement.

21. PROFESSIONAL REGISTRATION

The Consultant shall endorse, as required by law, plans and reports prepared under this Agreement, and shall affix thereto his or her seal of professional registration, showing that he or she is licensed to practice in the State of Wyoming, if necessary.

22. ENTIRETY OF AGREEMENT

This Agreement consisting of six pages, and Exhibit A consisting of three pages, Exhibit B consisting of one pages, and the SLEMP consisting of 74 pages, contains the entire understanding of the parties. There are no other terms or conditions, written or oral, concerning or controlling this matter.

IN WITNESS WHEREOF, the parties have hereunto set their hands and seals the day and year first above written.

City of Cheyenne

(SEAL)

Attest:

Carol Intlekofer
Carol Intlekofer, City Clerk

By: M. J. Orr
Marian J. Orr, Mayor

Date: 3.13.13

Consultant

By: Deby L. Forry

Print Name: Deby L. Forry, ESQ

Date: March 26, 2018

EXHIBIT A

SCOPE OF WORK

This Exhibit acts as a supplemental information document and is made a part of the Professional Services Agreement (Agreement) entered into by and between the City of Cheyenne, Wyoming ("City") and Trihydro Corporation ("Consultant"). The provisions of the Exhibit shall be incorporated into the Agreement. The Consultant's proposal shall be attached as a reference and shall be considered subsidiary to this agreement.

A.1 GROUNDWATER SAMPLING AND ANALYSIS

Consultant shall perform environmental monitoring as described in the April 12, 2017 Sanitary Landfill Environmental Monitoring Plan (SLEMP), prepared by Trihydro Corporation. The Consultant will follow all the steps outlined in the SLEMP section 2.3 Groundwater Monitoring Procedures. As stated in SLEMP section 2.3.2.1 Well Evacuation using a dedicated pump, eight (8) of the wells in the Semi-Confined Aquifer have dedicated pneumatically driven pumps. The City does own, and the Consultant will have access to, a towed compressor but the compressor should not be considered reliable and the Consultant is responsible for obtaining the samples if the City owned compressor is not working. All other wells will be sampled using bailers. Table 3 of the SLEMP outlines the sampling schedule.

Sampling shall be performed in the field on the following months:

- | | |
|-----------|------|
| 1) April | 2018 |
| 2) August | 2018 |
| 3) April | 2019 |
| 4) August | 2019 |
| 5) April | 2020 |
| 6) August | 2020 |

A.2 GROUNDWATER DATA STATISTICAL ANALYSES

Consultant shall follow the statistical analysis outlined in the SLEMP section 4.0 Groundwater Data Statistical Analyses. The Consultant will receive the Groundwater monitoring results back to September 1982 once the Professional Services Agreement has been signed by both parties.

A.3 FLUID GAUGING

Consultant shall perform groundwater and leachate gauging for the wells, sumps and other locations as outlined in the SLEMP Table 4. Fluid gauging shall be performed in the field every sampling event, beginning April 2018, and continuing through and including August 2020.

A.4 EXPANSION WELL BASELINE MONITORING

As a variance condition for the expansion of the Landfill the City has agreed to install and monitor four (4) new monitoring wells. The construction of these wells may or may not be constructed during this contract, but in the event the wells are constructed during this contract period, the Consultant will establish a WDEQ approved monitoring plan and perform eight (8) quarterly sampling events as outlined in the SLEMP section 2.4 Future Expansion Area Wells. A second variance condition that needs to be met is outlined in the SLEMP section 2.5 Off-site Water Well Monitoring Program. Table 8 of the SLEMP identifies the known off-site water wells that will need to be tested. Once the City authorizes the Consultant to enact the off-site monitoring outlined in the SLEMP section 2.5.1 Baseline Monitoring Program, the Consultant will create and submit a sampling protocol to the WDEQ prior to initiating the first sampling event.

A.5 METHANE MONITORING

Consultant shall perform methane gas monitoring in the Construction Demolition Recycling Facility, Landfill Equipment Shelter Shop Area and methane monitoring wells (construction date to be determined) as outlined in the referenced SLEMP. There will be fourteen (14) methane monitoring wells surrounding the permitted boundary of the Landfill. Methane monitoring shall be performed in the field during the following months:

March	2018
June	2018
September	2018
December	2018
March	2019
June	2019
September	2019
December	2019
March	2020
June	2020
September	2020
December	2020

A.6 IMMEDIATE NOTIFICATION

Consultant shall immediately, upon becoming aware of any one or more of the following, notify the City by phone call and written, hand delivered letter if:

- 1) Methane concentration above 25% Lower Explosion Limit (LEL) in facility buildings or above 100% LEL at compliance points at the landfill boundary.
- 2) Groundwater detection above Groundwater Protection Standards.
- 3) Statistically significant increases in contaminant concentrations in groundwater.

All requirements are explained in the SLEMP section 6.0 Notification Requirements.

A.7 EVENT REPORTS

Consultant shall submit a draft copy (electronic is acceptable) of a summary event report as outlined in section 7.0 Reporting Requirements of the SLEMP within 30-days of receipt of complete and accurate analytical results for each April and August monitoring events. Consultant shall incorporate City comments and submit one paper and one electronic copy of the final report (if paper draft, page changes are acceptable) to the City and paper and electronic deliverables to WDEQ per their requirements not later than 30-days following submission of draft.

A.8 ANNUAL REPORTS

Consultant shall submit a draft copy (electronic is acceptable) of a comprehensive annual report to the City by March 1 of each contract year. Each annual report shall be outlined in Section 7.0 Reporting Requirements of the SLEMP. Consultant shall incorporate City comments and submit one electronic and one paper copy of final report (if paper draft, page changes are acceptable) to the City and paper and electronic deliverables to WDEQ per their requirements by April 1 of each contract year.

EXHIBIT B
COMPENSATION

This Exhibit acts as a supplemental information document and is made a part of the Professional Services Agreement (Agreement) entered into by and between the City of Cheyenne, Wyoming (CITY) and Trihydro Corporation.

Items (below) are to be paid for on a unit price/time and materials (T&M) or lump sum (LS) basis, as designated in the following schedule. For all unit price/T&M work, the CONSULTANT shall charge and the CITY shall pay for actual materials and services rendered at the unit prices listed in this Exhibit or standard Schedule of Charges (attached). In no event shall total contract costs exceed \$171,912 without a written change order signed by both parties.

B.1 Detection Monitoring

April Well Gauging, Sampling (Bill per LS/Well/Event):

\$ 562.46 per well x 26 Appendix A wells = \$ 14,624

\$ 591.21 per well x 8 wells, Appendix B (VOCs + Inorganics) = \$ 4,730

April Event Report (Bill per Unit Price/T&M): \$ 4,125

August Well Gauging, Sampling (Bill per LS/Well/Event):

\$ 279.03 per well x 33 Appendix A, C wells = \$ 9,208

August Event Report (Bill per Unit Price/T&M): \$ 4,061

April and August Fluid Level

\$ 14.66 per well x up to 50 wells x 2 events (Bill per LS/Well/Event) \$ 1,466

B.2 Expansion Well Baseline Monitoring

Quarterly Well Gauging, Sampling (Bill per LS/Well/Event):

\$ 899.42 per well x 12 Appendix A, C wells = \$ 10,793

B.3 Methane Monitoring

February, April, August, & December, Methane Monitoring

\$ 23.80 per well x 14 well x 4 events (Bill per LS/Well/Event): \$ 1,333

B.4 Annual Report

Statistical Analysis per WDEQ requirements,

Preparation, printing, and delivery of annual report (Bill per Unit Price/T&M): \$ 6,964

Annual Subtotal: \$ 57,304

TOTAL NOT-TO-EXCEED CONTRACT AMOUNT: \$ 171,912
(3x annual total)



**SANITARY LANDFILL ENVIRONMENTAL MONITORING PLAN
(SLEMP)**

**HAPPY JACK LANDFILL
CHEYENNE, WYOMING**

April 12, 2017

Revised August 23, 2017, January 30, 2018

Project #: 47A-001-001

PREPARED BY: Trihydro Corporation

1252 Commerce Drive, Laramie, WY 82070



ENGINEERING SOLUTIONS. ADVANCING BUSINESS.

Home Office | 1252 Commerce Drive | Laramie, WY 82070 | phone 307/745.7474 | fax 307/745.7729 | www.trihydro.com

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1.0 INTRODUCTION

This Sanitary Landfill Environmental Monitoring Plan (SLEMP) has been prepared to address groundwater, leachate, and methane monitoring requirements at the City of Cheyenne Sanitary Landfill, also known as the Happy Jack Landfill (Landfill). The plan has been prepared as part of the facility's 2017 lifetime permit application, primarily to fulfill monitoring requirements contained within the Wyoming Department of Environmental Quality/Solid and Hazardous Waste Division (WDEQ/SHWD) municipal solid waste landfill regulations. Additional monitoring requirements have also been incorporated pursuant to the WDEQ November 25, 2015 Location Standards Variance Approval for the expansion area. This SLEMP has been prepared in accordance with revised Chapter 2 of the WDEQ/SHWD Rules and Regulations dated January 3, 2017.

The primary objective of the environmental monitoring program at the Landfill is early identification of impacts to the subsurface as a result of landfill activities. Additionally, the hydrogeologic and analytical data generated during routine monitoring may be used for assessment and/or remedial design purposes, if impacts are identified in subsurface soils or groundwater beneath the landfill.

The remainder of the SLEMP is organized into the following sections:

- Section 2.0 – Groundwater and Leachate Monitoring Plan: provides a description for each of the aquifers beneath the landfill, leachate monitoring points, different phases of monitoring, frequency of sampling, as well as fluid gauging and groundwater sampling procedures. This section also addresses the off-site water well monitoring program for wells within a half mile of the expansion area boundary that are permitted or certificated for domestic or stock use.
- Section 3.0 – Groundwater Data Quality Assurance/Quality Control: presents the quality assurance objectives for this project and the methodology for reviewing field and laboratory data.
- Section 4.0 – Groundwater Data Statistical Analyses: includes a description of the statistical methods used to evaluate the groundwater analytical results from each monitoring event.
- Section 5.0 – Landfill Methane Monitoring Plan: describes the monitoring system, frequency, and procedures for monitoring methane in landfill structures and in the subsurface.
- Section 6.0 – Notification Requirements: outlines the notification requirements and follow up actions to statistically significant increases in contaminant concentrations in groundwater, detections above groundwater protection standards, and methane concentration measuring more than 25% or 100% of the lower explosive limit (LEL).

- Section 7.0 – Reporting Requirements: details the frequency and content of the environmental monitoring reports submitted to the WDEQ/SHWD.
- Section 8.0 – References.

2.0 GROUNDWATER AND LEACHATE MONITORING PLAN

A total of 32 groundwater monitoring wells/piezometers are in the current groundwater monitoring network at the Landfill. In addition, 4 leachate wells (installed in the southern portion of the landfill at the refuse-soil interface), and 5 leachate sumps (installed in the northern trenches) are currently in the leachate monitoring program. Figure 1 shows the locations of each of the groundwater monitoring wells and piezometers, as well as the leachate wells and sumps. Hydrogeologic zones of completion are noted on Figure 1 where applicable.

Table 1 provides a summary of the construction history of monitoring wells and piezometers installed at the Landfill, dating back to 1975, including a list of associated reference documents. Construction details for each of the active monitoring wells and piezometers is included in Table 2 with the available associated boring logs/construction diagrams included in Attachment A. Each well and piezometer at the landfill was constructed with a protective surface casing and locked to prevent tampering.

2.1 PHASES OF GROUNDWATER MONITORING

Groundwater monitoring is divided into two phases at the Landfill: detection and assessment monitoring, following the general guidelines established in Chapter 2 of the WDEQ/SHWD Rules and Regulations (WDEQ 2017). Routine groundwater monitoring including gauging and sampling activities will be conducted in April and August of each year. Tables 3 and 4 present summaries of the sampling and fluid level monitoring plans for this landfill. Tables 5, 6, and 7 show the Solid Waste Rules and Regulations (SWRR) Chapter 2 Appendix A, Appendix B, and Appendix C Constituents, as required for detection and/or assessment monitoring.

2.1.1 DETECTION MONITORING

Detection monitoring includes the collection and analysis of groundwater samples where statistically significant impacts to groundwater have not been identified. Detection monitoring requires the analysis of a select list of constituents, followed by statistical analysis of the results to determine if there are increasing contaminant concentrations above background or baseline water quality. Groundwater samples collected during detection monitoring are analyzed for WDEQ/SHWD Appendix A (Table 5) and Appendix C (Table 7) constituents.

For routine detection monitoring, samples will be analyzed for Appendix A constituents on an annual basis and for Appendix C constituents on a semi-annual basis. However, for new monitoring well installations (see Section 2.4), baseline data will be developed by collecting samples quarterly for the first two years. Samples will be analyzed for

both Appendix A and C constituents at each sampling event. After the first two years, the sampling schedule will return to routine detection monitoring as described above.

2.1.2 ASSESSMENT MONITORING

Assessment monitoring is implemented whenever a statistically significant increase over background water quality has been detected for an Appendix A constituent, indicating potential impacts to groundwater quality. Assessment monitoring specifies a much broader range of analyses within selected wells to characterize the nature of groundwater impacts. Historically, a subset of wells developed in consultation with the WDEQ Administrator were sampled under assessment monitoring and analyzed for Appendix B (Table 6) constituents at least annually.

When an Appendix B or C constituent is detected at a statistically significant level above the applicable groundwater protection standard for that constituent, additional monitoring wells may be required to characterize the nature and extent of the release. The EE-series wells and PIE-9 are used to help define the nature and extent of VOC impacts first identified in compliance monitoring well SRM-6. Historically, samples collected from these wells were analyzed annually for the WDEQ/SHWD Appendix A constituent list plus the constituent dichlorodifluoromethane from the Appendix B list. Dichlorodifluoromethane is no longer sampled alongside Appendix A constituents; the current monitoring plan does not deviate from constituents listed in WDEQ/SHWD Appendices A, B, and C.

All wells scheduled for assessment monitoring (the subset of compliance wells described above) will be sampled semiannually for Appendix C parameters and annually for Appendix B (VOCs and Inorganics). This group of wells also includes up-gradient wells in the Perched Water Bearing Zone and the Principal Unconfined Aquifer for statistical comparison. The assessment monitoring wells will be sampled for a reduced list of Appendix B constituents (only VOCs and Inorganics) based on an extensive data record that showed no significant detections of SVOCs or pesticides, herbicides or PCBs. The down-gradient wells scheduled for assessment monitoring will be sampled once every five years for the full Appendix B list, to confirm the lack of detection of these constituents.

The City will collect and analyze samples for the Appendix B (VOCs and Inorganics) constituent list during the April sampling event. However, if there is insufficient liquid to complete the analyses, a second attempt will be made during the August event. Table 3 shows which wells are scheduled for assessment monitoring and Table 6 presents the Appendix B constituent list.

2.2 GROUNDWATER BEARING ZONES, AQUIFERS AND LEACHATE MONITORING POINTS

Four (4) saturated units have been identified beneath the Landfill. These include two (2) Perched Water Bearing Zones, the Principal Unconfined Aquifer, and the Semi-Confined (formerly called Confined) Aquifer. The perched



zones and the aquifers are not considered to be hydraulically connected beneath the Landfill. These water bearing zones and aquifers and the associated monitoring networks are described briefly in the following sections. Leachate wells are screened across the refuse-soil interface and are utilized to measure and characterize leachate generated beneath the unlined areas of the landfill. Leachate sumps are also monitored to assess the volume and characteristics of leachate generated from the lined areas. Additional monitoring wells, sumps, and piezometers may be installed at the landfill as part of investigation and monitoring activities.

2.2.1 PERCHED 1 WATER BEARING ZONE

The Perched 1 Water Bearing Zone has been identified as the uppermost water-bearing unit beneath the landfill. Intermittent saturated conditions have been encountered in the Spring Creek alluvium along the southern portions of the landfill extending into the Ogallala Formation to the east of Well SRM-6. Depth to water in the Perched 1 Water Bearing Zone in the southern portions of the landfill ranges between six (6) and 45 feet below ground surface (ft-bgs). East of the landfill, the perched zone is encountered at depths between 55 and 75 ft-bgs, and is limited at depth by a sandy clay layer in the Ogallala Formation. This zone is of limited extent both horizontally and vertically. Flow through the Perched 1 Water Bearing Zone is likely the result of channelized subsurface flow such as a sandy braided stream depositional environment.

In order to limit the amount of water entering the landfill from the Spring Creek drainage, a cut-off wall was constructed near the southern boundary where the creek enters the landfill. In addition, Spring Creek was re-routed to the south of the landfill. As part of that project, a pump was installed in an up-slope well to dewater any saturated alluvium in front of the cut-off wall. These efforts appear to be successful with minimal amounts of water entering the landfill through leakage around the cut-off wall or vertical infiltration through the landfill cap. In fact, many of the wells/piezometers screened across the Perched 1 Water Bearing Zone only intermittently contain groundwater, with limited water level data often precluding delineation of potentiometric surface mapping.

The Perched 1 monitoring wells will be gauged in April and August, of each year (Table 4). Monitoring wells and piezometers screened in the Perched 1 Water Bearing Zone are sampled under either assessment or detection monitoring, as outlined below and in Table 3:

- Detection Monitoring: Piezometers P-1, P-3, P-4, P-5, P-8, P-10, P-11 and Well S-9 are scheduled to be sampled for the Appendix A and C constituent lists in April and for the Appendix C list in August of each year.
- Assessment Monitoring: Groundwater samples are collected from Well SRM-6, PIE-9, EE-04 and EE-07 and analyzed for the Appendix B (VOCs and Inorganics) and Appendix C lists in April, and the Appendix C list in August of each year.

The Perched 1 Water Bearing Zone is generally referred to as the Perched Water Bearing Zone (without numbering) throughout the documentation post-2003, since wells in the Perched 2 Water Bearing Zone were abandoned in 2003 as described below.

2.2.2 PERCHED 2 WATER BEARING ZONE

A Perched 2 Water Bearing Zone was identified in two Wells EE-01 and EE-02 installed along the eastern edge of the landfill. Monitoring Wells EE-01 and EE-02 were plugged and abandoned on June 9, 2003, because refuse was scheduled to be placed in close proximity to these two wells due to completion of planned development of the waste disposal area approved by the WDEQ/SHWD. The monitoring wells were plugged and abandoned in accordance with the provisions of Chapter 26, Section 11 of the Water Quality Division Rules and Regulations. The concrete pads were destroyed and the protective casings removed. Exposed portions of the well casings were cut off below the ground surface. The wells were then sealed with a bentonite grout from the total depth of the well to ground surface. A permanent marking was set into a concrete surface monument indicating the well identification and date of abandonment. The abandonment notices filed with the Wyoming State Engineer's Office were included in the May through August Environmental Monitoring Report submitted to the WDEQ/SHWD on October 28, 2003.

2.2.3 PRINCIPAL UNCONFINED AQUIFER

The entire landfill is underlain by an unconfined aquifer approximately 150 -160 ft-bgs. Thickness in the Principal Unconfined Aquifer varies from five (5) to 50 feet; groundwater flows from the southwest to the northeast at a gradient of approximately 0.01 to 0.02 feet per foot.

There are twelve (12) monitoring wells screened across the Principal Unconfined Aquifer at the landfill. Fluid levels from each of the wells screened in the Principal Unconfined Aquifer will be gauged in April, and August each year (Table 4). These wells will be sampled following the detection and/or assessment phase of groundwater monitoring, as outlined below and in Table 3:

- Detection Monitoring: Wells PU-1, PU-2, PU-3, PU-4, PU-5, and P3-1, P3-2 and P3-3 are scheduled to be sampled for the Appendix A and C constituent lists in April and for the Appendix C list in August each year.
- Assessment Monitoring: Groundwater samples will be collected and analyzed from Wells SRM-4, SRM-5, SRM-7 and SRM-8 for the Appendix B (VOCs and Inorganics) and Appendix C constituent lists in April, and the Appendix C list in August each year.

2.2.4 SEMI-CONFINED AQUIFER

In 2013, a series of groundwater monitoring wells (GMW series) was installed to help define the subsurface hydrogeology specifically in the landfill expansion area. WDEQ asked for confirmation as to whether the GMW-series wells are confined and if they monitor the same water bearing zone as the C-series wells, a set of wells that had been described as monitoring a confined zone within the Ogallala Aquifer identified when the wells were drilled in 1999. The confined zone was said to be primarily composed of sand with minor amounts of clay, silt, and gravel which lies below the Principal Unconfined Aquifer unit at approximately 170 ft-bgs.

A transducer study was then completed to shed more light on the local subsurface hydrology as a condition of the WDEQ Location Standards Variance Approval. The results of the study indicated that the Confined Aquifer would be more appropriately called a Semi-Confined aquifer. The wells monitor different intervals but may represent a series of “stacked” water bearing zones that locally are connected to varying degrees. Some may meet the definition of confined (C-series) because water levels in certain wells are consistently above the screened zone, but the wells as a group do not appear to represent confined aquifer conditions. The Ogallala Aquifer in this part of the country is very complex and the degree of interconnectivity is difficult to accurately assess. It was recommended that the GMW and C series wells be plotted together to represent groundwater flow conditions below the Principal Unconfined Aquifer set of wells. USGS mapping shows a general northeasterly groundwater flow gradient that begins to change to the northwest consistent with what is also observed in the potentiometric surface maps constructed based on the C-series and GMW-series wells.

Approximately ten (10) municipal water wells operated by the Cheyenne Board of Public Utilities (BOPU) are located within a one-mile radius of the landfill site. These wells are installed at depths between 200 and 400 ft-bgs. Pumping from the wells operated by the Board of Public Utilities is intermittent as necessary to augment the City of Cheyenne’s water supply. Each of the water supply wells has the capacity to produce between 150 to 600 gallons per minute from the Ogallala Aquifer. The transducer study results indicated that even at high late-summer pumping rates, the BOPU pumping wells did not alter the aquifer flow direction from the regional flow pattern. The Cheyenne Board of Public Utilities intermittently collects groundwater samples from these wells and analyzes the samples for the Appendix A constituent list. This data is provided to the City and shared with WDEQ through the annual report. Three (3) C-series and 5 GMW-series wells will be gauged and sampled in April and August of each year according to the detection monitoring program (Tables 3 and 4).

2.2.5 LEACHATE WELLS

There are four active leachate wells screened into the refuse across the southern portions of the Landfill. Figure 1 shows the locations of each of the leachate wells. Historically, water had been sporadically present in over 11 leachate wells, based on fluid gauging results in April and August of each year. In the past, grab samples had been collected annually from Leachate Wells P1E-1, S-7A, and S-4 and analyzed for the Appendix A constituent list in order to quantify constituent concentrations detected in the leachate along the southern portion of the landfill. Constituent concentrations are primarily for information purposes to characterize the leachate quality and are not related to any compliance criteria. The City has been collecting leachate data for many years and additional sampling was not likely to provide added value. As a result, the four active leachate wells (P1E-1, S-1, S-4 and S-7A) are not being sampled but fluid levels are gauged during each sampling event (Table 4).

2.2.6 LEACHATE SUMPS

There are five (5) sumps installed to monitor and evacuate leachate generated beneath the northern trenches at the landfill. Figure 1 shows the locations of each of the leachate sumps installed at the landfill. Fluid levels in sumps NT-1 and NT-2 are manually gauged semi-annually in concert with the routine sampling events to the nearest 0.1-foot relative to the top of each sump. NT-1 and NT-2 are not pumped unless leachate depths approach 12-inches of accumulation above the liner. Fluid levels in these two sumps have been either dry or declining such that future pumping is unlikely.

Fluid levels in NT-3 through NT-5 are recorded using pressure transducers installed in each sump. Landfill staff record fluid levels and leachate pumping volumes read from the electronic display on a monthly basis. The pressure transducers in Sumps NT-3 through NT-5 will be verified every two years. The pressure transducers will be verified to the nearest 0.1-foot utilizing a water level indicator. If the measurement between the water level indicator and the pressure transducer differ by greater than 0.3 foot, the transducer will be recalibrated. The sumps are evacuated using dedicated submersible pumps programmed to prevent more than 12-inches of leachate from accumulating above the cell liner. Grab samples will be collected annually from the NT-5 leachate sump (if sufficient volume is present) and analyzed for the Appendix B (VOC) and C constituent lists in order to quantify constituent concentrations detected in the leachate (Table 3). Sump NT-5 was selected because it collects leachate from the most recently constructed cell and will produce leachate for the longest future time period. In addition, sample collection can be accomplished easily because of a sample port location on the discharge line and no confined space entry is required. Again, constituent concentrations are primarily for information purposes to characterize the leachate quality and are not related to any compliance criteria. As new landfill cells are constructed, the City proposes to sample one leachate collection area for the new cells for Appendix A and C constituents annually.

2.3 GROUNDWATER MONITORING PROCEDURES

This section details the procedures and techniques that will be followed for groundwater monitoring at the Landfill. Due to the heterogeneity of the hydrogeologic units encountered beneath the landfill, it may be necessary to modify these procedures as sampling progresses to ensure that representative groundwater samples are collected and submitted to the laboratory for analysis.

2.3.1 FLUID LEVEL MONITORING

Upon arriving at a well location, sampling personnel will check the well for damage and remove the lock and well cap. Any damage to the wellhead or casing, or any items requiring maintenance will be noted in the field sampling record and reported to the project manager so that the necessary repairs can be scheduled. Field personnel will wear new, sterile gloves during all preparation and sampling activities.

Fluid levels within each water bearing zone will be measured within a 24-hour period during the two scheduled sampling events (April and August). Fluid level measurements will be performed using a water level indicator accurate to 0.01-feet. The fluid level probe will be lowered down the center of the well casing. When the probe encounters water, an audio alarm is triggered. The measurement will be made from the pre-marked (surveyed) measuring point on the north side of the well casing. Manufacturer's instructions will be followed to ensure proper care of the fluid level probe. The total depth of the well will be measured annually to determine if the well is silting in and needs to be rehabilitated or replaced.

If non-aqueous phase liquid (NAPL) is observed in any well, the thickness will be measured with an interface probe. A typical interface probe gives an audible beeping sound when submerged in water and a solid tone when submerged in NAPL. The fluid level measurements will be recorded on the fluid level summary form (or similar) included in Attachment B.

All measurements during a particular monitoring event are made to the reference points with the same instrument, to ensure internal consistency. The vertical reference points (i.e. north side top of casing) are established by a licensed surveyor or under supervision of a professional engineer in relation to the nearest permanent benchmark. The horizontal well coordinates are established by marking distances in reference to the project site plan in triangulation or by surveying in reference to the nearest permanent benchmark. The exposed portion of the tape and the probe will be decontaminated before performing measurements at each monitoring well.

2.3.2 WELL EVACUATION PROCEDURES

Depending on the depth to water and the yield of a well, a dedicated pneumatic pump with dedicated tubing or a disposable bailer will be employed to purge groundwater from selected wells and piezometers. Field personnel shall note the approximate evacuation rate on the groundwater sampling forms (Attachment B).

Conventional sampling techniques will be used to evacuate and sample monitoring wells screened in the Perched Water Bearing Zone, as well as the leachate wells and sumps. The column of standing fluid in each well (casing volume) will be calculated based on the measured fluid level, total well depth, and well diameter. Where possible, the calculated well casing volume will be evacuated from the well a minimum of three times. Some wells may be pumped or bailed “dry” before three casing volumes are evacuated. If this occurs, groundwater samples will be collected following sufficient recharge to the well.

2.3.2.1 WELL EVACUATION USING A DEDICATED PUMP

There are currently eight wells with dedicated pumps at this facility. This includes pumps at the Semi-Confined Aquifer Wells C-2 and C-3 as well as pumps at the Principal Unconfined Aquifer Wells PU-1, PU-2, SRM-4, SRM-5, SRM-6, and SRM-8. These pumps are operated at speeds varying from 0.19 gpm to 0.29 gpm during evacuation and then allowed to pump a minimum of three well casing volumes. This is done by using an air compressor and discharge controller in collaboration with the dedicated pumps. Wells that have had low rechargeability during previous events will be purged to the extent possible; however, enough volume will be left in the well to ensure sufficient water is available to fill the laboratory required volume for analyses. The only well likely affected by this scenario is SRM-5 and will be determined through the volume of water in the well and recharge rate during evacuation.

2.3.2.2 GROUNDWATER EVACUATION USING BAILERS

Some wells will be evacuated using bailers due to the minimal volume of groundwater in the well. When collecting groundwater using a bailer, the bailer will be slowly lowered using twine or cable into the well to minimize agitation of the water table. The bailer will be raised to the surface manually or using an electric or battery operated auto bailer downrigger system. Wells that require an auto bailer downrigger system are C-1, GMW-03, GMW-04, GMW-06, and GMW-07. Disposable polyethylene bailers will be used to purge wells. Wells P3-1, P3-3, PU-3, and P1E-1 have had low rechargeability during previous events and will be bailed to the extent possible; however, enough volume will be left in the well to ensure sufficient water is available to fill the laboratory required volume for analyses.

2.3.2.3 ALTERNATIVE SAMPLING TECHNIQUES

In the future, the City of Cheyenne may opt to perform the sampling at the landfill using alternative sampling techniques. At this time, no decision has been made as to the alternative method or methods that will be used. If the City of Cheyenne decides to use an alternative sampling method in the future, detailed sampling methods will be submitted to the WDEQ/SHWD prior to any changes in sampling techniques at the landfill. The following is one alternative sampling techniques that may be considered by the City of Cheyenne:

- A low-flow “minimal drawdown” sampling methodology to evacuate groundwater and sample wells screened in the Principal Unconfined and the Confined Aquifers. Low-flow sampling involves removing water at the same rate as it flows through the formation minimizing the entrainment of sediment. Low-flow sampling will conform to USEPA recommended procedures outlined in the document, *Low-Flow (Minimal Drawdown) Groundwater Sampling Procedures* (Puls and Barcelona, 1995). The pumps would be installed so that the intake height is located in the interval of interest as determined in conjunction with the City of Cheyenne and the WDEQ/SHWD.

2.3.3 FIELD ANALYSES

Field parameters including specific conductance, pH, and temperature will be measured during groundwater evacuation and at the completion of sample collection. Field parameters will be recorded on the groundwater sampling form for each well. Physical characteristics of the groundwater will be noted and recorded during sampling (e.g. sediment, hydrocarbon sheen, color, odor, etc.).

2.3.4 SAMPLE WITHDRAWAL

Groundwater samples will be collected from wells using dedicated pump/tubing or from disposable bailers. In accordance with SWRR Chapter 2, Section 9.b.i.C.II, no samples will be field filtered prior to laboratory analysis. Sample agitation will be minimized during sampling and groundwater will be transferred directly from the sampling device into the sample containers, intermediary containers will not be used for sample collection. Sample bottles shall be filled as appropriate to minimize headspace. Volatile organic analysis (VOA) sample vials will be filled with zero-headspace. Field personnel will wear disposable gloves to prevent contamination of samples and equipment.

Groundwater samples will be submitted for analysis in accordance with the WDEQ/SHWD Appendix A list (Table 5), which includes VOCs by USEPA Method 8260B and total metals via USEPA Method 6020 or the Appendix B list (Table 6) which includes additional VOCs and total metals, as well as, semi-volatile organic compounds (SVOCs) by USEPA Method 8270C; and pesticides, herbicides/fungicides, and polychlorinated biphenyls (PCBs) via USEPA Methods 8081A, 8151A, and 8082, respectively. In addition, the City of Cheyenne will monitor the Appendix C list (Table 7) of geochemical parameters from each of the wells included in the sampling network.



Sample bottles will be preserved in the laboratory before being sent to the site. The samples will be collected and analyzed in general accordance with the Test Methods for Evaluating Solid Waste, USEPA, SW-846 (EPA 2007). The collection order for water quality samples is:

1. VOCs
2. SVOCs
3. Extractable organic compounds
4. Total metals
5. Inorganic compounds and nutrients

A trip blank prepared by the laboratory will be shipped with the groundwater samples. The samples will be labeled and placed in an opaque cooler and stored on ice after collection. Glass containers shall be protected against breakage during transport to the laboratory. A chain-of-custody form will accompany the samples to the laboratory. The samples will be submitted to TestAmerica in Denver, Colorado or an equivalent laboratory.

2.3.5 FIELD MEASUREMENTS, CALIBRATION, AND DECONTAMINATION

Field instruments will be used to analyze air and/or water samples during monitoring activities at the landfill. Field measurements provide information critical to assessing health and safety issues, evaluating the nature, degree, and extent of impacts, and ensuring representative samples are selected for laboratory analyses. Therefore, calibration and maintenance of field instruments is important to achieving the overall project objectives.

Numerous types and brands of field instruments are currently available for use during environmental assessment projects. Maintenance and calibration procedures vary depending on the type, manufacturer, model, and age of the instruments. Therefore, specifying calibration and maintenance procedures for one or many types of instruments is not considered to be practicable in the SLEMP.

Instruments and equipment used to gather, generate, or measure environmental data will be calibrated with sufficient frequency and in a manner that is consistent with the manufacturers' specifications. A record of calibration will be maintained documenting calibration results and will include the date, standards, personnel, and calibration results.

Sampling equipment and containers will be decontaminated prior to sampling, either in the laboratory and/or in the field. The sampling devices will be decontaminated in the field immediately prior to and after sampling each site. In general, decontamination will consist of a detergent/water wash, tap water rinse, and distilled water final rinse followed

by drying before subsequent use or storage. Decontaminated sampling devices and containers will be stored in contaminant free locations or containers until use. Glass jars, etc. may be stored until use in clean sample coolers.

2.3.6 DOCUMENTATION REQUIREMENTS

All field activities and samples collected will be documented in order to verify compliance with the SLEMP. Field forms will be used to document the monitoring activities. Information generated from sampling activities will be documented on the appropriate forms including:

- Fluid Level Summary
- Groundwater Sampling Form
- Record of Calibration
- Chain-of-Custody

Incoming documents will be dated and filed. If distribution is necessary, the documents will be distributed to project personnel. Examples of all of these forms are included in Attachment B.

2.3.7 CHAIN-OF-CUSTODY

A sample is under custody if it is in the field personnel's possession or if the sample was in the field personnel's possession and then locked up, or placed in a designated secure area. The following procedures will be used in the field:

1. The field personnel will be responsible for the care and custody of the samples collected until they are delivered to the analytical laboratory or entrusted carrier. As few people as possible will handle the samples.
2. Field forms will be used to document the data collection activities such as the date, start and end time of activities, field personnel name, weather conditions, sample location, equipment used to collect sample, depth of sample, time of collection, sample description, sample identification number, and the volume and number of containers. Entries will be made in ink with no erasures. A single stroke will be used to cross out incorrect information; corrections will be dated and initialed.
3. A chain-of-custody will be completed in the field to the fullest extent possible prior to sample shipment. It will include the following information: sample number, time collected, source of sample and location, depth from which sample was taken, preservative, and name of field personnel. The chain-of-custody will be filled out in a legible manner, using ink, and signed by field personnel. Similar information will be provided on the sample label,



which will be securely attached to each sample bottle/container. In addition, sampling forms will be used to document collection, filtration (if required), and preparation procedures.

2.3.8 CUSTODY FOR SHIPMENT

The following procedures will be used when transferring custody of samples:

1. Samples will always be accompanied by a chain-of-custody completed according to the protocols outlined in this section. When transferring samples, the individuals relinquishing and receiving them will sign, indicate the date, and note the time on the chain-of-custody. Upon arrival at the laboratory, internal custody procedures will be followed, in accordance with standard laboratory procedures.
2. Samples will be packaged properly for shipment and dispatched to the laboratory for analysis with individual custody records accompanying each shipment. Shipping containers will be sealed for shipment to the laboratory. The method of shipment, courier name, and other pertinent information will be entered in the remarks section of the chain-of-custody.
3. All shipments will be accompanied by the chain-of-custody form identifying the contents. The original record will accompany the shipment, and a copy will be retained by the field sampler.
4. Proper documentation will be maintained for shipments by the common carrier.

2.3.9 SAMPLE SHIPMENT PROCEDURES

Sample containers will be packed securely in a cooler to minimize the potential for breakage during shipment. Packing materials, such as poly-foam or bubble wrap, shall be used to protect sample containers in the cooler.

The following procedures will be followed when shipping samples for laboratory analysis:

1. Samples requiring refrigeration will be promptly chilled with ice to a temperature of four (4) °C and packed in an insulated cooler for transport to the laboratory.
2. Only shipping containers that meet all applicable state and federal standards for safe shipment will be used for this project.
3. To provide a means for detecting any potential tampering during shipment, custody seals will be affixed to cooler and tape wrapped entirely around the cooler.

4. The chain-of-custody will be placed inside the shipping container in a sealed plastic envelope.
5. Shipment will be made by overnight courier or picked up by laboratory personnel within 24 hours of sampling or as soon as practicable.

2.3.10 SYSTEM AUDITS

Performance and system audits of both field and laboratory activities will be performed on a periodic basis, as appropriate, to assure that the sampling and analysis are performed in accordance with the procedures established in the SLEMP. Examples of audits that may be performed are as follows:

- Field inspections to confirm that the groundwater monitoring program was conducted correctly; field measurements were made accurately; equipment was thoroughly decontaminated; samples were collected and handled properly; and field work was accurately documented.
- A comprehensive review of the analytical report submitted by the laboratory to verify all requested analyses were performed; sample holding times were met; the data was generated using the appropriate methodology, the level of QC effort and reporting; and the analytical results were in conformance with the prescribed acceptance criteria.
- Oversight of the field and data validation activities to ensure data management proceeds in an organized and expeditious manner.

2.3.11 CORRECTIVE MEASURES

Any person identifying an unacceptable condition will notify the project manager. The project manager will be responsible for developing and initiating appropriate corrective action and verifying that the corrective action has been effective. Corrective actions may include repeating measurements, re-sampling and/or re-analyzing samples, and amending or adjusting sampling procedures. If warranted by the severity of the problem (e.g., if monitoring wells require re-sampling or if the project schedule may be affected), the WDEQ/SHWD will be notified.

2.4 FUTURE EXPANSION AREA WELLS

Variance condition #1 requires the installation of additional groundwater monitoring wells downgradient of the expansion area at a spacing not to exceed 500 feet. Four additional monitoring wells will be installed prior to waste placement in the expansion area to meet this requirement. Well locations will be provided to WDEQ prior to installation.

Eight quarterly sampling events for Appendix A and C constituents will be conducted to establish baseline conditions for the new expansion area wells. At a minimum, four of the eight quarters of baseline monitoring will occur prior to waste placement. The sampling frequency will be reduced to semiannual thereafter to be consistent with the detection monitoring sampling schedule, as described in Section 2.1.1 above.

2.5 OFF-SITE WATER WELL MONITORING PROGRAM

Variance Conditions # 9 and #10 specify that the City of Cheyenne implement a groundwater monitoring program for water wells permitted or certificated for domestic or stock use located within ½ mile of the expansion area facility boundary, provided that the City is given access to the specified water wells by the property owner.

2.5.1 BASELINE MONITORING PROGRAM

In accordance with Variance Condition #10, the City will implement a baseline water quality monitoring program for off-site water wells. Following lifetime permit approval, the City will contact the permit holders listed in Table 8 via certified mail, taking care to check for any new wells installed since the initial tabulation. In the event landowners do not respond to the first certified mailing, a second mailing will be sent to landowners seeking permission. Landowners will be asked to provide written permission to allow the City to inspect water well integrity and take water quality samples for Appendix A and C constituents for four quarterly baseline sampling events and potential future expanded monitoring events. The City will maintain records of the mailings and signed approvals for each property contacted.

Every effort will be made to take samples from the subject wells upstream of any treatment system that may have been installed. Sampling procedures for each property will be on a location by location basis and dependent upon the conditions and use of the property well. Therefore, once initial inspection is completed at each location, sampling protocol will be determined and submitted to WDEQ prior to initiating the first sampling event. Field technicians will document sampling locations and any modified procedures in the associated field notes. Following completion of sampling, monitoring data will be submitted to the WDEQ.

Note that Table 8 does show a well owned by the City and a well owned by BOPU. Consent for locating the landfill in proximity to these wells was previously provided by these two entities. While the City and BOPU signed letters of consent for the location standards in the expansion area, this does not mean that the City and BOPU also waived the additional sampling on their permitted wells. These two entities will be included as part of the baseline monitoring program and will be given the opportunity to decline and/or accept this monitoring. In addition, water quality analyses is routinely completed for the BOPU well (Koppes #6) and is already provided to WDEQ as part of the Landfill annual environmental monitoring report. Please also note that the King Ranch well, while listed as being within ½ mile of the

expansion area, is actually closer to the existing landfill and not located hydraulically down-gradient of the expansion area. After consultation with WDEQ, it was not determined to be necessary to include the King Ranch well in the off-site monitoring program.

2.5.2 EXPANDED MONITORING PROGRAM

In accordance with Variance Condition #9, if constituents in any groundwater monitoring well located downgradient of the expansion area exceed the EPA Maximum Contaminant Level, the City will institute a monitoring program for the off-site water wells within 90 days, and annually thereafter, for Appendix A, B and C constituents. The variance language states Appendix B and baseline constituents, however, given the January 3, 2017 revisions to Chapter 2 of the SWRR, the Appendix A, B and C constituent lists will be used.

3.0 GROUNDWATER DATA QUALITY ASSURANCE/QUALITY-CONTROL

The overall quality assurance objectives for the environmental monitoring at the Happy Jack Landfill is to provide quantitative and qualitative measures to produce legally defensible data used to evaluate subsurface quality beneath the landfill. The objectives of the quality assurance/quality control (QA/QC) program are to:

- Verify that all procedures are documented, including any changes from the protocols established in the SLEMP.
- Verify that all sampling and analytical procedures are conducted according to sound scientific principles.
- Monitor the performance of the field sampling team and laboratory with a systematic audit program and provide corrective actions necessary to assure quality.
- Evaluate the quality of the analytical data through a system of quantitative and qualitative criteria.
- Verify that all data and observations are properly recorded and archived.

Previous sections of this report have documented the procedures for field sampling, chain-of-custody, and laboratory analysis in order to satisfy many of the aforementioned objectives. This section addresses the qualitative and quantitative criteria which will be used to evaluate the quality of the analytical data collected during field activities and laboratory analysis.

3.1 QUALITY CONTROL PARAMETERS

The level of quality control effort will generally be consistent with the United States Environmental Protection Agency (USEPA) Contract Laboratory Program (CLP) National Functional Guidelines (USEPA 2014a, 2014b, 2013, 2004, and 1999). The data quality elements that will be evaluated include:

1. Blind Duplicate Samples: blind duplicate (field replicate) samples will be collected to evaluate precision associated with the reproducibility of sampling techniques and the homogeneity of sample matrices. Replicate samples will be collected for each matrix at a frequency of ten percent (10%) or one (1) per every ten (10) samples collected and submitted for laboratory analysis.

If less than ten (10) samples are collected during a particular sampling event, one blind duplicate sample will be collected. Since the replicate will be “blind” to the laboratory, it will have a coded identity on its label and on the chain-of-custody record form. The actual sampling location and identification will be recorded on the sampling forms.

2. Trip Blanks: one (1) trip blank will be submitted for VOC analysis with each cooler containing aqueous sample requiring VOC analyses. Trip blanks are supplied by the laboratory with the sampling containers at the start of field activities and accompany the sample containers throughout the project.
3. Equipment Blanks: equipment blanks will be prepared and submitted for laboratory analysis to determine if cross-contamination has occurred during sampling, and to verify that equipment decontamination procedures are effective. Equipment blanks are submitted for analysis of the same constituent list required for the associated field samples. Equipment blanks will be preserved in the same manner as the associated samples. Equipment blanks will be shipped with the field sample containers. Equipment blanks will be prepared at a minimum of one per sampling event if required.
4. Method Blanks: method blanks will be prepared and analyzed by the laboratory at a frequency of one per every 20 samples or one per sample batch. The method blank is a “clean” matrix similar sample prepared and analyzed by the laboratory. Analysis of the method blank is used to identify laboratory derived contaminants introduced during sample preparation, extraction, and analysis.
5. Matrix Spike/Matrix Spike Duplicates (MS/MSD): matrix spike/matrix spike duplicates will be prepared and analyzed by the laboratory at a frequency of five percent or one per every 20 samples submitted to the laboratory. MS/MSDs are samples in which known spike concentrations of analytes are added to the collected sample prior to extraction and analyses. The recoveries for spiked compounds can be used to assess how well the method for analysis recovers target compounds within the submitted matrix. Evaluation of the MS/MSD is used to evaluate laboratory accuracy and precision, as well as to determine any potential matrix related interferences within the analytical results.
6. Laboratory Control Samples (Performance Evaluation Samples): laboratory control samples (LCS) are “clean” matrix similar samples prepared by the laboratory and spiked with a known concentration of constituents prior to extraction and analysis. The LCS is used to evaluate laboratory accuracy and method compliance. The LCS will be prepared and analyzed by the laboratory at a frequency of one (1) per every 20 samples analyzed or one per sample batch.
7. Surrogate Spiking: surrogate compounds consist of laboratory derived compounds that are introduced to each sample submitted to and prepared by the laboratory prior to extraction and analysis. The surrogate compounds spiked to each sample are specific to the laboratory analytical method. Quantification of the surrogate compounds allows for determination of matrix effects and laboratory performance on individual samples.

3.2 QUALITY ASSURANCE AND QUALITY CONTROL MEASURES

Four QA/QC measures will be evaluated for samples collected as part of routine monitoring and assessment activities:



- Precision
- Accuracy
- Method compliance
- Completeness

A discussion of each quantitative and qualitative measure is outlined below.

3.2.1 PRECISION

Precision is the measure of variability of sample measurements. Measurements of data precision are necessary to demonstrate the reproducibility of the analytical data. Field precision is determined by a comparison of field duplicate sample results. Laboratory precision is determined by examining the laboratory duplicate results. Evaluation of both the field and laboratory duplicates for precision is accomplished using the relative percent difference (RPD). The RPD is defined as the difference between the primary and duplicate samples divided by the mean and expressed as a percentage. Field duplicate RPD limits are set at 0-30% and laboratory RPD limits reference published or laboratory control charted limits.

3.2.2 ACCURACY

Accuracy is the relationship of the reported data to the accepted or “true” value. Accuracy is a measure of sampling and analysis bias. Biased results generally arise from personnel, instrument, or analytical method influences. The accuracy of the methods used for the analyses of the samples will be evaluated through the analysis of calibration standards, MS/MSD recoveries, surrogate spike recoveries, and blank results. Analytical accuracy is expressed as the percent recovery of a spiked constituent reported in the surrogate, LCS, or MS/MSD; and is expressed as the concentration of the primary sample (zero in the case of the LCS) subtracted from the total concentration divided by the spiked concentration.

$$\text{Accuracy} = (C^T - C^P) / C^S$$

Where: C^P = concentration of primary sample

C^T = total reported concentration

C^S = concentration of spike constituent

The spiked concentration is specified by the laboratory quality control requirements. The compounds used and their accuracy limits are specified in the respective methods listed in SW-846.

3.2.3 METHOD COMPLIANCE

Method compliance is a qualitative evaluation of the field and laboratory procedures used to collect routine monitoring data against the prescribed methods outlined in the SLEMP, SW-846, and the laboratory quality assurance project plan. Laboratory method compliance is determined by reviewing the holding times, detection limits, surrogate recoveries, methods blanks, and laboratory control samples against method specific requirements. Method compliance for sample collection is evaluated by reviewing sampling forms, chain-of-custody, and field notes.

3.2.4 COMPLETENESS

Completeness is a measure of the percentage of analytical data that meet all the acceptance criteria for precision, accuracy, and method compliance. Completeness is the overall ratio of the number of samples planned versus the number of samples with valid analyses.

The laboratory shall provide data that meet QC acceptance criteria for 90% or more of the requested determinations, which is standard industry practice. The ability to meet or exceed a completeness objective is dependent on the nature of samples submitted for analysis. Completeness goals can still be achieved if the data requires qualification as estimated above or below laboratory reporting limits (i.e. data of known quality even if not flawless, can be suitable for specific project goals). If the percent completion limits cannot be met because data is qualified as rejected or non-useable for the project objectives, the laboratory or environmental compliance contractor may be responsible for re-sampling and/or re-analysis dependent on the nature of the errors.

4.0 GROUNDWATER DATA STATISTICAL ANALYSES

Annually, statistical analysis will be completed on the analytical results collected over the year following receipt and validation of the analytical data. Multiple statistical tests will be performed to determine if there is statistically significant evidence of impacts to any of the aquifers/groundwater bearing zones. Groundwater monitoring results dating back to September 1982 will be included in the statistical analyses. Statistical analyses will be completed using selected up-gradient (background) and down-gradient (compliance) wells screened in the Semi-Confined Aquifer, Principal Unconfined Aquifer, and Perched Water Bearing Zones.

The analytical procedures used to evaluate groundwater quality beneath the landfill meet and exceed the requirements stipulated in SWRR Chapter 2, Section 9.b.i.C.VI (WDEQ 2017). The statistical procedures include: determining the distribution of the data, calculating basic summary statistics, interwell prediction interval testing, intrawell trend analysis and compliance limit comparisons to groundwater protection standards. Statistical analysis is not required when all concentrations of constituents are reported below the practical quantitation limit (PQL), aka reporting limit (RL). Statistical analysis will be completed in general accordance with EPA guidance (EPA 2009).

4.1 GENERAL STATISTICS

Prior to conducting the statistical analysis, the distribution of the data for each constituent will be determined. The Shapiro-Wilk test for normality is the default test and is used when the sample size is less than 50. There are often less than 50 data points available for groundwater analytical results for each of the wells included within the monitoring well network. Once the sample size exceeds 50, the Shapiro-Francia test for normality will be used.

After the distribution of the data is determined for each constituent, the data set will be examined for possible outliers. In accordance with the Statistical Analysis for Groundwater at Resource Conservation and Recovery Act (RCRA) Facilities, Unified Guidance (USEPA 2009) observed outliers will not be excluded from the data set unless it can be demonstrated that the outlier results from a data entry or laboratory error.

Summary statistics, to include the mean, median, standard deviation, and variance will then be calculated for each of the constituents detected in groundwater collected from each monitoring well.

4.1.1 NON-DETECT DATA

Laboratory results reported as non-detect (ND) will be entered into the statistical analysis as a value of one-half the PQL. Estimated values reported below the PQL but above the method detection limit (MDL) will be treated as valid measurements for statistical purposes.

4.2 INTERWELL STATISTICS

Interwell prediction limits (or intervals) will be calculated to identify constituents with statistically significant differences in concentrations between up-gradient and down-gradient wells. Upper prediction limits will be constructed to contain, with $(1-\alpha)$ probability ($\alpha = 0.05$), the next measured concentration. The interwell procedure calculates an upper prediction limit using the background wells and then compares the most recent concentration reported in each compliance well to the upper prediction limit.

Table 9 provides the compliance and background monitoring wells utilized for interwell analysis in each of the aquifers/water bearing zones. The upgradient/background wells were chosen in consultation with WDEQ staff based on joint review of historical data. Background wells were chosen that represent the same aquifer/water-bearing unit as the compliance wells in each case.

In the Perched Water Bearing Zone, wells P-5 and S-9 were chosen as background wells as they appeared to be located upgradient of landfill activities, however, background conditions in a perched water zone may be difficult to determine as hydraulic conditions are not well defined. Please note that some of the Perched Water Bearing Zone wells have been dry for an extended period of time and will not always be included in the statistical analysis. This is sometimes the case for wells EE-04, EE-07, S-9 and P1E-9. Deviations from the standard analysis will be noted in future reports.

4.3 INTRAWELL STATISTICS

Intrawell compliance tolerance interval analyses will be performed for down-gradient wells for Appendix A and B constituents which exceed the interwell prediction limits. This approach focuses intrawell compliance statistical analyses on those down-gradient wells and constituents with a statistically significant increase over background water quality.

The Mann-Kendall/Sen's Slope trend analysis will be used to determine if there is a significantly increasing trend in constituent concentrations reported in groundwater samples collected from any of the wells included in the monitoring network. An increasing slope is indicated by a positive Mann-Kendall statistic and an estimated Sen's Slope greater than zero. Time series plots and the data will be examined to ensure that a significant slope is not an artifact of changes

in detection limits over time. The trend analysis will not be used for compliance evaluations, but may provide important information regarding changes in groundwater chemistry, both up-gradient and down-gradient.

4.4 COMPARISONS TO GROUNDWATER PROTECTION STANDARDS

A 95% upper tolerance limit will be calculated for constituents detected in groundwater samples collected from monitoring wells that exhibit significantly higher concentrations compared to those reported in background wells (interwell analysis). The calculated 95% upper tolerance limit will be compared to the appropriate Groundwater Protection Standard (GWPS).

4.4.1 TOLERANCE LIMIT ANALYSIS

Tolerance limit analysis is a component of compliance limit testing. SWRR Chapter 2 Section 9.b.i.C.VI.3 (WDEQ 2017) recommends the use of tolerance limits or prediction intervals as a method to compare background wells to compliance wells. The 95% upper tolerance limit is currently being used to compare constituent concentrations measured in compliance wells to groundwater protection standards. When variation in the data set is large, the upper tolerance limit is often estimated well above actual constituent concentrations, which is the case with many constituents reported in groundwater samples collected from monitoring wells installed at the City of Cheyenne Sanitary Landfill.

In addition, if the distribution of analytical data for a constituent reported within a monitoring well is not normally distributed, the upper tolerance limit is estimated as the highest or the second highest concentration detected for that constituent, whichever provides the best estimate of the 90th percentile to the dataset with 95% confidence. This approach allows some flexibility when the data do not conform to an identifiable distribution and standard mathematical models cannot be used. If the highest or second highest reported concentration is greater than the groundwater protection standard, then the tolerance limit will consistently predict an exceedance above the groundwater protection standard, even if the remaining results are non-detects at the reporting limit. Historical statistical analysis has shown that many of the constituent concentrations reported in monitoring wells at the City of Cheyenne Landfill are not normally distributed and therefore the tolerance limit analysis could produce positive results driven by a single historic extreme value, potentially falsely indicating an exceedance above the groundwater protection standard.

5.0 LANDFILL METHANE MONITORING PLAN

A methane venting system was installed in the filled areas at the south end of the landfill before placing the final cap in 1991. Quarterly monitoring of methane concentrations in these vents was conducted from March 1992 through September 1993. Methane concentrations over this period were reduced from greater than 1,000 parts per million (ppm) to less than 1.0 ppm. Methane monitoring was discontinued along the south end of the landfill in September 1993.

A methane monitoring well (MMW-1) was installed between the North Trench and the newly constructed equipment building located north of the disposal cells. Methane concentrations have been measured in Well MMW-1 since September 1996, with concentrations sporadically exceeding 25% of the LEL. At that time, action limits for the perimeter of the landfill were established at 25% of the LEL. In correspondence dated December 7, 1998, the WDEQ/SHWD stated that additional measures to address the methane detections in MMW-1 were not necessary. Methane concentrations at MMW-1 had been more recently reported exceeding 100% of the LEL. This monitoring point is not located at the landfill boundary and is not considered a compliance methane monitoring point. The City has scheduled to plug and abandon MMW-1 in 2017 in favor of proposed methane monitoring probes along the facility boundary.

5.1 METHANE MONITORING AT THE LANDFILL BOUNDARY

Refer to the Addendum to the Sanitary Landfill Environmental Monitoring Plan (SLEMP), Methane Gas Sampling and Analysis Plan, Happy Jack Landfill, prepared by Golder and Associates, Inc. (2017; Appendix H-2 in Lifetime Permit), for the monitoring procedures for proposed gas monitoring probes located along the facility boundary. Methane monitoring at the facility boundary will be conducted quarterly in February, April, August, and December each year.

5.2 METHANE MONITORING OF EQUIPMENT BUILDINGS

The equipment building is constructed on a slab on grade beneath the shop area and on grade in the equipment storage bays. The building has a flexible membrane liner buried beneath the earth floor and concrete slab. The building is not equipped with a methane venting system. However, the building is equipped with two wind propelled turbine ventilators over the garage area and exhaust fans over the garage and shop area. The shop area and recycle building will also be monitored. Overhead doors on the buildings being monitored will be closed for a minimum of one-hour before testing.

A portable gas meter capable of monitoring oxygen and methane concentrations and % LEL will be utilized. Before collecting readings, the meter will be allowed to warm up and calibrated using methane as the calibration standard. Ambient air concentrations shall be recorded over a five-minute interval. The meter will be placed at a height of approximately three feet above the ground and allowed to equilibrate. Methane, oxygen, and percent LEL will be recorded every minute over a five-minute interval. After completing monitoring in the equipment area, the technician will proceed to the shop area and recycle building and repeat the sampling in those locations. Ambient air concentrations will be recorded on the methane monitoring field forms or similar forms included in Attachment B.

6.0 NOTIFICATION REQUIREMENTS

This section outlines the notification requirements and follow up actions for:

- Methane concentration above 25% LEL in facility buildings or above 100% LEL at compliance points at the landfill boundary
- Groundwater detections above GWPS
- Statistically significant increases in contaminant concentrations in groundwater

6.1 METHANE MONITORING NOTIFICATION REQUIREMENTS

If a methane level exceeding 25% of the LEL is detected in the landfill equipment building, shop building and/or recycle building, the building should be evacuated immediately, electricity shut off, contact made with the City of Cheyenne Public Works Director's office, the City of Cheyenne Fire Department, and the WDEQ/SHWD. Within seven (7) days of the exceedance, a copy of the methane test data will be placed in the operating record, along with a written description of the steps taken to protect human health. Within 60 days of the exceedance, a remediation plan will be implemented, which has been approved by the WDEQ/SHWD, and a copy of that plan placed in the operating record.

Please refer to the "Addendum to the Sanitary Landfill Environmental Monitoring Plan (SLEMP), Methane Gas Sampling and Analysis Plan, Happy Jack Landfill", prepared by Golder and Associates, Inc. (2017; Appendix H-2 in Lifetime Permit), for an explanation of the notification requirements should methane levels exceed the LEL at the gas monitoring probes to be located along the facility boundary.

6.2 DETECTION MONITORING NOTIFICATION REQUIREMENTS

If the concentration of an Appendix A constituent in any downgradient well located in the Perched Water Bearing Zone, Principal Unconfined Aquifer, or Semi-Confined Aquifer represents a statistically significant increase over background water quality during an annual review of available data, the City of Cheyenne Public Work's Office will be immediately notified. The WDEQ/SHWD will be contacted and a written report placed in the facility operating record within 14 days. Assessment Monitoring will begin within 90 days or in an acceptable time frame approved by the Administrator. Alternatively, the City may demonstrate to the WDEQ/SHWD that the statistically significant water quality difference is not due to the solid waste disposal facility, but that the difference is due to another source of pollution, error in sampling, analysis, or statistical evaluation, or natural variation in groundwater quality. The City of Cheyenne shall prepare a report documenting this demonstration, and following approval by the WDEQ/SHWD place

the report in the operating record for the facility. If the report is approved, the City of Cheyenne shall continue Detection Monitoring. If, after 90 days, a successful demonstration is not made, the City of Cheyenne will initiate an Assessment Monitoring program as required in SWRR Chapter 2, Section 9 unless an alternate timeframe is approved by the Administrator.

The WDEQ/SHWD will also be notified in writing and a notice placed within the operating record (SWRR Chapter 2, Section 9.b.i.D) within 14 days if the concentration of an Appendix A or Appendix C constituent with an MCL or a class of use based limit in the Water Quality Rules and Regulations (WDEQ 2015) is found to be a statistically significant increase over background water quality. The City will request in writing that WDEQ/SHWD establish a groundwater protection standard (WDEQ 2009) for that constituent. Within 30 days of the establishment of that standard, the City will determine whether the concentrations represent a statistical exceedance of the groundwater protection standard and notify the WDEQ/SHWD in writing within 14 days should that be the case.

6.3 ASSESSMENT MONITORING NOTIFICATION REQUIREMENTS

If there is a detection of any Appendix B constituent as part of Assessment Monitoring, the City of Cheyenne Public Works Director's Office will be notified. The WDEQ/SHWD will be notified within 14 days and a notice will be placed in the operating record identifying the Appendix B constituents that have been detected.

If the concentrations of all Appendix B constituents are at or below background values using the approved statistical procedures for two (2) or more consecutive sampling events, the City of Cheyenne must notify the WDEQ/SHWD and may return to Detection Monitoring.

If one or more Appendix B constituents are detected at statistically significant levels above the GWPS during an annual review of available data, the City of Cheyenne Public Works Director's Office will be notified. Within 14 days of this finding, a written report will be placed in the operating record identifying the Appendix B constituents (SWRR Chapter 2, Section 9.b.i.E). The WDEQ/SHWD Administrator, and appropriate local government officials as determined by the Administrator, will be notified. The City is also required to notify all persons who own or reside on the land that directly overlies any part of the plume of contamination, if that plume has migrated off-site.

7.0 REPORTING REQUIREMENTS

This section details the frequency and content of the routine progress reports submitted to the WDEQ/SHWD.

7.1 EVENT REPORT

A summary event report will be submitted to WDEQ/SHWD for the April and August monitoring events through the operating life of the landfill and through the post-closure period. Paper copies of each event report will be submitted within 60 days of receipt of complete and accurate analytical results along with electronic data, in the format specified by the WDEQ/SHWD Administrator. The event report will include:

- List of monitoring wells included in the Groundwater Monitoring Program
- Appendices containing field data, laboratory analytical data, quality control data, and chain of custody forms
- Fluid level summary tables and electronic data
- Groundwater analytical data tables and electronic data
- Potentiometric surface maps for the Semi-Confined and Principal Unconfined Aquifers
- Gradient, flow direction and velocity of groundwater in the Semi-Confined and Principal Unconfined Aquifers
- Results of quarterly methane monitoring
- Results of off-site monitoring events, if applicable

7.2 ANNUAL REPORT

Paper copies of a comprehensive annual report will be submitted to the WDEQ/SHWD prior to April 1 of the following year, along with required electronic data in the format specified by the WDEQ/SHWD Administrator. The annual monitoring period is January 1 through December 31. The annual reports will include:

- Summary of field and laboratory work completed during the previous year
- Documentation of operating procedures and field methods utilized
- Analytical data summary tables including methane monitoring results at both the facility boundary and the facility structures in accordance with May 6, 2008 Memorandum (Groundwater Monitoring Data Reporting Format for Municipal Solid Waste Landfills) and Section 3.2 of the Methane Gas Sampling and Analysis Plan (Golder and Associates 2017; Appendix H-2 in Lifetime Permit)
- Annual monitoring leachate data from NT-5

- Description of statistical analyses employed
- A summary of results from the statistical analyses and significant statistical increases
- Data disk of analytical results
- Summary of off-site monitoring data, if applicable
- Conclusions regarding the potential impact of landfill activities upon each aquifer and water bearing zone
- Recommendations for modifications to the Environmental Monitoring Program

8.0 REFERENCES

- Golder Associates, Inc. 2017. Addendum to the Sanitary Landfill Environmental Plan (SLEMP), Methane Gas Sampling and Analysis Plan, Happy Jack Landfill, Cheyenne, Wyoming. Lakewood, Colorado.
- Puls, R. and Barcelona, M. 1995. Low-Flow (minimal drawdown) ground-water sampling procedures. Washington D.C.: U.S. Environmental Protection Agency, Office of Research and Development, Office of Solid Waste and Emergency Response. 12 p.
- United States Environmental Protection Agency (USEPA). 2014a. USEPA Contract Laboratory Program (CLP) National Functional Guidelines for Superfund Organic Methods Data Review. Document Number EPA-540-R-014-002 Office of Superfund Remediation and Technology Innovation, Washington, D.C. August 2014.
- USEPA. 2014b. USEPA CLP National Functional Guidelines for Inorganic Superfund Data Review. Document Number EPA-540-R-013-001, Office of Superfund Remediation and Technology Innovation, Washington, D.C. August 2014.
- USEPA. 2013. USEPA New England Environmental Data Review Supplement for Regional Data Review Elements and Superfund Specific Guidance/Procedures, EQADR-Supplement0, April 2013.
- USEPA. 2012. 2012 Edition of the Drinking Water Standards and Health Advisories. Office of Water, Washington, D.C. (EPA 822-S-12-001). April 2012.
- USEPA. 2009. Statistical Analysis of Groundwater Monitoring Data at RCRA Facilities, Unified Guidance. Office of Resource Conservation and Recovery, Washington, D.C. (EPA530-R-09-007).
- USEPA. 2007. Test Methods for Evaluating Solid Waste, SW-846, Revision G. Office of Solid Waste, Washington, D.C.
- USEPA. 2004. USEPA CLP National Functional Guidelines for Inorganic Data Review. Document Number EPA 540-R-04-004, Office of Superfund Remediation and Technology Innovation, Washington, D.C. October 2004.

USEPA. 1999. USEPA CLP National Functional Guidelines for Organic Data Review. Document Number EPA 540/R-99/008, Office of Superfund Remediation and Technology Innovation, Washington, D.C. October 1999.

Wyoming Department of Environmental Quality (WDEQ), Solid and Hazardous Waste Division. 2017. Solid Waste Rules & Regulations, Chapter 2, Sanitary Landfill Regulations. Cheyenne, WY. January 3.

Wyoming Department of Environmental Quality (WDEQ), Solid and Hazardous Waste Division. 2009. Memorandum: Groundwater Protection Standards for Municipal Solid Waste Disposal Facilities. Cheyenne, Wyoming. March 15.

Wyoming Department of Environmental Quality (WDEQ), Water Quality Division. 2015. Water Quality Rules & Regulations, Chapter 8, Quality Standards for Wyoming Groundwaters. Cheyenne, WY. September 15.

TABLES

**TABLE 1. MONITORING WELL HISTORY
HAPPY JACK LANDFILL, CHEYENNE, WYOMING**

Monitoring Well/Piezometer	Date Drilled	Current Aquifer Designation	Report Reference
C-1	1999	Semi-Confined	Hydrogeologic Study 2000, Terracon
C-2	1999	Semi-Confined	Hydrogeologic Study 2000, Terracon
C-3	1999	Semi-Confined	Hydrogeologic Study 2000, Terracon
D-1	1982	Semi-Confined	Environmental Monitoring Plan 2000, IME
D-1A	1982	Principal Unconfined	Environmental Monitoring Plan 2000, IME
D-2	1982	Semi-Confined	Environmental Monitoring Plan 2000, IME
D-2A	1982	Principal Unconfined	Environmental Monitoring Plan 2000, IME
D-3	1982	Semi-Confined	Environmental Monitoring Plan 2000, IME
D-4	1982	Principal Unconfined	Environmental Monitoring Plan 2000, IME
Eagle #2	1988	Unknown	1995 Renewal Permit, Exhibit 20B
Eagle #3	1988	Unknown	1995 Renewal Permit, Exhibit 20B
Eagle #4	1988	Unknown	1995 Renewal Permit, Exhibit 20B
EE-01	1996	Perched 2	Environmental Monitoring Plan 2000, IME
EE-02	1996	Perched 2	Environmental Monitoring Plan 2000, IME
EE-04	1996	Perched 1	1998 Permit Renewal, ref East End Groundwater Study Feb 199
EE-05	1996	Perched 1	1998 Permit Renewal, ref East End Groundwater Study Feb 199
EE-06	1996	Perched 1	1998 Permit Renewal, ref East End Groundwater Study Feb 199
EE-07	1996	Perched 1	1998 Permit Renewal, ref East End Groundwater Study Feb 199
GMW-01	2013	Semi-Confined	Site Characterization Report, HJ Landfill Expansion 2014, Golder
GMW-03	2013	Semi-Confined	Site Characterization Report, HJ Landfill Expansion 2014, Golder
GMW-04	2013	Semi-Confined	Site Characterization Report, HJ Landfill Expansion 2014, Golder
GMW-06	2013	Semi-Confined	Site Characterization Report, HJ Landfill Expansion 2014, Golder
GMW-07	2013	Semi-Confined	Site Characterization Report, HJ Landfill Expansion 2014, Golder
No. 4	1975	Perched 1	1998 Permit, Exhibit 21
No. 7	1975	Perched 1	Environmental Monitoring Plan 2000, IME
No. 8	1977	Perched 1	1998 Permit, Exhibit 21
P-1	1985	Perched 1	1998 Permit, Exhibit 21
P-2	1985	Perched 1	Environmental Monitoring Plan 2000, IME
P-3	1985	Perched 1	1998 Permit, Exhibit 21
P-4	1985	Perched 1	1998 Permit, Exhibit 21
P-5	1985	Perched 1	1998 Permit, Exhibit 21
P-7	1990	Leachate	1998 Permit, Exhibit 21
P-8	1990	Perched 1	1998 Permit, Exhibit 21

**TABLE 1. MONITORING WELL HISTORY
HAPPY JACK LANDFILL, CHEYENNE, WYOMING**

Monitoring Well/Piezometer	Date Drilled	Current Aquifer Designation	Report Reference
P-9	1990	Perched 1	1998 Permit, Exhibit 21
P-10	1990	Perched 1	1998 Permit, Exhibit 21
P-11	1990	Perched 1	1998 Permit, Exhibit 21
P-12	1990	Perched 1	Environmental Monitoring Plan 2000, IME
P1-2	NA	Perched 2	Environmental Monitoring Plan 2000, IME
P1E-1	1999	Leachate	Hydrogeologic Study 2000, Terracon
P1E-2	1999	Perched 1	Hydrogeologic Study 2000, Terracon
P1E-9	1999	Perched 1	Hydrogeologic Study 2000, Terracon
P3-1	1999	Principal Unconfined	Hydrogeologic Study 2000, Terracon
P3-2	1999	Principal Unconfined	Hydrogeologic Study 2000, Terracon
P3-3	1999	Principal Unconfined	Hydrogeologic Study 2000, Terracon
PU-1	1999	Principal Unconfined	Hydrogeologic Study 2000, Terracon
PU-2	1999	Principal Unconfined	Hydrogeologic Study 2000, Terracon
PU-3	1999	Principal Unconfined	Hydrogeologic Study 2000, Terracon
PU-4	2008	Principal Unconfined	PUA Hydrogeologic Investigation 2009, Trihydro
PU-5	2008	Principal Unconfined	PUA Hydrogeologic Investigation 2009, Trihydro
PU-6	2008	Principal Unconfined	PUA Hydrogeologic Investigation 2009, Trihydro
S-1	1982	Leachate	1998 Permit, Exhibit 21
S-2	1982	Leachate	1998 Permit, Exhibit 21
S-3	1982	Leachate	1998 Permit, Exhibit 21
S-4	1982	Leachate	1998 Permit, Exhibit 21
S-5	1982	Leachate	1998 Permit, Exhibit 21
S-6	1982	Perched 1	1998 Permit, Exhibit 21
S-7	1982	Leachate	1998 Permit, Exhibit 21
S-7A	1982	Leachate	1998 Permit, Exhibit 21
S-8	1982	Leachate	1998 Permit, Exhibit 21
S-9	1982	Perched 1	1998 Permit, Exhibit 21
S-9A	1982	Perched 1	1998 Permit, Exhibit 21
S-10	1982	Perched 1	Environmental Monitoring Plan 2000, IME
S-11	1982	Perched 1	1998 Permit, Exhibit 21
S-11A	1982	Perched 1	1998 Permit, Exhibit 21
S-11B	1982	Leachate	1998 Permit, Exhibit 21
S-12	1982	Perched 1	Environmental Monitoring Plan 2000, IME
S-13	1982	Perched 1	1998 Permit, Exhibit 21

**TABLE 1. MONITORING WELL HISTORY
HAPPY JACK LANDFILL, CHEYENNE, WYOMING**

Monitoring Well/Piezometer	Date Drilled	Current Aquifer Designation	Report Reference
S-14	1982	Perched 1	Environmental Monitoring Plan 2000, IME
SRM-2	1989	Perched 1	1998 Permit, Exhibit 21
SRM-4	1990	Principal Unconfined	1998 Permit, Exhibit 21
SRM-5	1990	Principal Unconfined	1998 Permit, Exhibit 21
SRM-6	1990	Perched 1	1998 Permit, Exhibit 21
SRM-7	1990	Principal Unconfined	1998 Permit, Exhibit 21
SRM-8	1990	Principal Unconfined	1998 Permit, Exhibit 21

**TABLE 2. CURRENT MONITORING WELL, SUMP, AND PIEZOMETER CONSTRUCTION SUMMARY
HAPPY JACK LANDFILL, CHEYENNE, WYOMING**

Monitoring Well/ Piezometer	Classification	Hydraulic Location	Casing Inside Diameter	Ground Surface Elevation (ft-amsl)	Top of Casing Elevation (ft-amsl)	Total Depth Aug-16 (ft-btoc)	Depth Au (ft-
Semi-Confined Aquifer Wells							
C-1	Detection	Up-gradient	2-inch	6621.289	6623.329	260.00	21
C-2	Detection	Down-gradient	2-inch	6548.208	6550.468	259.50	14
C-3	Detection	Down-gradient	2-inch	6600.104	6601.416	247.00	19
GMW-01	Detection	Down-gradient	2-inch	6596.867	6599.371	218.38	[
GMW-03	Detection	Down-gradient	2-inch	6579.348	6581.368	217.17	20
GMW-04	Detection	Down-gradient	2-inch	6582.018	6584.19	218.55	20
GMW-06	Detection	Down-gradient	2-inch	6560.635	6563.047	188.95	16
GMW-07	Detection	Down-gradient	2-inch	6565.371	6567.913	238.53	20
Principal Unconfined Aquifer Wells							
PU-1	Detection	Up-gradient	2-inch	6618.595	6620.804	194.59	15
PU-2	Detection	Down-gradient	2-inch	6636.789	6638.386	178.89	16
PU-3	Detection	Up-gradient	2-inch	6647.535	6649.63	182.95	17
PU-4	Detection	Down-gradient	2-inch	6621.354	6623.728	189.75	18
PU-5	Detection	Down-gradient	2-inch	6547.811	6550.108	147.95	[
P3-1	Detection	Up-gradient	2-inch	6575.244	6577.528	153.25	14
P3-2	Detection	Down-gradient	2-inch	6539.942	6542.242	168.32	[
P3-3	Detection	Up-gradient	2-inch	6592.15	6595.243	182.84	17
SRM-4	Assessment	Down-gradient	5-inch	6558.571	6560.907	174.95	16
SRM-5	Assessment	Down-gradient	5-inch	6572.817	6574.097	177.25	15
SRM-7	Assessment	Down-gradient	5-inch	6600.087	6601.778	207.45	20
SRM-8	Assessment	Down-gradient	5-inch	6619.662	6621.465	203.40	18
Perched Water Bearing Zone Wells							
EE-04	Assessment	Down-gradient	2-inch	6558.006	6560.34	58.68	48
EE-07	Assessment	Down-gradient	2-inch	6547.747	6550.295	76.41	[
P-1	Detection	Up-gradient	2-inch	6603.55	6605.263	12.00	[
P-3	Detection	Up-gradient	2-inch	6599.325	6598.073	15.10	11
P-4	Detection	Up-gradient	2-inch	6592.134	6592.02	13.55	12
P-5	Detection	Up-gradient	2-inch	6594.964	6597.02	10.87	8
P-8	Detection	Up-gradient	2-inch	6591.297	6593.121	9.10	[
P-10	Detection	Up-gradient	2-inch	6589.189	6591.331	10.65	[
P-11	Detection	Up-gradient	2-inch	6586.945	6589.826	7.52	[
P1E-9	Assessment	Down-gradient	2-inch	6557.593	6559.985	55.45	50
S-9	Detection	Up-gradient	2-inch	6595.177	6596.493	15.57	[
SRM-6	Assessment	Down-gradient	5-inch	6574.928	6577.524	65.05	46

**TABLE 2. CURRENT MONITORING WELL, SUMP, AND PIEZOMETER CONSTRUCTION SUMMARY
HAPPY JACK LANDFILL, CHEYENNE, WYOMING**

Monitoring Well/ Piezometer	Classification	Hydraulic Location	Casing Inside Diameter	Ground Surface Elevation (ft-amsl)	Top of Casing Elevation (ft-amsl)	Total Depth Aug-16 (ft-btoc)	Depth Au (ft-
Leachate Wells							
P1E-1	Leachate	NA	2-inch	6589.566	6591.955	62.37	60
S-1	Leachate	NA	3-inch	6615.051	6617.73	23.55	1
S-4	Leachate	NA	3-inch	6612.125	6613.684	24.61	2
S-7A	Leachate	NA	2-inch	6604.273	6607.771	49.53	49
Leachate Sumps							
NT-1	Leachate	NA	6-inch	6643.908	6649.061	64.46	64
NT-2	Leachate	NA	6-inch	6636.99	6634.98	62.98	63
NT-3	Leachate	NA	10-inch	6623.384	6627.984	NM	NM
NT-4	Leachate	NA	10-inch	6601.426	6606.761	NM	NM
NT-5	Leachate	NA	16-inch	6552.013	6552.004	NM	NM

Notes:

ft - feet

ft-amsl - feet above mean sea level

ft-btoc - feet below top of casing

Information in this table was based on the best known data available

NM - not gauged manually (electronic display in use)

**TABLE 3. SAMPLING SCHEDULE SUMMARY
HAPPY JACK LANDFILL, CHEYENNE, WYOMING**

Revised January 2018

Monitoring Location	Sampling Requirements ¹	
	April	August
Semi-Confined Aquifer Wells		
C-1	Appendix A, C	Appendix C
C-2	Appendix A, C	Appendix C
C-3	Appendix A, C	Appendix C
GMW-01	Appendix A, C	Appendix C
GMW-03	Appendix A, C	Appendix C
GMW-04	Appendix A, C	Appendix C
GMW-06	Appendix A, C	Appendix C
GMW-07	Appendix A, C	Appendix C
Principal Unconfined Aquifer Wells		
PU-1	Appendix A, C	Appendix C
PU-2	Appendix A, C	Appendix C
PU-3	Appendix A, C	Appendix C
PU-4	Appendix A, C	Appendix C
PU-5	Appendix A, C	Appendix C
P3-1	Appendix A, C	Appendix C
P3-2	Appendix A, C	Appendix C
P3-3	Appendix A, C	Appendix C
SRM-4	Appendix B (VOCs & Inorganics), C ²	Appendix C
SRM-5	Appendix B (VOCs & Inorganics), C ²	Appendix C
SRM-7	Appendix B (VOCs & Inorganics), C ²	Appendix C
SRM-8	Appendix B (VOCs & Inorganics), C ²	Appendix C
Perched Water Bearing Zone Wells		
EE-04	Appendix B (VOCs & Inorganics), C ²	Appendix C
EE-07	Appendix B (VOCs & Inorganics), C ²	Appendix C
P1E-9	Appendix B (VOCs & Inorganics), C ²	Appendix C
P-5	Appendix A, C	Appendix C
S-9	Appendix A, C	Appendix C
SRM-6	Appendix B (VOCs & Inorganics), C ²	Appendix C
P-1	Appendix A, C	Appendix C
P-3	Appendix A, C	Appendix C
P-4	Appendix A, C	Appendix C
P-5	Appendix A, C	Appendix C
P-8	Appendix A, C	Appendix C
P-10	Appendix A, C	Appendix C
P-11	Appendix A, C	Appendix C
Leachate Sumps		
NT-5	Appendix A, C	N/A

Abbreviations:

N/A = Not applicable

Notes:

1. If an annual sample is required and cannot be collected in April, attempt will be made in August.
2. Full Appendix B parameters will be collected once every five years.

**TABLE 4. FLUID MONITORING SCHEDULE SUMMARY
HAPPY JACK LANDFILL, CHEYENNE, WYOMING**

Revised August 23, 2017

Monitoring Location	Fluid Monitoring Requirements and Event	
	April	August
Semi-Confined Aquifer Wells		
C-1	Required	Required
C-2	Required	Required
C-3	Required	Required
GMW-01	Required	Required
GMW-03	Required	Required
GMW-04	Required	Required
GMW-06	Required	Required
GMW-07	Required	Required
Principal Unconfined Aquifer Wells		
PU-1	Required	Required
PU-2	Required	Required
PU-3	Required	Required
PU-4	Required	Required
PU-5	Required	Required
P3-1	Required	Required
P3-2	Required	Required
P3-3	Required	Required
SRM-4	Required	Required
SRM-5	Required	Required
SRM-7	Required	Required
SRM-8	Required	Required
Perched Water Bearing Zone Wells		
EE-04	Required	Required
EE-07	Required	Required
P1E-9	Required	Required
S-9	Required	Required
SRM-6	Required	Required
P-1	Required	Required
P-3	Required	Required
P-4	Required	Required
P-5	Required	Required
P-8	Required	Required
P-10	Required	Required
P-11	Required	Required

**TABLE 4. FLUID MONITORING SCHEDULE SUMMARY
HAPPY JACK LANDFILL, CHEYENNE, WYOMING**

Revised August 23, 2017

Monitoring Location	Fluid Monitoring Requirements and Event	
	April	August
Leachate Wells		
P1E-1	Required	Required
S-1	Required	Required
S-4	Required	Required
S-7A	Required	Required
Leachate Sumps¹		
NT-1	Required	Required
NT-2	Required	Required

Note:

1. Fluid levels for NT-3, NT-4 and NT-5 read from electronic display monthly by City staff.

**TABLE 5. WDEQ/SHWD APPENDIX A CONSISTUENT LIST
HAPPY JACK LANDFILL, CHEYENNE, WYOMING**

Revised August 23, 2017

Common name ²	CAS RN ³	Chemical abstracts service index name ⁴	Suggested methods ^{1,5}	PQL (ug/L) ^{1,6}
Inorganics (15)				
Antimony	(Total)	Antimony	6010 7040 7041	300 2000 30
Arsenic	(Total)	Arsenic	6010 7060 7061	500 10 20
Barium	(Total)	Barium	6010 7080	20 1000
Beryllium	(Total)	Beryllium	6010 7090 7091	3 50 2
Cadmium	(Total)	Cadmium	6010 7130 7131	40 50 1
Chromium	(Total)	Chromium	6010 7190 7191	70 500 10
Cobalt	(Total)	Cobalt	6010 7200 7201	70 500 10
Copper	(Total)	Copper	6010 7210 7211	60 200 10
Lead	(Total)	Lead	6010 7420 7421	400 1000 10
Nickel	(Total)	Nickel	6010 7520	150 400
Selenium	(Total)	Selenium	6010 7740 7741	750 20 20
Silver	(Total)	Silver	6010 7760	70 100
Thallium	(Total)	Thallium	6010 7840 7841	400 1000 10
Vanadium	(Total)	Vanadium	6010 7910 7911	80 2000 40
Zinc	(Total)	Zinc	6010 7950 7951	20 50 0.5
Volatiles (47)				
Acetone	67-64-1	2-Propanone	8260	100
Acrylonitrile	107-13-1	2-Propenenitrile	8030 8260	5 200
Benzene	71-43-2	Benzene	8020 8021 8260	2 0.1 5
Bromochloromethane; Chlorobromomethane	74-97-5	Methane, bromochloro-	8021 8260	0.1 5
Bromodichloromethane; Dibromochloromethane	75-27-4	Methane, bromodichloro-	8010 8021 8260	1 0.2 5
Bromoform; Tribromomethane	75-25-2	Methane, tribromo-	8010 8021 8260	2 15 5

**TABLE 5. WDEQ/SHWD APPENDIX A CONSISTENT LIST
HAPPY JACK LANDFILL, CHEYENNE, WYOMING**

Revised August 23, 2017

Common name ²	CAS RN ³	Chemical abstracts service index name ⁴	Suggested methods ^{1,5}	PQL (ug/L) ^{1,6}
Carbon disulfide	75-15-0	Carbon disulfide	8260	100
Carbon tetrachloride	56-23-5	Methane, tetrachloro-	8010 8021 8260	1 0.1 10
Chlorobenzene	108-90-7	Benzene, chloro-	8010 8020 8021 8260	2 2 0.1 5
Chloroethane; Ethyl chloride	75-00-3	Ethane, chloro-	8010 8021 8060	5 1 10
Chloroform; Trichloromethane	67-66-3	Methane, trichloro-	8010 8021 8260	0.5 0.2 5
Dibromochloromethane; Chlorodibromomethane	124-48-1	Methane, dibromochloro-	8010 8021 8260	1 0.3 5
1,2-Dibromo-3-chloropropane; DBCP	96-12-8	Propane, 1,2-dibromo-3-chloro-	8011 8021 8260	0.1 30 25
1,2-Dibromoethane; Ethylene dibromide; EDB	106-93-4	Ethane, 1,2-dibromo-	8011 8021	0.1 10
o-Dichlorobenzene	95-50-1	Benzene, 1,2-dichloro-	8010 8020 8021 8120 8260 8270	2 5 0.5 10 5 10
p-Dichlorobenzene; 1,4 Dichlorobenzene	106-46-7	Benzene, 1,4-dichloro-	8010	2
trans-1,4-Dichloro-2-butene	110-57-6	2-Butene, 1,4-dichloro-, (E)-	8260	100
1,1-Dichloroethane; Ethylidene chloride	75-34-3	Ethane, 1,1-dichloro-	8010 8021 8260	1 0.5 5
1,2-Dichloroethane; Ethylene dichloride	107-06-2	Ethane, 1,1-dichloro-	8010 8021 8260	0.5 0.3 5
1,1-Dichloroethylene; 1,1- Dichloroethene; Vinylidene chloride	75-35-4	Ethene, 1,1-dichloro-	8010 8021 8260	1 0.5 5
cis-1,2-Dichloroethylene; cis-1,2- Dichloroethene	156-59-2	Ethene, 1,2-dichloro-, (Z)-	8021 8260	0.2 5
trans-1,2-Dichloroethylene trans-1,2- Dichloroethene	156-60-5	Ethene, 1,2-dichloro-, (E)-	8010 8021 8260	1 0.5 5
1,2-Dichloropropane; Propylene dichloride	78-87-5	Propane, 1,2-dichloro-	8010 8021 8260	0.5 0.05 5
cis-1,3-Dichloropropene	10061-01-5	1-Propene, 1,3-dichloro-, (Z)-	8010 8260	20 10
trans-1,3-Dichloropropene	10061-02-6	1-Propene, 1,3-dichloro-, (E)-	8010	5
Ethylbenzene	100-41-4	Benzene, ethyl-	8020 8221 8260	2 0.05 5
2-Hexanone; Methyl butyl ketone	591-78-6	2-Hexanone	8260	50
Methyl bromide; Bromomethane	74-83-9	Methane, bromo-	8010 8021	20 10
Methyl chloride; Chloromethane	74-87-3	Methane, chloro-	8010 8021	1 0.3

**TABLE 5. WDEQ/SHWD APPENDIX A CONSISTENT LIST
HAPPY JACK LANDFILL, CHEYENNE, WYOMING**

Revised August 23, 2017

Common name ²	CAS RN ³	Chemical abstracts service index name ⁴	Suggested methods ^{1,5}	PQL (ug/L) ^{1,6}
Methylene bromide; Dibromomethane	74-95-3	Methane, dibromo-	8010 8021 8260	15 20 10
Methylene chloride; Dichloromethane	75-09-2	Methane, dichloro-	8010 8021 8260	5 0.2 10
Methyl ethyl ketone; MEK; 2-Butanone	78-93-3	2-Butanone	8015 8260	10 100
Methyl iodide; Iodomethane	74-88-4	Methane, iodo-	8010 8260	40 10
4-Methyl-2-pentanone; Methyl isobutyl ketone	108-10-1	2-Pentanone, 4-methyl-	8015 8260	5 100
Styrene	100-42-5	Benzene, ethenyl-	8020 8021 8260	1 0.1 10
1,1,1,2-Tetrachloroethane	630-20-6	Ethane, 1,1,1,2-tetrachloro-	8010 8021 8260	5 0.05 5
1,1,2,2-Tetrachloroethane	79-34-5	Ethane, 1,1,2,2-tetrachloro-	8010 8021 8260	0.5 0.1 5
Tetrachloroethylene; Tetrachloroethene; Perchloroethylene	127-18-4	Ethene, tetrachloro-	8010 8021 8260	0.5 0.5 5
Toluene	108-88-3	Benzene, methyl-	8020 8021 8260	2 0.1 5
1,1,1-Trichloroethane; Methylchloroform	71-55-6	Ethane, 1,1,1-trichloro-	8010 8021 8260	0.3 0.3 5
1,1,2-Trichloroethane	79-00-5	Ethane, 1,1,2-trichloro-	8010 8260	0.2 5
Trichloroethylene; Trichloroethene	79-01-6	Ethene, trichloro-	8010 8021 8260	1 0.2 5
Trichlorofluoromethane; CFC- 11	75-69-4	Methane, trichlorofluoro-	8010 8021 8260	10 0.3 5
1,2,3-Trichloropropane	96-18-4	Propane, 1,2,3-trichloro-	8010 8021 8260	10 5 15
Vinyl acetate	108-05-4	Acetic acid, ethenyl ester	8260	50
Vinyl chloride; Chloroethene	75-01-4	Ethene, chloro-	8010 8021 8260	2 0.4 10
Xylene (total)	See Note 11	Benzene, dimethyl-	8020 8021 8260	5 0.2 5

Notes:

¹The regulatory requirements pertain only to the list of substances; the right hand columns (Methods and PQL) are given for informational purposes only. See also footnotes 5 and 6.

²Common names are those widely used in government regulations, scientific publications, and commerce; synonyms exist for many chemicals.

³Chemical Abstracts Service registry number. Where "Total" is entered, all species in the groundwater that contain this element are included.

⁴CAS index names are those used in the 9th Collective Index.

⁵Suggested Methods refer to analytical procedure numbers used in EPA Report SW-846 "Test Methods for Evaluating Solid Waste", third edition, November 1986, as revised, December 1987. Analytical details can be found in SW-846 and in documentation on file at the Department. CAUTION: The methods listed are representative SW-846 procedures and may not always be the most suitable method(s) for monitoring an analyte under the regulations.

⁶Practical Quantitation Limits (PQLs) are the lowest concentrations of analytes in groundwaters that can be reliably determined within specified limits of precision and accuracy by the indicated methods under routine laboratory operating conditions. The PQLs listed are generally stated to one significant figure. PQLs are based on 5 mL samples for volatile organics and 1 L samples for semivolatile organics. CAUTION: The PQL values in many cases are based only on a general estimate for the method and not on a determination for individual compounds; PQLs are not a part of the regulation.

**TABLE 6. WDEQ/SHWD APPENDIX B CONSTITUENT LIST
HAPPY JACK LANDFILL, CHEYENNE, WYOMING**

Revised August 23, 2017

Common name ²	CAS RN ³	Chemical abstracts service index name ⁴	Suggested methods ^{1,5}	PQL (ug/L) ^{1,6}
Inorganics (19)				
Antimony	(Total)	Antimony	6010 7040 7041	300 2000 30
Arsenic	(Total)	Arsenic	6010 7060 7061	500 10 20
Barium	(Total)	Barium	6010 7080	20 1000
Beryllium	(Total)	Beryllium	6010 7090 7091	3 50 2
Cadmium	(Total)	Cadmium	6010 7130 7131	40 50 1
Chromium	(Total)	Chromium	6010 7190 7191	70 500 10
Cobalt	(Total)	Cobalt	6010 7200 7201	70 500 10
Copper	(Total)	Copper	6010 7210 7211	60 200 10
Cyanide	57-12-5	Cyanide	9010	200
Lead	(Total)	Lead	6010 7420 7421	400 1000 10
Mercury	(Total)	Mercury	7470	2
Nickel	(Total)	Nickel	6010 7520	150 400
Selenium	(Total)	Selenium	6010 7740 7741	750 20 20
Silver	(Total)	Silver	6010 7760	70 100
Sulfide	18496-25-8	Sulfide	9030	4000
Thallium	(Total)	Thallium	6010 7840 7841	400 1000 10
Tin	(Total)	Tin	6010	40
Vanadium	(Total)	Vanadium	6010 7910 7911	80 2000 40
Zinc	(Total)	Zinc	6010 7950 7951	20 50 0.5

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Common name ²	CAS RN ³	Chemical abstracts service index name ⁴	Suggested methods ^{1,5}	PQL (ug/L) ^{1,6}
Volatiles (64)				
Acetone	67-64-1	2-Propanone	8260	100
Acetonitrile; Methyl cyanide	75-05-8	Acetonitrile	8015	100
Acrolein	107-02-8	2-Propenal	8030	5
Acrylonitrile	107-13-1	2-Propenenitrile	8260	100
			8030	5
			8260	200
Allyl chloride	107-05-1	1-Propene, 3-chloro-	8010	5
			8260	10
Benzene	71-43-2	Benzene	8020	2
			8021	0.1
			8260	5
Bromochloromethane; Chlorobromomethane	74-97-5	Methane, bromochloro-	8021	0.1
			8260	5
Bromodichloromethane; Dibromochloromethane	75-27-4	Methane, bromodichloro-	8010	1
			8021	0.2
			8260	5
Bromoform; Tribromomethane	75-25-2	Methane, tribromo-	8010	2
			8021	15
			8260	5
Carbon disulfide	75-15-0	Carbon disulfide	8260	100
Carbon tetrachloride	56-23-5	Methane, tetrachloro-	8010	1
			8021	0.1
			8260	10
Chlorobenzene	108-90-7	Benzene, chloro-	8010	2
			8020	2
			8021	0.1
			8260	5
Chloroethane; Ethyl chloride	75-00-3	Ethane, chloro-	8010	5
			8021	1
			8060	10
Chloroform; Trichloromethane	67-66-3	Methane, trichloro-	8010	0.5
			8021	0.2
			8260	5
Chloroprene	126-99-8	1,3-Butadiene, 2-chloro-	8010	50
			8260	20
Dibromochloromethane; Chlorodibromomethane	124-48-1	Methane, dibromochloro-	8010	1
			8021	0.3
			8260	5
1,2-Dibromo-3-chloropropane; DBCP	96-12-8	Propane, 1,2-dibromo-3-chloro-	8011	0.1
			8021	30
			8260	25
1,2-Dibromoethane; Ethylene dibromide; EDB	106-93-4	Ethane, 1,2-dibromo-	8011	0.1
			8021	10
o-Dichlorobenzene	95-50-1	Benzene, 1,2-dichloro-	8010	2
			8020	5
			8021	0.5
			8120	10
			8260	5
m-Dichlorobenzene; 1,3-Dichlorobenzene	541-73-1	Benzene, 1,3-dichloro-	8270	10
			8010	5
			8020	5
			8021	0.2
			8120	10
p-Dichlorobenzene; 1,4-Dichlorobenzene	106-46-7	Benzene, 1,4-dichloro-	8260	5
			8120	10
			8260	5
			8270	10
			8010	2
p-Dichlorobenzene; 1,4-Dichlorobenzene	106-46-7	Benzene, 1,4-dichloro-	8010	2
trans-1,4-Dichloro-2-butene	110-57-6	2-Butene, 1,4-dichloro-, (E)-	8260	100
Dichlorodifluoromethane	75-71-8		8260	--
1,1-Dichloroethane; Ethylidene chloride	75-34-3	Ethane, 1,1-dichloro-	8010	1
			8021	0.5
			8260	5

**TABLE 6. WDEQ/SHWD APPENDIX B CONSTITUENT LIST
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Revised August 23, 2017

Common name ²	CAS RN ³ 10061-02-6	Chemical abstracts service index name ⁴	Suggested methods ^{1,5}	PQL (ug/L) ^{1,6}
1,2-Dichloroethane; Ethylene dichloride	107-06-2	Ethane, 1,1-dichloro-	8010 8021 8260	0.5 0.3 5
1,1-Dichloroethylene; 1,1-Dichloroethene; Vinylidene chloride	75-35-4	Ethene, 1,1-dichloro-	8010 8021 8260	1 0.5 5
cis-1,2-Dichloroethylene; cis-1,2-Dichloroethene	156-59-2	Ethene, 1,2-dichloro-, (Z)-	8021 8260	0.2 5
trans-1,2-Dichloroethylene trans-1,2-Dichloroethene	156-60-5	Ethene, 1,2-dichloro-, (E)-	8010 8021 8260	1 0.5 5
1,2-Dichloropropane; Propylene dichloride	78-87-5	Propane, 1,2-dichloro-	8010 8021 8260	0.5 0.05 5
1,3-Dichloropropane; Trimethylene dichloride	142-28-9	Propane, 1,3-dichloro-	8021 8260	0.3 15
2,2-Dichloropropane; Isopropylidene chloride	594-20-7	Propane, 2,2-dichloro-	8021 8260	0.5 5
1,1-Dichloropropene;	563-58-6	1-Propene, 1,1-dichloro-	8021 8260	0.2 5
cis-1,3-Dichloropropene	10061-01-5	1-Propene, 1,3-dichloro-, (Z)-	8010 8260	20 10
trans-1,3-Dichloropropene	10061-02-6	1-Propene, 1,3-dichloro-, (E)-	8010 8260	5 5
Ethylbenzene	100-41-4	Benzene, ethyl-	8020 8221 8260	2 0.05 5
Ethyl methacrylate	97-63-2	2-Propenoic acid, 2-methyl-, ethyl ester	8015 8260 8270	5 10 10
2-Hexanone; Methyl butyl ketone	591-78-6	2-Hexanone	8260	50
Isobutyl alcohol	78-83-1	1-Propanol, 2-methyl-	8015 8240	50 100
Methacrylonitrile	126-98-7	2-Propenenitrile, 2-methyl-	8015 8260	5 100
Methyl bromide; Bromomethane	74-83-9	Methane, bromo-	8010 8021	20 10
Methyl chloride; Chloromethane	74-87-3	Methane, chloro-	8010 8021	1 0.3
Methylene bromide; Dibromomethane	74-95-3	Methane, dibromo-	8010 8021 8260	15 20 10
Methylene chloride; Dichloromethane	75-09-2	Methane, dichloro-	8010 8021 8260	5 0.2 10
Methyl ethyl ketone; MEK; 2-Butanone	78-93-3	2-Butanone	8015 8260	10 100
Methyl iodide; Iodomethane	74-88-4	Methane, iodo-	8010 8260	40 10
Methyl methacrylate	80-62-6	2-Propenoic acid, 2-methyl-, methyl ester	8015 8260	2 30
4-Methyl-2-pentanone; Methyl isobutyl ketone	108-10-1	2-Pentanone, 4-methyl-	8015 8260	5 100
Naphthalene	91-20-3	Naphthalene	8021 8100 8260 8270	0.5 200 5 10
Propionitrile; Ethyl cyanide	107-12-0	Propanenitrile	8015 8260	60 150
Styrene	100-42-5	Benzene, ethenyl-	8020 8021 8260	1 0.1 10
1,1,1,2-Tetrachloroethane	630-20-6	Ethane, 1,1,1,2-tetrachloro-	8010 8021 8260	5 0.05 5

**TABLE 6. WDEQ/SHWD APPENDIX B CONSTITUENT LIST
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Common name ²	CAS RN ³ 10061-02-6	Chemical abstracts service index name ⁴	Suggested methods ^{1,5}	PQL (ug/L) ^{1,6}
1,1,2,2-Tetrachloroethane	79-34-5	Ethane, 1,1,2,2-tetrachloro-	8010 8021 8260	0.5 0.1 5
Tetrachloroethylene; Tetrachloroethene; Perchloroethylene	127-18-4	Ethene, tetrachloro-	8010 8021 8260	0.5 0.5 5
Toluene	108-88-3	Benzene, methyl-	8020 8021 8260	2 0.1 5
1,2,4-Trichlorobenzene	120-82-1	Benzene, 1,2,4-trichloro-	8021 8120 8260 8270	0.3 0.5 10 10
1,1,1-Trichloroethane; Methylchloroform	71-55-6	Ethane, 1,1,1-trichloro-	8010 8021 8260	0.3 0.3 5
1,1,2-Trichloroethane	79-00-5	Ethane, 1,1,2-trichloro-	8010 8260	0.2 5
Trichloroethylene; Trichloroethene	79-01-6	Ethene, trichloro-	8010 8021 8260	1 0.2 5
Trichlorofluoromethane; CFC- 11	75-69-4	Methane, trichlorofluoro-	8010 8021 8260	10 0.3 5
1,2,3-Trichloropropane	96-18-4	Propane, 1,2,3-trichloro-	8010 8021 8260	10 5 15
Vinyl acetate	108-05-4	Acetic acid, ethenyl ester	8260	50
Vinyl chloride; Chloroethene	75-01-4	Ethene, chloro-	8010 8021 8260	2 0.4 10
Xylene (total)	See Note 11	Benzene, dimethyl-	8020 8021 8260	5 0.2 5

**TABLE 6. WDEQ/SHWD APPENDIX B CONSTITUENT LIST
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Common name ²	CAS RN ³	Chemical abstracts service index name ⁴	Suggested methods ^{1,5}	PQL (ug/L) ^{1,6}
Semi-Volatiles (108)				
Acenaphthene	83-32-9	Acenaphthylene, 1,2-dihydro-	8100 8270	200 10
Acenaphthylene	208-96-8	Acenaphthylene	8100 8270	200 10
Acetophenone	98-86-2	Ethanone, 1-phenyl-	8270	10
2-Acetylaminofluorene; 2-AAF	53-96-3	Acetamide, N-9H-fluoren-2-yl-	8270	20
4-Aminobiphenyl	92-67-1	[1,1'-Biphenyl]-4-amine	8270	20
Anthracene	120-12-7	Anthracene	8100 8270	200 10
Benzo[a]anthracene; Benzanthracene	56-55-3	Benz[a]anthracene	8100 8270	200 10
Benzo[b]fluoranthene	205-99-2	Benz[e]acephenanthrylene	8100 8270	200 10
Benzo[k]fluoranthene	207-08-9	Benzo[k]fluoranthene	8100 8270	200 10
Benzo[ghi]perylene	191-24-2	Benzo[ghi]perylene	8100 8270	200 10
Benzo[a]pyrene	50-32-8	Benzo[a]pyrene	8100 8270	200 10
Benzyl alcohol	100-51-6	Benzenemethanol	8270	20
Bis(2-chloroethoxy)methane	111-91-1	Ethane, 1,1'-[methylenebis (oxy)]bis[2-chloro-	8110 8270	5 10
Bis(2-chloroethyl)ether; Dichloroethyl ether	111-44-4	Ethane, 1,1'-oxybis[2-chloro-	8110 8270	3 10
Bis(2-chloro-1-methylethyl) ether; 2,2'- Dichlorodiisopropyl ether; DCIP, See note 7	108-60-1	Propane, 2,2'-oxybis[1-chloro-	8110 8270	10 10
Bis(2-ethylhexyl) phthalate	117-81-7	1,2-Benzenedicarboxylic acid, bis(2-ethylhexyl)ester	8060	20
4-Bromophenyl phenyl ether	101-55-3	Benzene, 1-bromo-4-phenoxy-	8110 8270	25 10
Butyl benzyl phthalate; Benzyl butyl phthalate	85-68-7	1,2-Benzenedicarboxylic acid, butyl phenylmethyl ester	8060 8270	5 10
p-Chloroaniline	106-47-8	Benzenamine, 4-chloro-	8270	20
Chlorobenzilate	510-15-6	Benzenecetic acid, 4-chloro- α-(4-chlorophenyl)-α-hydroxy-, ethyl ester	8270	10
p-Chloro-m-cresol; 4-Chloro-3-methylphenol	59-50-7	Phenol, 4-chloro-3-methyl-	8040 8270	5 20
2-Chloronaphthalene	91-58-7	Naphthalene, 2-chloro-	8120 8270	10 10
2-Chlorophenol	95-57-8	Phenol, 2-chloro-	8040 8270	5 10
4-Chlorophenyl phenyl ether	7005-72-3	Benzene, 1-chloro-4-phenoxy-	8110 8270	40 10
Chrysene	218-01-9	Chrysene	8100 8270	200 10
m-Cresol; 3-methylphenol	108-39-4	Phenol, 3-methyl-	8270	10
o-Cresol; 2-methylphenol	95-48-7	Phenol, 2-methyl-	8270	10
p-Cresol; 4-methylphenol	106-44-5	Phenol, 4-methyl-	8270	10
Diallate	2303-16-4	Carbamothioic acid, bis(1-methylethyl)-, S- (2,3-dichloro-2-propenyl) ester	8270	10
Dibenz[a,h]anthracene	53-70-3	Dibenz[a,h]anthracene	8100 8270	200 10
Dibenzofuran	132-64-9	Dibenzofuran	8270	10
3,3'-Dichlorobenzidine	91-94-1	[1,1'-Biphenyl]-4,4'-diamine, 3,3'-dichloro-	8270	20
2,4-Dichlorophenol	120-83-2	Phenol, 2,4-dichloro-	8040 8270	5 10
2,6-Dichlorophenol	87-65-0	Phenol, 2,6-dichloro-	8270	10
Diethyl phthalate	84-66-2	1,2-Benzenedicarboxylic acid, diethyl ester	8060 8270	5 10

**TABLE 6. WDEQ/SHWD APPENDIX B CONSTITUENT LIST
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Common name ²	CAS RN ³ 10061-02-6	Chemical abstracts service index name ⁴	Suggested methods ^{1,5}	PQL (ug/L) ^{1,6}
O,O-Diethyl O-2-pyrazinyl phosphorothioate; Thionazin	297-97-2	Phosphorothioic acid, O,O- diethyl O-pyrazinyl ester	8141 8270	5 20
Dimethoate	60-51-5	Phosphorodithioic acid, O,O- dimethyl S-[2-(methylamino)-2- oxoethyl] ester	8141 8270	3 20
p-(Dimethylamino)azobenzene	60-11-7	Benzenamine, N,N-dimethyl-4- (phenylazo)-	8270	10
7,12- Dimethylbenz[a]anthracene	57-97-6	Benz[a]anthracene, 7,12- dimethyl-	8270	10
3,3'-Dimethylbenzidine	119-93-7	[1,1'-Biphenyl]-4,4'-diamine, 3,3'- dimethyl-	8270	10
2,4-Dimethylphenol; m-Xylenol	105-67-9	Phenol, 2,4-dimethyl-	8040 8270	5 10
Dimethyl phthalate	131-11-3	1,2-Benzenedicarboxylic acid, dimethyl ester	8060 8270	5 10
m-Dinitrobenzene	99-65-0	Benzene, 1,3-dinitro-	8270	20
4,6-Dinitro-o-cresol; 4,6-Dinitro-2-methylphenol	534-52-1	Phenol, 2-methyl-4,6-dinitro-	8040 8270	150 50
2,4-Dinitrophenol	51-28-5	Phenol, 2,4-dinitro-	8040 8270	150 50
2,4-Dinitrotoluene	121-14-2	Benzene, 1-methyl-2,4-dinitro-	8090 8270	0.2 10
Di-n-butyl phthalate	84-74-2	1,2-Benzenedicarboxylic acid, dibutyl ester	8060 8270	5 10
2,6-Dinitrotoluene	606-20-2	Benzene, 2-methyl-1,3-dinitro-	8090 8270	0.1 10
Dinoseb; DNBP; 2-sec-Butyl-4,6-dinitrophenol	88-85-7	Phenol, 2-(1-methylpropyl)- 4,6-dinitro-	8150 8270	1 20
Di-n-octyl phthalate	117-84-0	1,2-Benzenedicarboxylic acid, dioctyl ester	8060 8270	30 10
Diphenylamine	122-39-4	Benzenamine, N-phenyl-	8270	10
Disulfoton	298-04-4	Phosphorodithioic acid, O,O- diethyl S-[2- (ethylthio)ethyl]ester	8140 8141 8270	2 0.5 10
Ethyl methanesulfonate	62-50-0	Methanesulfonic acid, ethyl ester	8270	20
Famphur	52-85-7	Phosphorothioic acid, O-[4- [(dimethylamino)sulfonyl]phenyl]- O,O-dimethyl ester	8270	20
Fluoranthene	206-44-0	Fluoranthene	8100 8270	200 10
Fluorene	86-73-7	9H-Fluorene	8100 8270	200 10
Hexachlorobenzene	118-74-1	Benzene, hexachloro-	8120 8270	0.5 10
Hexachlorobutadiene	87-68-3	1,3-Butadiene, 1,1,2,3,4,4- hexachloro-	8021 8120 8260 8270	0.5 5 10 10
Hexachlorocyclopentadiene	77-47-4	1,3-Cyclopentadiene, 1,2,3,4,5,5-hexachloro-	8120 8270	5 10
Hexachloroethane	67-72-1	Ethane, hexachloro-	8120 8260 8270	0.5 10 10
Hexachloropropene	1888-71-7	1-Propene, 1,1,2,3,3,3- hexachloro-	8270	10
Indeno(1,2,3-cd)pyrene	193-39-5	Indeno[1,2,3-cd]pyrene	8100 8270	200 10
Isodrin	465-73-6	1,4,5,8- Dimethanonaphthalene, 1,2,3,4,1 0,10-hexachloro-1,4,4a,5,8,8a hexahydro- (1α,4α,4aβ,5β,8β,8aβ)-	8270 8260	20 10
Isophorone	78-59-1	2-Cyclohexen-1-one, 3,5,5- trimethyl-	8090 8270	60 10

**TABLE 6. WDEQ/SHWD APPENDIX B CONSTITUENT LIST
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Common name ²	CAS RN ³ 10061-02-6	Chemical abstracts service index name ⁴	Suggested methods ^{1,5}	PQL (ug/L) ^{1,6}
Isosafrole	120-58-1	1,3-Benzodioxole, 5-(1-propenyl)-	8270	10
Kepone	143-50-0	1,3,4-Metheno-2H-cyclobuta- [cd]pentalen-2-one, 1,1a,3,3a,4,5,5a,5b,6- decachloro-octahydro-	8270	20
Methapyrilene	91-80-5	1,2-Ethanediamine, N,N- dimethyl- N'-2-pyridinyl-N'-(2-thienylmethyl)-	8270	100
3-Methylcholanthrene	56-49-5	Benz[<i>jj</i>]aceanthrylene, 1,2- dihydro- 3-methyl-	8270	10
Methyl methanesulfonate	66-27-3	Methanesulfonic acid, methyl ester	8270	10
2-Methylnaphthalene	91-57-6	Naphthalene, 2-methyl-	8270	10
Methyl parathion; Parathion methyl	298-00-0	Phosphorothioic acid, O,O- dimethyl O-(4-nitrophenyl) ester	8140 8141 8270	0.5 1 10
1,4-Naphthoquinone	130-15-4	1,4-Naphthalenedione	8270	10
1-Naphthylamine	134-32-7	1-Naphthalenamine	8270	10
2-Naphthylamine	91-59-8	2-Naphthalenamine	8270	10
o-Nitroaniline; 2- Nitroaniline	88-74-4	Benzenamine, 2-nitro-	8270	50
m-Nitroaniline; 3- Nitroaniline	99-09-2	Benzenamine, 3-nitro-	8270	50
p-Nitroaniline; 4-Nitroaniline	100-01-6	Benzenamine, 4-nitro-	8270	50
Nitrobenzene	98-95-3	Benzene, nitro-	8090 8270	40 10
o-Nitrophenol; 2-Nitrophenol	88-75-5	Phenol, 2-nitro-	8040 8270	5 10
p-Nitrophenol; 4-Nitrophenol	100-02-7	Phenol, 4-nitro-	8040 8270	10 50
N-Nitrosodiethylamine	55-18-5	Ethanamine, N-ethyl-N-nitroso-	8270	20
N-Nitrosodimethylamine	62-75-9	Methanamine, N-methyl-N- nitroso-	8070	2
N-Nitrosodi-n-butylamine	924-16-3	1-Butanamine, N-butyl-N- nitroso-	8270	10
N-Nitrosodiphenylamine	86-30-6	Benzenamine, N-nitroso-N- phenyl-	8070	5
N-Nitrosodipropylamine; N-Nitroso-N-dipropylamine; Di-n-propylnitrosamine	621-64-7	1-Propanamine, N-nitroso-N- propyl-	8070	10
N-Nitrosomethylethylamine	10595-95-6	Ethanamine, N-methyl-N- nitroso-	8270	10
N-Nitrosomorpholine	59-89-2	Morpholine, 4-nitroso-	8270	10
N-Nitrosopiperidine	100-75-4	Piperidine, 1-nitroso-	8270	20
N-Nitrosopyrrolidine	930-55-2	Pyrrolidine, 1-nitroso-	8270	40
5-Nitro-o-toluidine	99-55-8	Benzenamine, 2-methyl-5-nitro-	8270	10
Pentachlorophenol	87-86-5	Phenol, pentachloro-	8040 8270	5 50
Phenanthrene	85-01-8	Phenanthrene	8100 8270	200 10
Phenol	108-95-2	Phenol	8040	1
p-Phenylenediamine	106-50-3	1,4-Benzenediamine	8270	10
Pentachlorobenzene	608-93-5	Benzene, pentachloro-	8270	10
Pentachloronitrobenzene	82-68-8	Benzene, pentachloronitro-	8270	20
Phenacetin	62-44-2	Acetamide, N-(4-ethoxyphenyl)	8270	20

**TABLE 6. WDEQ/SHWD APPENDIX B CONSTITUENT LIST
HAPPY JACK LANDFILL, CHEYENNE, WYOMING**

Revised August 23, 2017

Common name ²	CAS RN ³ 10061-02-6	Chemical abstracts service index name ⁴	Suggested methods ^{1,5}	PQL (ug/L) ^{1,6}
Phorate	298-02-2	Phosphorodithioic acid, O,O-diethyl S-[(ethylthio)methyl] ester	8140 8141 8270	2 0.5 10
Pronamide	23950-58-5	Benzamide, 3,5-dichloro-N- (1,1-dimethyl-2-propynyl)-	8270	10
Pyrene	129-00-0	Pyrene	8100 8270	200 10
Safrole	94-59-7	1,3-Benzodioxole, 5-(2-propenyl)-	8270	10
1,2,4,5-Tetrachloro-benzene	95-94-3	Benzene, 1,2,4,5-tetrachloro-	8270	10
2,3,4,6-Tetrachlorophenol	58-90-2	Phenol, 2,3,4,6-tetrachloro-	8270	10
o-Toluidine	95-53-4	Benzenamine, 2-methyl-	8270	10
2,4,5-Trichlorophenol	95-95-4	Phenol, 2,4,5-trichloro-	8270	10
2,4,6-Trichlorophenol	88-06-2	Phenol, 2,4,6-trichloro-	8040 8270	5 10
O,O,O-Triethyl phosphorothioate	126-68-1	Phosphorothioic acid, O,O,O-triethyl ester	8270	10
sym-Trinitrobenzene	99-35-4	Benzene, 1,3,5-trinitro-	8270	10

**TABLE 6. WDEQ/SHWD APPENDIX B CONSTITUENT LIST
HAPPY JACK LANDFILL, CHEYENNE, WYOMING**

Revised August 23, 2017

Common name ²	CAS RN ³ 10061-02-6	Chemical abstracts service index name ⁴	Suggested methods ^{1,5}	PQL (ug/L) ^{1,6}
Pesticides (20)				
Aldrin	309-00-2	1,4:5,8-Dimethanonaphthalene, 1,2,3,4,10,10-hexachloro- 1,4,4a,5,8,8a-hexahydro- (1α,4α,4aβ,5α, 8α,8aβ)-	8080 8270	0.05 10
alpha-BHC	319-84-6	Cyclohexane, 1,2,3,4,5,6- hexachloro-, (1α, 2α,3β,4α,5β,6β)-	8080 8270	0.05 10
beta-BHC	319-85-7	Cyclohexane, 1,2,3,4,5,6- hexachloro-, (1α,2β, 3α,4β,5α,6β)-	8080 8270	0.05 20
delta-BHC	319-86-8	Cyclohexane, 1,2,3,4,5,6- hexachloro-, (1α, 2α,3α,4β,5α,6β)-	8080 8270	0.1 20
gamma-BHC; Lindane	58-89-9	Cyclohexane, 1,2,3,4,5,6- hexachloro-, (1α,2α, 3β,4α,5α,6β)-	8080 8270	0.05 20
Chlordane	See Note 8	4,7-Methano-1H-indene, 1,2,4,5,6,7,8,8-octachloro- 2,3,3a,4,7,7a-hexahydro-	8080 8270	0.1 50
4,4'-DDD	72-54-8	Benzene, 1,1'-(2,2- dichloroethylidene)bis[4- chloro-	8080 8270	0.1 10
4,4'-DDE	72-55-9	Benzene, 1,1'-(dichloroethenylidene)bis[4- chloro-	8080 8270	0.05 10
4,4'-DDT	50-29-3	Benzene, 1,1'-(2,2,2- trichloroethylidene)bis[4- chloro-	8080 8270	0.1 10
Dieldrin	60-57-1	2,7:3,6-Dimethanonaphth[2,3- b]oxirene, 3,4,5,6,9,9- hexachloro- 1a,2,2a,3,6,6a,7,7a-octahydro- , (1α,2β,2α,3β,6β,6α,7β,7α)-	8080 8270	0.05 10
Endosulfan I	959-98-8	6,9-Methano-2,4,3- benzodioxathiepin, 6,7,8,9,10,10-hexachloro- 1,5,5a,6,9,9a-hexahydro-, 3- oxide, (3α,5αβ,6α,9α,9aβ)-	8080 8250	0.1 10
Endosulfan II	33213-65-9	6,9-Methano-2,4,3- benzodioxathiepin, 6,7,8,9,10,10-hexachloro- 1,5,5a,6,9,9a-hexahydro-, 3- oxide, (3α,5α,6β,9β,9α)-	8080 8270	0.05 20
Endosulfan sulfate	1031-07-8	6,9-Methano-2,4,3- benzodioxathiepin, 6,7,8,9,10,10-hexachloro- 1,5,5a,6,9,9a-hexahydro-, 3,3- dioxide	8080 8270	0.5 10
Endrin	72-20-8	2,7:3,6-Dimethanonaphth[2,3- b]oxirene, 3,4,5,6,9,9- hexachloro- 1a,2,2a,3,6,6a,7,7a-octahydro- , (1α, 2β,2αβ,3α,6α, 6aβ,7β,7α)-	8080 8270	0.1 20
Endrin aldehyde	7421-93-4	1,2,4- Methenocyclopenta[cd]pentalene -5-carboxaldehyde, 2,2a,3,3,4,7- hexachlorodecahydro-, (1α,2β,2αβ,4β,4aβ,5β,6β,,6bβ,7 R*)-	8080 8270	0.2 10

**TABLE 6. WDEQ/SHWD APPENDIX B CONSTITUENT LIST
HAPPY JACK LANDFILL, CHEYENNE, WYOMING**

Revised August 23, 2017

Common name ²	CAS RN ³ 10061-02-6	Chemical abstracts service index name ⁴	Suggested methods ^{1,5}	PQL (ug/L) ^{1,6}
Heptachlor	76-44-8	4,7-Methano-1H-indene, 1,4,5,6,7,8,8-heptachloro- 3a,4,7,7a-tetrahydro-	8080 8270	0.05 10
Heptachlor epoxide	1024-57-3	2,5-Methano-2H-indeno[1,2- b]oxirene, 2,3,4,5,6,7,7- heptachloro-1a,1b,5,5a,6,6a,- hexahydro-, (1aα,1bβ,2α,5α,5aβ,6β,6aα)	8080 8270	1 10
Methoxychlor	72-43-5	Benzene, 1,1'- (2,2,2-trichloroethylidene)bis [4-methoxy-	8080 8270	2 10
Parathion	56-38-2	Phosphorothioic acid, O,O- diethyl-O-(4-nitrophenyl) ester	8141 8270	0.5 10
Toxaphene	See Note 10	Toxaphene	8080	2
Herbicides (3)				
2,4-D; 2,4-Dichlorophenoxy-acetic acid	94-75-7	Acetic acid, (2,4- dichlorophenoxy)-	8150	10
2,4,5-T; 2,4,5- Trichlorophenoxyacetic acid	93-76-5	Acetic acid, (2,4,5- trichlorophenoxy)-	8150	2
Silvex; 2,4,5-TP	93-72-1	Propanoic acid, 2-(2,4,5- trichlorophenoxy)-	8150	2
PCBs (7)				
Polychlorinated biphenyls; PCBs; Aroclors	See Note 9	1,1'-Biphenyl, chloro derivatives	8080 8270	50 200

Notes:

¹The regulatory requirements pertain only to the list of substances; the right hand columns (Methods and PQL) are given for informational purposes only. See also footnotes 5 and 6.

²Common names are those widely used in government regulations, scientific publications, and commerce; synonyms exist for many chemicals.

³Chemical Abstracts Service registry number. Where "Total" is entered, all species in the groundwater that contain this element are included.

⁴CAS index names are those used in the 9th Collective Index.

⁵Suggested Methods refer to analytical procedure numbers used in EPA Report SW-846 "Test Methods for Evaluating Solid Waste", third edition, November 1986, as revised, December 1987. Analytical details can be found in SW-846 and in documentation on file at the Department. CAUTION: The methods listed are representative SW-846 procedures and may not always be the most suitable method(s) for monitoring an analyte under the regulations.

⁶Practical Quantitation Limits (PQLs) are the lowest concentrations of analytes in groundwaters that can be reliably determined within specified limits of precision and accuracy by the indicated methods under routine laboratory operating conditions. The PQLs listed are generally stated to one significant figure. PQLs are based on 5 mL samples for volatile organics and 1 L samples for semivolatile organics. CAUTION: The PQL values in many cases are based only on a general estimate for the method and not on a determination for individual compounds; PQLs are not a part of the regulation.

⁷This substance is often called Bis(2-chloroisopropyl) ether, the name Chemical Abstracts Service applies to its noncommercial isomer, Propane, 2,2"-oxybis[2-chloro- (CAS RN 39638-32-9)

⁸Chlordane: This entry includes alpha-chlordane (CAS RN 5103-71-9), beta-chlordane (CAS RN 5103-74-2), gamma-chlordane (CAS RN 5566-34-7), and constituents of chlordane (CAS RN 57-74-9 and CAS RN 12789-03-6). PQL shown is for technical chlordane. PQLs of specific isomers are about 20 ug/L by method 8270.

⁹Polychlorinated biphenyls (CAS RN 1336-36-3); this category contains congener chemicals, including constituents of Aroclor 1016 (CAS RN 12674-11-2), Aroclor 1221 (CAS RN 11104-28-2), Aroclor 1232 (CAS RN 11141-16-5), Aroclor 1242 (CAS RN 53469-21-9), Aroclor 1248 (CAS RN 12672-29-6), Aroclor 1254 (CAS RN 11097-69-1), and Aroclor 1260 (CAS RN 11096-82-5). The PQL shown is an average value for PCB congeners.

¹⁰Toxaphene: This entry includes congener chemicals contained in technical toxaphene (CAS RN 8001-35-2), i.e., chlorinated camphene.

¹¹Xylene (total): This entry includes o-xylene (CAS RN 96-47-6), m-xylene (CAS RN 108-38-3), p-xylene (CAS RN. 106-42-3), and unspecified xylenes (dimethylbenzenes) (CAS RN 1330-20-7). PQLs for method 8021 are 0.2 for o-xylene, and 0.1 for m- or p-xylene. The PQL for m-xylene is 2.0 ug/L by method 8020 or 8260.

**TABLE 7. WDEQ/SHWD APPENDIX C CONSTITUENT LIST
HAPPY JACK LANDFILL, CHEYENNE, WYOMING**

Revised August 23, 2017

Constituents	Analytical Method
<u>Constituents for Detection Monitoring</u>	
pH ¹	
Ammonia as N ¹	350.1
Chloride ¹	300.0A
Manganese ¹	SW 6010B
Sulfate ¹	300.0A
Total Dissolved Solids (TDS) ¹	SM 2540C
Bicarbonate ²	SM 2320B
Fluoride ¹	300.0A
Nitrate as N ¹	EPA 353.2
Chemical Oxygen Demand (COD) ²	410.4
Calcium ²	SW 6010B
Iron ¹	SW 6010B
Potassium ²	SW 6010B
Total Organic Carbon (TOC) ²	SM 5310B
Carbonate ²	SM 2320B
Magnesium ²	SW 6010B
Sodium ²	SW 6010B

¹ Constituents that may have a class of use based limit in Chapter 8 of the Wyoming Water Quality Rules and Regulations and/or an MCL.

² Constituents that may be used to characterize and compare groundwater quality. These constituents are useful in determining the similarities and/or differences in the composition of water from specific hydrogeologic units and may help show whether particular units are hydraulically separate or connected. These constituents may be used to classify natural waters and help differentiate between natural variability and a release from a landfill.

**TABLE 8. GROUNDWATER WELLS PERMITTED FOR DOMESTIC OR STOCK WATERING PURPOSES
WITHIN 1/2 MILE OF EXPANSION AREA,
HAPPY JACK LANDFILL, CHEYENNE, WYOMING**

SEO PERMIT Number	Permittee	Application Year	Permittee Address		
22711	Gary J. & Linda Jo Denney	1973	1466 Happy Jack Road	Cheyenne, WY	82007
112671	Margaret M. Grayer	1998	1498 Happy Jack Road	Cheyenne, WY	82009
49990	Susan E. & Ellis E. Hart	1979	1010 Bade Road	Cheyenne, WY	82009
179101	William S. Walker	1995	1497 Happy Jack Road.	Cheyenne, WY	82009
36775	Carl Bade	1977	1016 Bade Road	Cheyenne, WY	82009
290	Board of Public Utilities	1946	Koppes No. 6	Cheyenne, WY	82009
23571	City of Cheyenne	1973	Happy Jack Landfill	Cheyenne, WY	82009
43659	Chris C. and Teresa A. Bade	1978	3703 Charles St	Cheyenne, WY	82001
231	Arthur L. King	1905	PO BOX 905	Cheyenne, WY	82003
74560	Board of Public Utilities	1986	PO Box 1469	Cheyenne, WY	82003

Notes:

Data derived from the Wyoming State Engineer's Office Website, December 12, 2016

ft-bgs - feet below ground surface

**TABLE 9. BACKGROUND AND COMPLIANCE MONITORING WELLS
FOR INTERWELL STATISTICAL COMPARISONS
HAPPY JACK LANDFILL, CHEYENNE, WYOMING**

Hydrogeologic Zone	Background (Up-gradient) Monitoring Wells	Compliance (Down-gradient) Monitoring Wells
Semi-Confined Aquifer	C-1	C-2 C-3 GMW-03 GMW-04 GMW-06 GMW-07
Principal Unconfined Aquifer	PU-1 PU-3 P3-1 P3-3	SRM-4 SRM-5 SRM-7 SRM-8 PU-2 PU-4
Perched 1 Water Bearing Zone	P-5 S-9	SRM-6 EE-04

CITY OF CHEYENNE
CONTRACT MODIFICATION

APPROVED

By Alessandra at 2:57 pm, Apr 20, 2020

Contract Modification: # 1
Project: Happy Jack Landfill Environmental Monitoring
Project Number: Contract # 6798
Contractor: Trihydro
Date Prepared: April 13, 2020

The above referenced contract is amended or modified as set forth in the following and these amendments and modifications of the plans and specifications are agreed to by the parties. All other aspects of the original agreement remain unchanged and in full force and effect.

I. Description of Changes

Modification would amend Contract to have contractor perform off-site environmental monitoring of surrounding property owners within a half-mile of the Landfill and extend the contract from March 1st, 2020, to December 31st, 2021

II. New Contract Items

See attached Exhibit A1 for description of work services and cost estimate.

III. Schedule

Date for Contract renewal, including additional work as described in Exhibit A1, to be completed by December 31st, 2021.

IV. Summary

Original Contract Amount.....	\$171,912.00
This Modification.....	\$58,795.00
New Contract Amount...	\$230,707.00

V. Approvals

City of Cheyenne

DATE: _____

Mayor Marian J. Orr / ATTEST: City Clerk Kristina Jones



Recommend approval by Project Manager

Trihydro



EXHIBIT A1

Contract #6798 Amendment
Happy Jack Landfill Environmental Monitoring

Abstract

Offsite Environmental Monitoring for Residential wells within ½ mile.
Cost Estimate.

Matt Theriault

mtheriault@cheyennecity.org

(307) 637-6279

Number of Pages

7



April 1, 2020

Mr. Matt Theriault, Landfill Manager
City of Cheyenne
2101 O'Neil Avenue, Rm. 206
Cheyenne, WY 82001

RE: Amendment to Professional Services Agreement, #6798: Happy Jack Landfill Environmental Monitoring

Dear Mr. Theriault:

Trihydro Corporation (Trihydro) is providing an amendment request for the City of Cheyenne Professional Services Agreement #6798 for the Happy Jack Landfill Environmental Monitoring project. As outlined in Trihydro's original proposal under Request for Proposal, RFP-10-18, anticipated work included assistance with off-site water well monitoring. Costs were not originally included in the proposal per the City's direction because of the uncertainty regarding the timing for commencement of the off-site monitoring program and details of the scope of work. Based on recent conversations with the City, Trihydro better understands the scope and has prepared this contract amendment request to assist with the upcoming task. The off-site water well monitoring program is required by the Wyoming Department of Environmental Quality. The scope includes four items:

- Task 1: Landowner well identification and notification. Trihydro will verify private and public drinking and stock water wells within one-half mile of the landfill expansion area boundary for inclusion in the program. Trihydro will prepare landowner notifications and information for the City's use to gauge landowner's interest in participating in the off-site water well monitoring program. Trihydro will track responses from potential program participants.
- Task 2: Landowner meeting. Trihydro will prepare a presentation and hold a landowner meeting (virtual or in-person) to discuss the off-site well monitoring program and register program participants.
- Task 3: Preparation of Sampling and Analysis Plan (SAP). Trihydro will tour program wells and develop a sampling methodology for each individual well location. A SAP will then be prepared and submitted to WDEQ for approval.
- Task 4: Quarterly sampling and reporting. Each well enrolled in the program will be sampled on a quarterly basis for one year. Program participants will be provided with water quality information for their respective wells. If necessary, a landowner meeting will be conducted at the conclusion of the sampling program to answer landowner questions regarding the data and to proactively address any concerns.



Mr. Matt Theriault
April 1, 2020
Page 2

A detailed cost estimate, including activities and assumptions, is provided as Attachment A. Our cost estimate for conducting the above tasks is \$58,795 on a time and materials, not-to-exceed basis. Trihydro will strive to complete the requested activities as efficiently and cost-effectively as possible, and we are available to start the work upon receipt of a signed contract amendment. The current contract expires on March 1st 2021. Trihydro also requests that the contract be extended to December 31st, 2021, to ensure completion of the off-site monitoring program scope acknowledging that there may be delays related to access given current social distancing advisements. If you have any questions or require additional information regarding this contract amendment request, please do not hesitate to contact us at (307) 745-7474.

Sincerely,
Trihydro Corporation

Caroline Brewer, P.G.
Project Manager

Marjorie Bedessem, P.E., Ph.D.
Project Director

427-010-007

Enclosure

ATTACHMENT A
COST ESTIMATE

ATTACHMENT A. COSTS AND ASSUMPTIONS FOR OFF-SITE WELL MONITORING PROGRAM SET-UP AND FIELDWORK, HAPPY JACK LANDFILL, CHEYENNE, WYOMING



Activity		Schedule of Charges	Hourly Rate or Charge	Hours or Unit Estimate	Subtotal
Task 6A	Landowner Well Identification and Notification				
Activities:					
- Conduct desktop review of down-gradient and cross-gradient domestic and stock wells within one-half mile of landfill expansion area boundary		Technical Specialist 1	\$201	3	\$603.00
- Prepare a form letter and survey for affected homeowners to identify those who want to be part of the program		Professional Level 7	\$131	10	\$1,310.00
- Conduct tracking regarding landowner interest in program		Professional Level 5	\$104	5	\$520.00
		Professional Level 4	\$94	0	\$0.00
		Technical Level 5	\$93	0	\$0.00
		Administrative 2	\$62	2	\$124.00
Assumptions:		Labor			\$2,557.00
- No more than 15 landowners are identified as wanting to participate in the program		Equipment		0	\$0.00
- The City can provide initial contact information for the identified landowners		Travel		0	\$0.00
- The City will conduct mailing and file related tracking documents		Per Diem		0	\$0.00
- Two mailings will be conducted by City, if needed, to reach individual landowners		Miscellaneous		0	\$0.00
		Expenses			\$0.00
		Other		0	\$0.00
		Subcontractors			\$0.00
		Activity Task 6A Subtotal			\$2,557.00

ATTACHMENT A. COSTS AND ASSUMPTIONS FOR OFF-SITE WELL MONITORING PROGRAM SET-UP AND FIELDWORK, HAPPY JACK LANDFILL, CHEYENNE, WYOMING



Task 6B		Activity	Schedule of Charges	Hourly Rate or Charge	Hours or Unit Estimate	Subtotal		
Task 6B	Landowner Meeting	Activities: - Present at one landowner meeting in Cheyenne to introduce to program - Prepare presentation and short document to facilitate meeting discussion	Technical Specialist 1	\$201	8	\$1,608.00		
			Professional Level 7	\$131	25	\$3,275.00		
			Professional Level 5	\$104	8	\$832.00		
			Professional Level 3	\$84	0	\$0.00		
			Technical Level 5	\$93	5	\$465.00		
			Administrative 2	\$62	4	\$248.00		
				Labor				\$6,428.00
				Equipment		0	\$0.00	
				Travel		0	\$0.00	
				Per Diem		0	\$0.00	
				Miscellaneous		0	\$0.00	
				Expenses			\$0.00	
				Other		0	\$0.00	
				Subcontractors			\$0.00	
Activity Task 6B Subtotal						\$6,428.00		

- Activities:
- Present at one landowner meeting in Cheyenne to introduce to program
 - Prepare presentation and short document to facilitate meeting discussion
- Assumptions:
- Only one trip to Cheyenne will be required for meeting
 - No significant follow-up correspondence will be required by landowners

ATTACHMENT A. COSTS AND ASSUMPTIONS FOR OFF-SITE WELL MONITORING PROGRAM SET-UP AND FIELDWORK, HAPPY JACK LANDFILL, CHEYENNE, WYOMING



Activity		Schedule of Charges	Hourly Rate or Charge	Hours or Unit Estimate	Subtotal
Task 6C	Preparation of Sampling and Analysis Plan (SAP)				
	Activities:				
	- Assist City with well site inspections and development of well-by-well specific sampling procedures	Technical Specialist 1	\$201	6	\$1,206.00
	- Develop and secure approved SAP for off-site well monitoring program	Professional Level 9	\$161	2	\$322.00
	- Update Health and Safety Plan depending on required off-site well activities	Professional Level 7	\$131	12	\$1,572.00
		Professional Level 5	\$104	0	\$0.00
		Professional Level 3	\$84	15	\$1,260.00
		Technical Level 5	\$93	23	\$2,139.00
		Technical Level 2	\$62	2	\$124.00
		Administrative 2	\$62	2	\$124.00
	Assumptions:				
	- SAP will require only one set of revisions through Wyoming Department of Environmental Quality				
	- No more than 15 landowner wells will require inspection and inclusion in SAP				
	- Fieldwork will take no more than three days				
	Labor				\$6,747.00
	Equipment			0	\$0.00
	Vehicle Use		\$86	3	\$258.00
	Per Diem		\$45	3	\$135.00
	Miscellaneous			0	\$0.00
	Expenses				\$393.00
	Other			0	\$0.00
	Subcontractors				\$0.00
	Activity Task 6C Subtotal				\$7,140.00

ATTACHMENT A. COSTS AND ASSUMPTIONS FOR OFF-SITE WELL MONITORING PROGRAM SET-UP AND FIELDWORK, HAPPY JACK LANDFILL, CHEYENNE, WYOMING



Task 6D		Activity	Schedule of Charges	Hourly Rate or Charge	Hours or Unit Estimate	Subtotal
	Quarterly Sampling and Reporting for One Year		Technical Specialist 1	\$201	16	\$3,216.00
			Professional Level 7	\$131	36	\$4,716.00
			Professional Level 5	\$104	0	\$0.00
			Professional Level 3	\$84	15	\$1,260.00
			Technical Level 5	\$93	120	\$11,160.00
			Administrative 2	\$62	12	\$744.00
			Labor			\$21,096.00
			Equipment	\$50	4	\$200.00
			Vehicle Use	\$86	16	\$1,376.00
			Per Diem	\$45	0	\$0.00
	Miscellaneous		0	\$0.00		
	Expenses			\$1,576.00		
		Laboratory	\$18,180	1	\$19,998.00	
		Subcontractors			\$19,998.00	
		Activity Task 6D Subtotal			\$42,670.00	

Trihydro Total	Add Client Info	TOTAL
\$2,557	\$0	\$2,557
\$6,428	\$0	\$6,428
\$7,140	\$0	\$7,140
\$42,670	\$0	\$42,670
Total	\$0	\$58,796

Notes:
- Information management will be billed at 3% of labor, and includes the cost for Project Direct and Cabinet archiving

Remember:
- If adding fees, remember to add a summary item to the task summary section.

APPROVED

By Alessandra at 8:56 am, Jan 20, 2021

CITY OF CHEYENNE
CONTRACT MODIFICATION

Contract Modification: # 2
Project: Happy Jack Landfill Environmental Monitoring
Project Number: Contract # 6798
Contractor: Trihydro
Date Prepared: January 19, 2021

The above referenced contract is amended or modified as set forth in the following and these amendments and modifications of the plans and specifications are agreed to by the parties. All other aspects of the original agreement remain unchanged and in full force and effect.

I. Description of Changes

The changes provide for the renewal of the Contract #6798, Happy Jack Environmental Monitoring. The Contract will be extended for an additional three (3) years, from March 1, 2021 through March 1, 2024. Trihydro agrees to perform all onsite well monitoring and testing as detailed in the original contract.

In addition, this contract modification will extend the timeframe for Trihydro to perform offsite well monitoring for an additional year, from March 1, 2021 to March 1, 2022.

This modification will increase the amount of the contract by \$230,707.00.

Original Contract Amount:	\$171,912.00
Modification #1:	\$ 58,795.00
Contract Amount Prior to this Modification:	\$230,707.00
This modification:	\$230,707.00
New Contract Amount:	\$461,414.00

II. New Contract Items

All items to remain the same as original contract.

III. Schedule

Groundwater Sampling and Analysis: Sampling shall be performed in the field on the following months:

April	2021
August	2021
April	2022
August	2022
April	2023
August	2023

Annual Reports: Consultant shall submit a draft copy (electronic is acceptable) of a comprehensive annual report to the City by March 1 of each contract year. Consultant shall incorporate City comments and submit one electronic and one paper copy of final report to the City and electronic deliverables to WDEQ per the requirements by April 1 of each contract year.

Methane Monitoring: Consultant shall perform methane monitoring in the Construction Demolition Recycling Facility, Landfill Equipment Shelter Shop Area and the fourteen (14) methane monitoring wells surrounding the permitted boundary. Methane monitoring shall be performed in the field during the following months:

March	2021
June	2021
September	2021
December	2021
March	2022
June	2022
September	2022
December	2022
March	2023
June	2023
September	2023
December	2023

IV. Summary

This modification increases the amount of the contract by \$230,707.00 and extends the contract for one additional three-year period. This modification includes groundwater sampling and analysis, groundwater gauging, methane monitoring and reports at the same frequency and number as included in original contract. The total cost of the contract will not exceed \$461,414.00 without a written change order signed by both parties.

V. Approvals

DATE: 02-11-2021

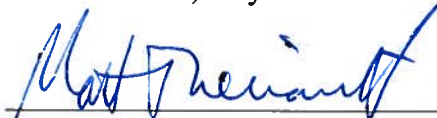
City of Cheyenne



Patrick Collins, Mayor



ATTEST: City Clerk Kristina Jones



Recommend approval by Project Manager



Accepted by Trihydro Corporation